

SingleRAN

Super Uplink Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Deleted descriptions of the mutually exclusive relationship between super uplink with SUL and NR FDD co-deployment and LNR uplink precise RB estimation. For details, see [5.3.2.2 Mutually Exclusive Functions](#).
Added descriptions of the cell requirements for configuring the **NRDUCellGigaBand.LnrUIRbEstCorrectCoeff** parameter. For details, see [5.3.5 Cells](#).
Revised descriptions of the mutually exclusive relationship between super uplink and PDCCH rate matching. For details, see [4.3.2.2 Mutually Exclusive Functions](#).
Added the impact relationships between super uplink with independent SUL deployment and the following functions: user-rate-based dynamic network slice resource management and user-rate-based dynamic QCI resource management. For details, see [4.3.2.3 Function Impacts](#).
Revised other descriptions in this document.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 07 (2025-11-07).

- Technical changes

Change Description	Parameter Change	Base Station Model
Added support for simultaneous use with UL and DL Decoupling when super uplink with SUL and NR FDD co-deployment is enabled. For details, see: • 5.1.3.4 Simultaneous Use with UL and DL Decoupling • 5.3.2.2 Mutually Exclusive Functions	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for 2CC SRS carrier switching for SUL and CA. For details, see: • 4.1.3.2.3 SRS Carrier Switching • 4.2.1 Benefits • 4.2.2 Impacts • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands • 5.2.1 Benefits • 5.2.2 Impacts	Modified parameter: Added the SUL_CA_2CC_SRS_SWITCHING_SW option to the NRDUCellCollabServ.SulAlgoSwitch parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Replaced the reserved parameter for configuration optimization for SUL UEs with a formal parameter in SUL and NR FDD co-deployment mode. For details, see: • 5.1.1.3 Carrier Management • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameters: • Added the RSVDSWPARAM1_BIT1 option of the NRDUCellRsrdExt01.RsrdSwParam1 parameter to the parameter disuse list. • Added the SUL_CONFIG_OPT_SW option to the NRDUCellCollabServ.SulAlgoSwitch parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added an impact relationship between super uplink and time-domain power saving BWP. For details, see: • 4.3.2.3 Function Impacts • 5.3.2.3 Function Impacts	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Revised the constraints affecting cell activation due to license insufficiency. For details, see 4.3.1 Licenses and 5.3.1 Licenses.	None	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between super uplink with SUL and NR FDD co-deployment and LNR uplink precise RB estimation. For details, see 5.3.2.2 Mutually Exclusive Functions.	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between super uplink and RF module fast deep dormancy in SUL and NR FDD co-deployment mode. For details, see 5.3.2.3 Function Impacts.	None	3900 and 5900 series base stations (macro base stations)
Modified the way in which the gNodeB measures random-access-related counters of NR TDD cells in SUL scenarios. For details, see 4.2.2 Impacts.	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added support for enhanced SRS carrier switching for SUL and downlink CA. For details, see: • 4.1.3.2.2 SUL and CA Optimal Carrier Selection • 4.2.1 Benefits • 4.2.2 Impacts • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands • 5.2.1 Benefits • 5.2.2 Impacts	Modified parameter: Added the SUL_CA_SRS_SWITCHING_EXT_SW option to the NRDUCellCollabServ.SulAlgoSwitch parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between super uplink and flexible experience differentiation. For details, see 4.3.2.3 Function Impacts and 5.3.2.3 Function Impacts.	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for fast SUL in SUL and NR FDD co-deployment mode. For details, see: • 5.1.2.7 Fast SUL • 5.2.1 Benefits • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the FAST_SUL_SW option to the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for SUL CoMP to take effect for UEs with specific QCs. For details, see: • 4.1.2.2 SUL CoMP • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands • 5.1.2.1 SUL CoMP • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Added the NRDUCellComp.CompConfigPol parameter. Modified parameter: Added the COMP_SW option to the gNBQciBearer.ServiceDiffAlgoSwitch parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Replaced the option of a reserved parameter controlling uplink optimal carrier selection for SUL and CA with an option of a formal parameter. For details, see: • 4.1.3.2.2 SUL and CA Optimal Carrier Selection • 4.2.2 Impacts • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands • 5.2.2 Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the RSVDSWPARAM1_BIT3 option of the NRCeRsvd.RsvdSwParam1 parameter to the parameter disuse list and replaced it with the SUL_CA_BASED_UL_CARR_SEL_SW option of the NRCeCaMgmtConfig.CaStrategySwitch parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

Moved the descriptions of simultaneous support for SUL and SRS carrier switching from section "SUL and CA Optimal Carrier Selection" to [4.1.3.2.3 SRS Carrier Switching](#).

Corrected the notes for deactivating the fast SUL function. For details, see [5.4.1.2 Using MML Commands](#).

Deleted descriptions of the mutually exclusive relationship between super uplink and smart PDCCH symbol allocation. For details, see [4.3.2.2 Mutually Exclusive Functions](#) and [5.3.2.2 Mutually Exclusive Functions](#).

Added descriptions of the mutually exclusive relationship between super uplink and downlink avoidance enhancement mode 2. For details, see [4.3.2.2 Mutually Exclusive Functions](#) and [5.3.2.2 Mutually Exclusive Functions](#).

Revised other descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to NR.

2.3 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-030211	Super Uplink	4 Independent SUL Deployment 5 SUL and NR FDD Co-Deployment

2.4 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Super uplink	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	4 Independent SUL Deployment 5 SUL and NR FDD Co-Deployment

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Super uplink	Supported only in low frequency bands	4 Independent SUL Deployment 5 SUL and NR FDD Co-Deployment

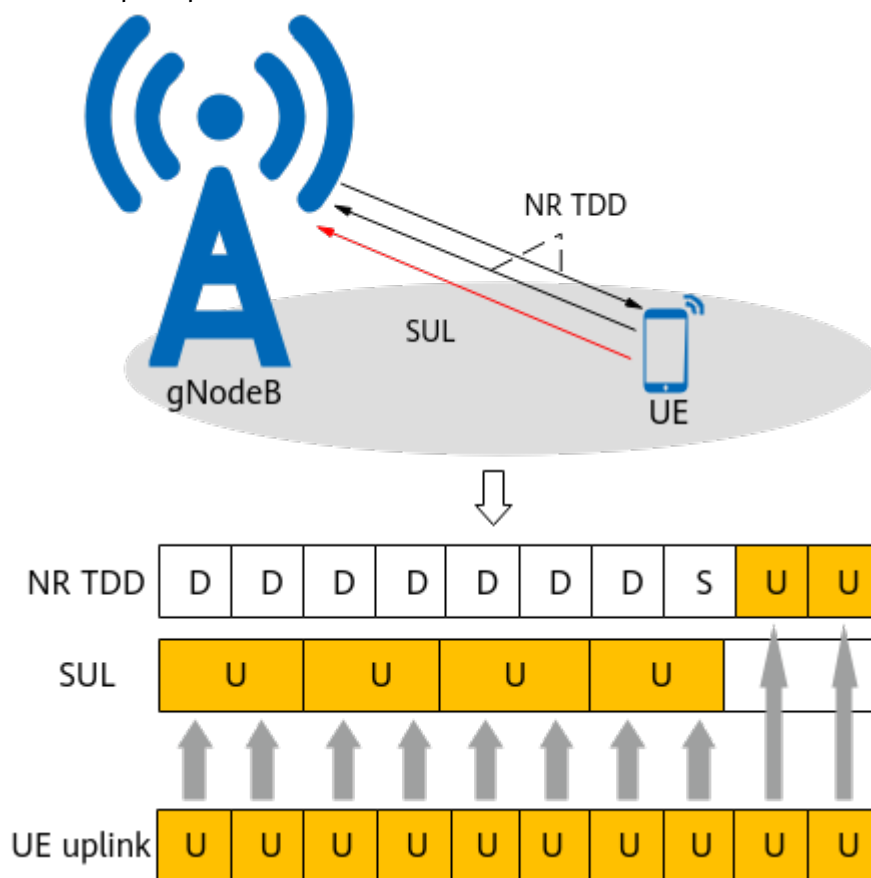
3 Overview

3.1 Background and Definition

Current 5G networks mainly use NR TDD mode, with uplink and downlink data transmitted using the same spectrum segment through time division multiplexing (TDM). However, the time-frequency resources used for uplink transmission are limited, thereby affecting uplink user experience.

Super uplink enables uplink data to be transmitted over the NR TDD spectrum and the supplementary uplink (SUL) spectrum in different slots, greatly increasing the time-frequency resources available to 5G UEs for uplink transmission. Specifically, uplink data can be transmitted in all possible slots, allowing UEs to send uplink data using uplink slots in the NR TDD spectrum and idle SUL spectrum resources that correspond to NR TDD downlink and self-contained slots. For details, see [Figure 3-1](#).

Figure 3-1 Super uplink



In the current version:

- The gNodeB supports SA networking or NSA and SA hybrid networking.
- The slot configurations of 4:1, single-period 8:2, and dual-period 8:2 are supported by NR TDD carriers.

3.2 Deployment Mode

Super uplink can be enabled with either independent SUL deployment or SUL and NR FDD co-deployment.

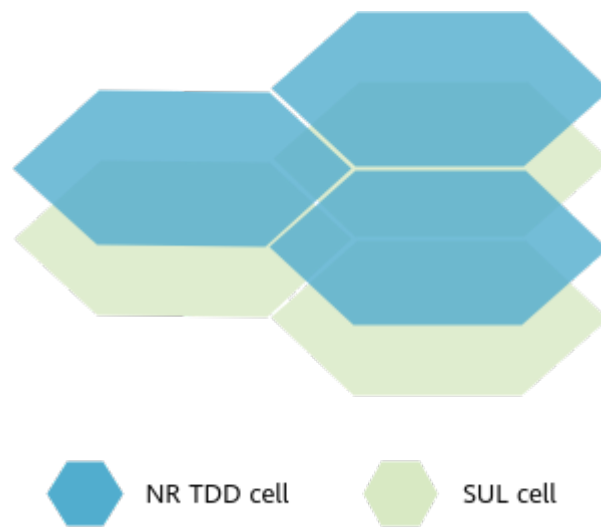
The main difference between the two deployment modes lies in the method of obtaining SUL spectrum resources.

Independent SUL Deployment

In this deployment mode, the SUL cell needs to be deployed independently, and NR TDD and SUL cells must be co-sited and cover the same area and they are always paired to implement the super uplink function, as shown in [Figure 3-2](#). This deployment mode is recommended when the operator has independent SUL spectrum resources or can obtain these resources through LTE FDD and NR Uplink Spectrum Sharing.

For details about LTE FDD and NR Uplink Spectrum Sharing, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

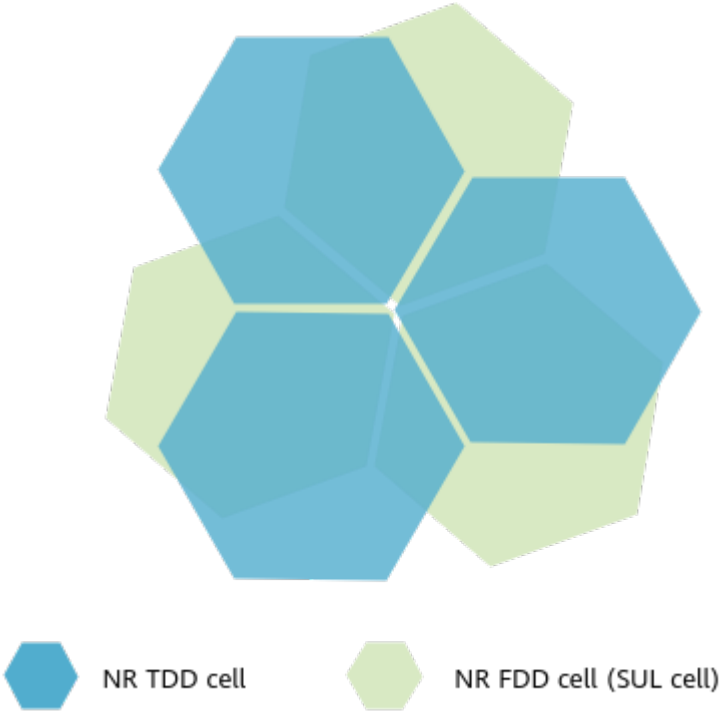
Figure 3-2 Independent SUL deployment



SUL and NR FDD Co-Deployment

In this deployment mode, the SUL link is provided by the NR FDD cell. That is, SUL and NR FDD are deployed in the same cell, as shown in [Figure 3-3](#). Super uplink is implemented based on flexible pairing between NR TDD cells and SUL cells (NR FDD cells). As such, neither the independent deployment of an SUL cell nor the co-coverage deployment of an NR TDD cell and an SUL cell (NR FDD cell) is required. In addition, an NR TDD cell and an SUL cell can be served by the same base station or different base stations. This deployment mode is recommended when the operator has spectrum resources available for both NR FDD and SUL and has established the NR FDD cell.

Figure 3-3 SUL and NR FDD co-deployment



4 Independent SUL Deployment

This chapter describes super uplink with independent SUL deployment. Super uplink described in this chapter refers to super uplink with independent SUL deployment.

4.1 Principles

[4.1.1 Basic Functions](#) and [4.1.2 Enhanced Functions](#) describe the basic and enhanced functions of super uplink, respectively. Enhanced functions are provided to solve issues in independent SUL deployment scenarios and further increase the benefits achieved by super uplink.

Super uplink can work together with other functions. [4.1.3 Simultaneous Use with Other Functions](#) describes function interaction.

[Table 4-1](#) lists the combinations of NR TDD and SUL frequency bands supported by the gNodeB. For details about the mapping between frequency bands and frequencies, see 3GPP TS 38.104. The bandwidth requirement of each frequency band is described in [RF Modules](#) in [4.3.3 Hardware](#).

Table 4-1 Combinations of NR TDD and SUL frequency bands for super uplink

NR TDD Frequency Band	SUL Frequency Band	Applicable Base Station Model
n78	n80	Macro base stations
	n82	Macro base stations
	n83	Macro base stations
	n84	Macro base stations
	n86	Macro base stations
n41	n80	Macro base stations
	n81	Macro base stations

NR TDD Frequency Band	SUL Frequency Band	Applicable Base Station Model
	n83	Macro base stations

4.1.1 Basic Functions

Super uplink is enabled for an NR TDD cell by selecting the **SUPER_UL_TDM_SCH_SW** option of **NRDUCellAlgoSwitch**. *SuperUplinkSwitch*.

If the gNodeB is shared by multiple operators (the **gNBSharingMode**.*gNBMultiOpSharingMode* parameter is set to a value other than **INDEPENDENT**), operators requiring super uplink must also select the **OP_SUL_SWITCH** option of the **NRDUCellOp**.*PolicySwitch* parameter for the NR TDD cell.

The differences between an SUL cell and a common NR TDD cell lie in the uplink, including cell configuration, carrier information, random access, uplink TDM scheduling, uplink power control, timing advance, and mobility management.

4.1.1.1 SUL Cell Configuration and Carrier Information

Before enabling super uplink, an SUL cell must be configured and associated with an NR TDD cell in order to form a super uplink cell. When UEs access the super uplink cell, the gNodeB must transmit SUL carrier information to UEs as well to ensure UEs' use of this carrier.

SUL Cell Configuration

[Table 4-2](#) describes the parameters that must be configured for an SUL cell.

Table 4-2 Mandatory parameters for an SUL cell

Parameter Name	Parameter ID	Description
Duplex Mode	NRDUCell . <i>DuplexMode</i>	This parameter indicates the duplex mode of a cell. For an SUL cell, this parameter must be set to CELL_SUL .
Frequency Band	NRDUCell . <i>FrequencyBand</i>	This parameter indicates the frequency band in which a cell operates. For the frequency bands supported by an SUL cell, see 4.1 Principles .
Uplink NARFCN	NRDUCell . <i>ULNarfcn</i>	For an SUL cell, only the uplink NR-ARFCN needs to be configured.

Parameter Name	Parameter ID	Description
Uplink Bandwidth	NRDUCell.UlBandwidth	Uplink bandwidth varies according to the operating frequency band. For details, see RF Modules in 4.3.3 Hardware .
Subcarrier Spacing	NRDUCell.SubcarrierSpacing	For an SUL carrier, this parameter must be set to 15KHZ .
Logical Root Sequence Index	NRDUCell.LogicalRootSequenceIndex	The parameter is used to generate a cell preamble sequence. For details about Logical Root Sequence Index, see <i>Cell Management</i> .

After the SUL cell is set up, it can be associated with an NR TDD cell using the **NRDUCellSulRelation.NrDuCellId** and **NRDUCellSulRelation.SulNrDuCellId** parameters. The **NRDUCellSulRelation.SulType** parameter must be set to **FIX_SUL** for the NR TDD cell. For details about NR TDD cell setup and operating mechanisms, see "Establishing a TDD Cell" in *Cell Management*.

SUL Carrier Information

SUL carrier information is carried in the following messages:

- System Information Block 1 (SIB1) for UEs in idle mode. For details, see section 5.2 "System information" in 3GPP TS 38.331 V15.4.0.
- RRCReconfiguration message for UEs in connected mode. For details, see section 5.3.5 "RRC reconfiguration" in 3GPP TS 38.331 V15.4.0.

4.1.1.2 Random Access

Random access is used to establish and recover the uplink synchronization between UEs and the gNodeB.

When super uplink with independent SUL deployment is enabled but UL and DL Decoupling is not enabled, the value of the *rsrp-ThresholdSSB-SUL* IE contained in SIB1 is set to the minimum value (–156 dBm). As the value of –156 dBm is definitely less than the camping threshold specified by the **NRDUCellSelConfig.MinimumRxLevel** parameter, UEs perform the random access procedure only on the NR TDD carrier.

Random access procedures for UEs in idle mode and connected mode are different. For details about NR TDD random access, see the sections describing random access in *5G Networking and Signaling* in *5G RAN Feature Documentation*.

For details about the random access procedure when both super uplink and UL and DL Decoupling are enabled, see [4.1.3.1 Simultaneous Use with UL and DL Decoupling](#).

For details about UL and DL Decoupling, see *UL and DL Decoupling*.

Random Access of a UE in Idle Mode

When a UE in idle mode triggers a random access procedure, the downlink SSB reference signal received power (RSRP) of the NR TDD carrier measured by the UE is larger than the value of the *rsrp-ThresholdSSB-SUL* IE. The UE initiates a random access procedure on the NR TDD carrier.

Random Access of a UE in Connected Mode

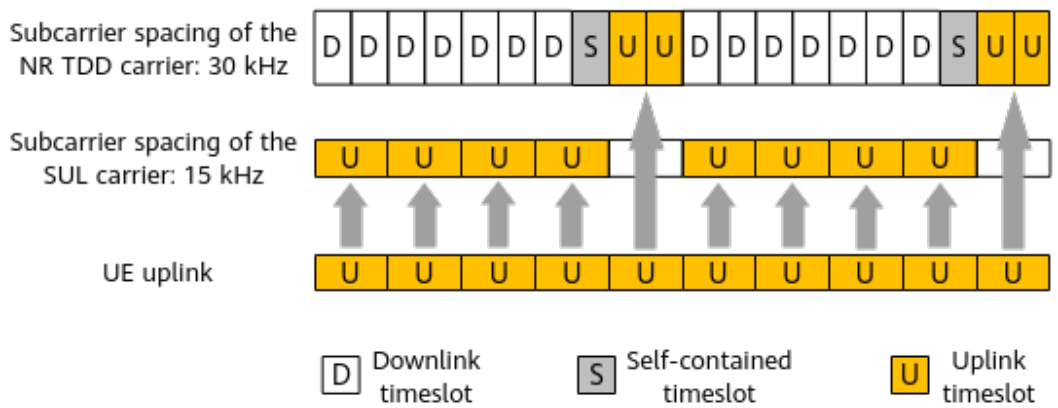
During a handover for a UE supporting super uplink, the gNodeB sends an RRCReconfiguration message to the UE in cases where only super uplink is enabled in the target cell, indicating the NR TDD carrier used for random access. The procedure is as follows:

- 1. The source gNodeB delivers intra-RAT event A3 measurement configurations to the UE, instructing it to measure the signal strength of neighboring NR cells.
- 2. After receiving a measurement report back from the UE, the source gNodeB forwards the RSRP of the target cell to the target gNodeB, which finds that the RSRP is larger than the value of the *rsrp-ThresholdSSB-SUL* IE. The target gNodeB determines that the UE initiates a random access procedure on the NR TDD carrier.
- 3. The target gNodeB notifies the source gNodeB of the NR TDD carrier information of the target cell, and the source gNodeB sends an RRCReconfiguration message to the UE.
- 4. The UE initiates a random access procedure based on the carrier information of the target cell in the RRCReconfiguration message.

4.1.1.3 Uplink TDM Scheduling

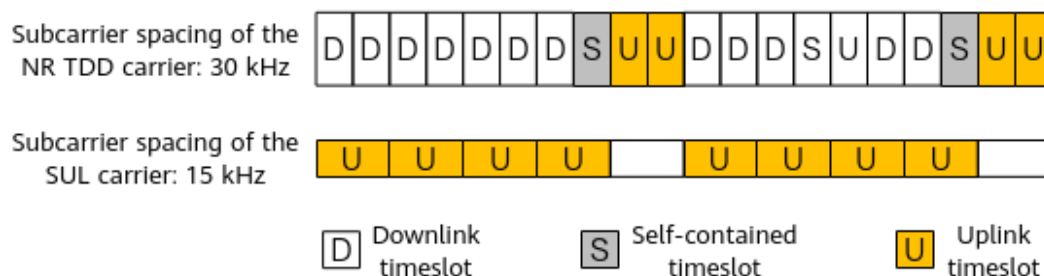
After super uplink takes effect, downlink data transmission is carried on the NR TDD carrier, and uplink data transmission can be carried on the NR TDD and SUL carriers. Specifically, uplink data is transmitted on the SUL carrier in the downlink timeslots of the NR TDD carrier and in the downlink symbols of the self-contained timeslots, and is transmitted on the NR TDD carrier in the uplink timeslots of the NR TDD carrier, as shown in [Figure 4-1](#).

Figure 4-1 TDM scheduling with super uplink enabled



The subcarrier spacing of the NR TDD carrier is 30 kHz, while that of the SUL carrier is 15 kHz. As such, the scheduling time sequence must be considered during the scheduling process. [Figure 4-2](#) uses a single-period 8:2 slot configuration as an example.

Figure 4-2 Scheduling time sequence when the SUL subcarrier spacing is 15 kHz and the NR TDD slot configuration is single-period 8:2

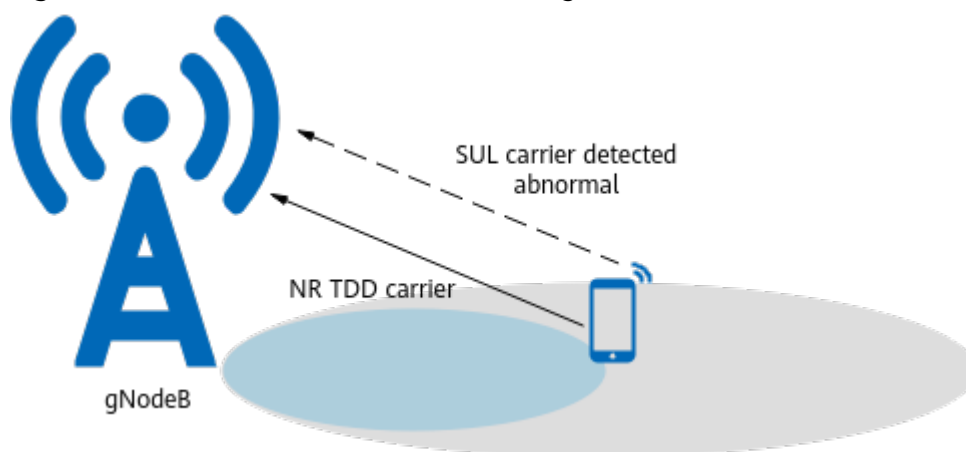


SUL Carrier Detection and Management

In scenarios where super uplink is enabled, gNodeB's scheduling on the SUL carrier for a UE will cause uplink bit errors if the SUL carrier is unavailable due to exceptions.

The SUL carrier is detected in real time, as shown in [Figure 4-3](#).

Figure 4-3 SUL carrier detection and management



- The gNodeB does not perform uplink scheduling on an abnormal SUL carrier, and the UE's uplink data is carried only on the NR TDD carrier.
- The gNodeB will resume uplink scheduling on the SUL carrier once the SUL carrier has recovered. The UE's uplink data is then carried on NR TDD and SUL carriers in different timeslots.

SUL carrier detection and management is controlled by the **SUL_FAULT_TOLERANCE_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

The SUL carrier for a UE is considered abnormal if either of the following conditions is met. Otherwise, the SUL carrier is considered normal.

- The gNodeB detects that the UE's SRS SINR is less than the result of the **NRDUCellCollabServ.SulSinrLowerLimit** parameter value minus the **NRDUCellCollabServ.SulSinrLowerLimitOffset** parameter value.
- The gNodeB detects that the UE's SRS SINR is greater than the **NRDUCellCollabServ.SulDtxSinrLowerLimit** parameter value and discontinuous transmission (DTX) is used on the SUL carrier.

The SUL carrier is considered recovered if the gNodeB detects that the UE's SRS SINR is greater than the sum of the **NRDUCellCollabServ.SulSinrLowerLimit** and **NRDUCellCollabServ.SulSinrLowerLimitOffset** parameter values, and DTX is not used on the SUL carrier.

4.1.1.4 Uplink Power Control

When super uplink is enabled, power control is performed separately on NR TDD and SUL carriers. The gNodeB delivers separate transmit power control (TPC) commands to inform a UE of the power adjustment amount of each carrier, and the UE sends separate power headroom reports (PHRs) to the gNodeB. For details about the power control principles for channels, see *Power Control*.

According to 3GPP TS 38.300 V15.0.0, the SUL carrier is not paired with a downlink carrier. As such, its path loss is estimated based on the downlink of the NR TDD carrier. The gNodeB adjusts the power of each channel on the SUL carrier based on the path loss difference between NR TDD and SUL carriers, and delivers the information to UEs through SI messages.

4.1.1.5 Timing Advance

Timing advance (TA) is introduced to adjust the uplink timing of UEs. This ensures an acceptable difference in time when uplink data from different UEs reaches the gNodeB and unaffected demodulation performance of uplink data.

If super uplink is enabled, NR TDD and SUL carriers share the same timing advance group (TAG). When CA is enabled on the NR TDD carrier, the primary component carrier (PCC), secondary component carrier (SCC), and SUL carrier can share the same TAG.

If the number of TAGs supported by a UE is less than that of CCs aggregated for the UE, some of the CCs have to share the TAs of other CCs. Inappropriate TA selection has a noticeable impact on the overall uplink demodulation performance of these CCs. To address this issue, flexible TA adjustment in uplink multi-CC scenarios is introduced. This function is controlled by the **UL_MULTI_CC_TA_FLEX_ADJ_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

With flexible TA adjustment in uplink multi-CC scenarios enabled, if the number of TAGs supported by a UE is less than that of CCs aggregated for the UE, a TA is selected for all CCs involved in a single TAG according to the following rules, which apply to all TAGs:

- If the absolute value of the TA difference between any two CCs involved in a TAG is less than the value of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on all the CCs involved in this TAG is selected as the TA for all these CCs.

- If the absolute value of the TA difference between any two CCs involved in a TAG is greater than or equal to the value of the **NRCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on the PCC is selected as the TA for all the CCs involved in this TAG.

4.1.1.6 Mobility Management

When a super uplink UE performs an inter-frequency or inter-RAT handover from a super uplink cell, an SUL threshold offset specified by the **NRCellMeasConfig.SulCovHoRsrpThldOffset** parameter is added to the original RSRP threshold for event A1 or A2 related to inter-frequency or inter-RAT handover to ensure that the super uplink UE keeps camping on the serving cell if possible.

- Inter-frequency and inter-RAT handovers:
 - Threshold for event A1: Original RSRP threshold for event A1 + **NRCellMeasConfig.SulCovHoRsrpThldOffset**
 - Threshold for event A2: Original RSRP threshold for event A2 + **NRCellMeasConfig.SulCovHoRsrpThldOffset**

For details about the original RSRP thresholds for events A1 and A2, see *SA Mobility Management in Connected Mode* and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

In other mobility scenarios, the mobility-related procedures for super uplink UEs and common UEs are the same.

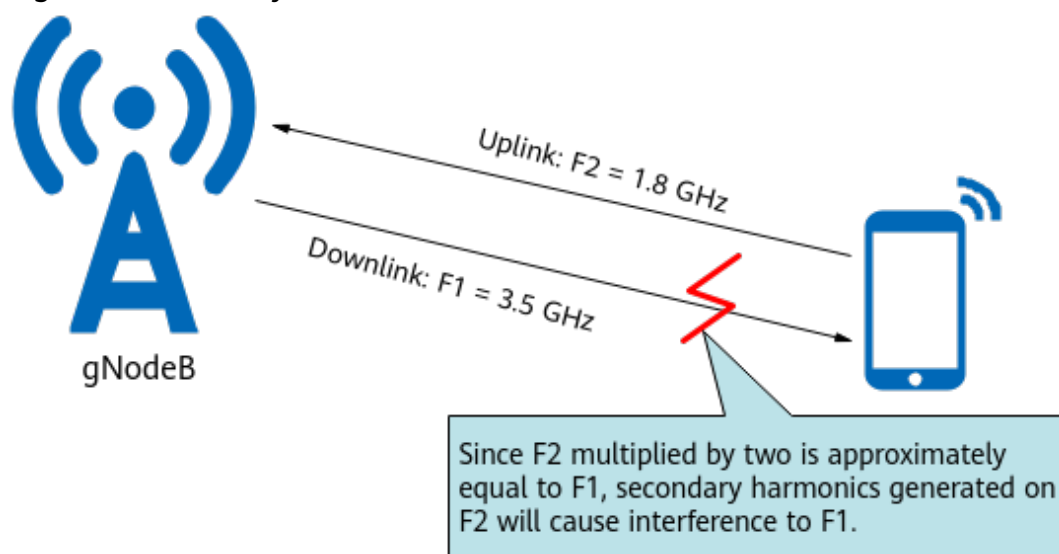
- For details about intra-frequency handovers, inter-frequency redirections, and inter-frequency blind redirections, see *SA Mobility Management in Connected Mode*.
- For details about inter-RAT redirections and inter-RAT blind redirections, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

4.1.2 Enhanced Functions

4.1.2.1 Secondary Harmonic Interference Avoidance

Secondary harmonics are the self-interference signals generated by UEs during uplink and downlink data transmission. When a UE's uplink data transmission is carried on the SUL carrier and its downlink data transmission is carried on the NR TDD carrier, the uplink transmit frequency multiplied by two falls into the downlink frequency band and interferes with the reception of downlink signals, as shown in [Figure 4-4](#).

Figure 4-4 Secondary harmonic interference



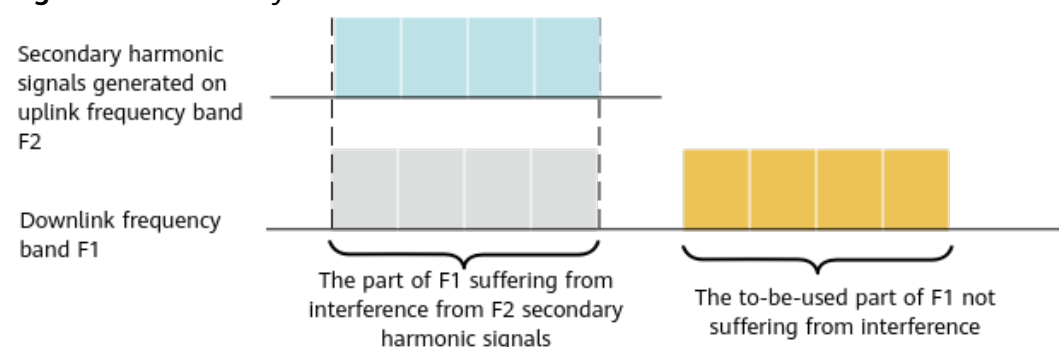
Secondary harmonic interference is generated when the following combinations of NR TDD and SUL frequency bands are used:

- NR n78+NR n80 (SUL)
- NR n78+NR n86 (SUL)

If either of the preceding frequency band combinations is used, it is recommended that the secondary harmonic interference avoidance switch be turned on. The switch is controlled by the **SEC_HARMONIC_INTRF_AVOID_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

Then, the gNodeB can schedule the downlink data for a super uplink UE in the part of the frequency band that does not suffer from interference, as shown in [Figure 4-5](#). This reduces interference from uplink signals to the downlink.

Figure 4-5 Secondary harmonic interference avoidance

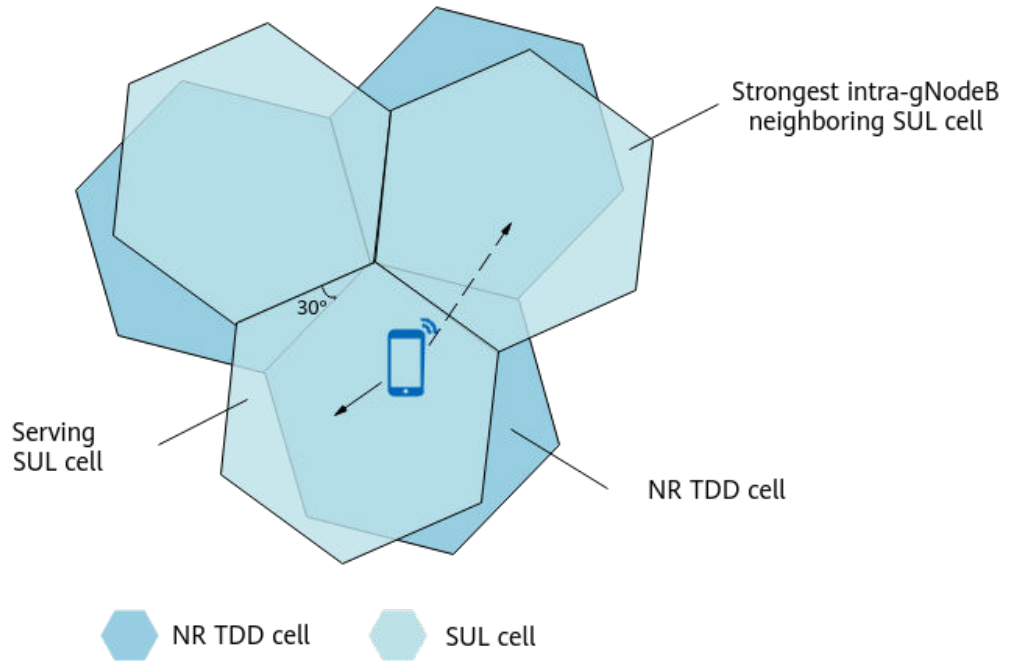


4.1.2.2 SUL CoMP

NR TDD and SUL carriers use different antennas, and any antenna azimuth inconsistency between these carriers during engineering implementation affects the benefits provided by super uplink. The SUL CoMP function is introduced to compensate for the gain loss caused by inconsistent antenna azimuths (with a difference less than 30°).

The SUL CoMP function is controlled by the **INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter. For cell edge users (CEUs), the gNodeB receives uplink signals from both the serving SUL cell and the strongest intra-gNodeB neighboring SUL cell, and then combines these signals, as shown in **Figure 4-6**. This improves the demodulation capability of the gNodeB.

Figure 4-6 SUL CoMP



The procedure is as follows:

1. The gNodeB selects UEs for synchronization signal and PBCH block (SSB) RSRP measurement and delivery of measurement configurations related to event A3 as follows:
 - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp.CompConfigPol** parameter is deselected, the gNodeB delivers measurement configurations to all UEs in the cell.
 - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp.CompConfigPol** parameter is selected, the gNodeB delivers measurement configurations to specific UEs in the cell. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter is selected.

The event A3 measurement control parameters (in the **NRCellComp** MO) are the same as those for 5G UL CoMP, including **NRCellComp.TimeToTrigger**, **NRCellComp.Hysteresis**, and **NRCellComp.UlCompRsrpOffset**. For details, see *CoMP in 5G RAN Feature Documentation*.

2. UEs perform event A3 measurement, and send A3 measurement reports, with NR TDD intra-frequency neighboring cells included.
3. The gNodeB determines whether the UEs are CEUs based on the received A3 measurement results, and determines the strongest intra-gNodeB neighboring NR TDD cell.

4. The gNodeB determines the strongest intra-gNodeB neighboring SUL cell based on the mapping between NR TDD and SUL carriers. The following configurations of this neighboring cell must be the same as those of the serving cell:
 - Transmit and receive mode: **NRDUCellTrp.TxRxMode**
 - Duplex mode: **NRDUCell.DuplexMode**
 - Downlink NR-ARFCN: **NRDUCell.DlNarfcn**
 - Uplink bandwidth: **NRDUCell.UlBandwidth**
 - Slot configuration: **NRDUCell.SlotAssignment**
 - Cyclic prefix length: **NRDUCell.CyclicPrefixLength**
 - Subcarrier spacing: **NRDUCell.SubcarrierSpacing**
 - PUSCH DMRS configuration: **NRDUCellPusch.UlDmrsType** and **NRDUCellPusch.UlAdditionalDmrsPos**
5. The strongest intra-gNodeB neighboring SUL cell receives uplink sounding reference signals (SRSs) from UEs. If the RSRP of the uplink SRSs sent by a UE in this neighboring cell is greater than the sum of the RSRP of uplink SRSs in the UE's serving cell and the value of **NRCellComp.UlCompRsrpOffset**, the gNodeB determines that SUL CoMP takes effect for the UE. Otherwise, SUL CoMP does not take effect for the UE.
6. For UEs for which SUL CoMP takes effect, the serving cell and the strongest intra-gNodeB neighboring SUL cell simultaneously receive the UEs' data on PUSCHs. The received data is combined and demodulated, improving the PUSCH performance of these UEs.

The strongest intra-gNodeB neighboring SUL cell preferentially processes the data from the UEs it serves. If this cell does not have sufficient capabilities (for example, the number of PRBs used by this cell plus the number of PRBs required for uplink joint reception exceeds the cell's total number of PRBs), it does not process the UL CoMP UE data.

In super uplink with independent SUL deployment scenarios with both SUL CoMP and LTE FDD and NR Flash Dynamic Spectrum Sharing enabled, SUL CoMP takes effect only when both the cooperating and serving cells are in the LTE FDD and NR spectrum sharing state.

For details about LTE FDD and NR Flash Dynamic Spectrum Sharing, see *LTE FDD and NR Dynamic Spectrum Sharing*.

4.1.2.3 Channel Decoupling

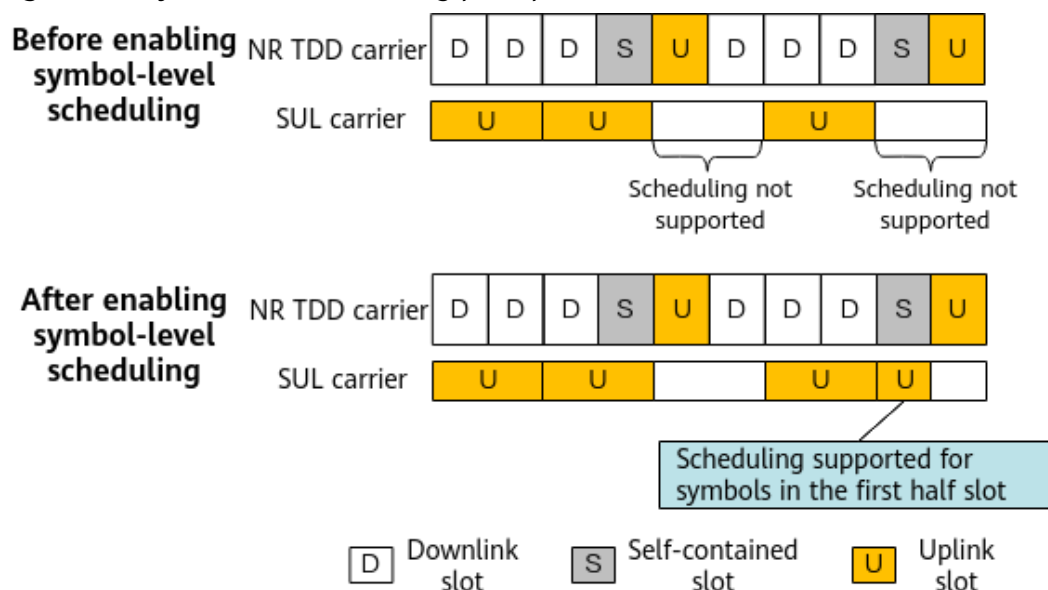
Channel decoupling allows control channels to be decoupled from data channels on the SUL carrier. With this function, only the PUSCH is carried on the SUL carrier, and both PRACH and PUCCH are carried on the NR TDD carrier. This function is controlled by the **SUL_CHANNEL_DECOUPLE_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter. It helps increase the uplink throughput.

Channel decoupling only takes effect for UEs that support the capability indicated by dynamicSwitchSUL whose value is supported in the UECapabilityInformation message. Channel decoupling is available only when super uplink is enabled and UL and DL Decoupling is disabled, as channel decoupling cannot work with UL and DL Decoupling.

4.1.2.4 Symbol-Level Scheduling on the SUL Carrier

Symbol-level scheduling on the SUL carrier applies to scenarios where the slot configuration is 4:1 in NR TDD. When this slot configuration is in use and uplink TDM scheduling is performed for a UE on NR TDD and SUL carriers, the slots on the SUL carrier corresponding to the slots with uplink data on the NR TDD carrier cannot be scheduled. With symbol-level scheduling, the gNodeB allows a UE to transmit data over symbols in the first half slot on the SUL carrier, which corresponds to the downlink or self-contained slot on the NR TDD carrier. This improves the uplink throughput of the SUL carrier. For details, see [Figure 4-7](#).

Figure 4-7 Symbol-level scheduling principle



Symbol-level scheduling is controlled by the **SCHEDULE_FIRST_HALF_SLOT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

4.1.2.5 SUL UE Power Saving

UE power saving is implemented for UEs in the super uplink state in the following ways:

- When the uplink traffic volume of the UEs is lower than a specified threshold, the gNodeB can trigger SUL carrier removal to implement SUL UE power saving.
- When the uplink traffic volume of the UEs is greater than the threshold, the gNodeB can trigger SUL carrier addition for UEs to enter the super uplink state.

SUL UE power saving is controlled by the **SUL_UE_POWER_SAVING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

4.1.2.6 Function of Updating the Uplink Transmit Power of Band Combinations

To increase the uplink throughput of CEUs that support R16-compliant power classes, the uplink transmit power of band combinations can now be updated. In

this case, the maximum uplink transmit power for NR TDD is 29 dBm. This function is controlled by the **BAND_COMB_UL_TX_PWR_UDT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

This function takes effect only when the **NRDUCellSelConfig.MaxUeTransmitPower** parameter is set to a value in the range of 29 to 33.

This function takes effect only when the value of the UE-reported **ue-PowerClass-v1610** or **powerClass-v1610** field is **pc1dot5**. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V17.10.0.

The NR TDD and SUL frequency band combination n78+n84 is currently supported.

4.1.3 Simultaneous Use with Other Functions

4.1.3.1 Simultaneous Use with UL and DL Decoupling

When super uplink is used together with UL and DL Decoupling, super uplink takes effect in cell center areas, and UL and DL Decoupling takes effect in cell edge areas. When both of these functions are enabled, random access and mobility management are different compared to when only super uplink is enabled. For details about UL and DL Decoupling, see *UL and DL Decoupling*. Super uplink can be enabled together with UL and DL Decoupling only on macro base stations, not on LampSite base stations.

Random Access

During initial RRC connection setup, a UE receives system information from the gNodeB, obtains the *rsrp-ThresholdSSB-SUL* IE in SIB1, measures the downlink RSRP of the NR TDD carrier, and determines the uplink carrier for random access. The value of *rsrp-ThresholdSSB-SUL* is specified by the **NRDUCellCollabServ.RsrpThld** parameter.

- If the RSRP of an NR cell is greater than or equal to the value of the **NRDUCellCollabServ.RsrpThld** parameter, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NR TDD carrier, and super uplink takes effect.
- If the RSRP of an NR cell is less than the value of the **NRDUCellCollabServ.RsrpThld** parameter, the UE is in an area with weak or no NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier, and UL and DL Decoupling takes effect.

Intra-Cell Mobility Management

When a UE moves in an NR cell, it uses super uplink in cell center areas, and UL and DL Decoupling in cell edge areas. These two functions are switched when certain conditions relating to either uplink quality or user experience are met. The **UL_CARR_SEL_FOR_USER_EXP_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter specifies whether uplink quality or user experience is used.

- When the **UL_CARR_SEL_FOR_USER_EXP_SW** option is selected, the uplink carrier is selected based on user experience. The gNodeB then determines whether super uplink or UL and DL Decoupling takes effect for a UE based on the selected carrier.
- When the **UL_CARR_SEL_FOR_USER_EXP_SW** option is deselected, the uplink carrier is selected based on uplink quality. The details are as follows:
 - If the SRS RSRP of a UE on the NR TDD carrier is greater than the sum of the **NRDUCellCollabServ.SrsRsrpThld** parameter value and hysteresis (2 dB), super uplink takes effect for the UE. In this scenario, the UE transmits uplink data on NR TDD and SUL carriers in TDM mode.
 - If the SRS RSRP of a UE on the NR TDD carrier is less than or equal to the result of the **NRDUCellCollabServ.SrsRsrpThld** parameter value minus hysteresis (2 dB), UL and DL Decoupling takes effect for the UE. In this scenario, the UE transmits uplink data on the SUL carrier only.

Inter-Cell Mobility Management

When a UE that supports super uplink moves between NR cells and a handover to the target cell for which super uplink and UL and DL Decoupling are enabled is triggered, the gNodeB selects the NR TDD or SUL carrier for the UE and notifies the UE of the uplink carrier through an RRCReconfiguration message. The specific selection procedure is as follows:

1. The gNodeB delivers intra-RAT event A3 measurement configurations to the UE, instructing it to measure the signal quality of neighboring NR cells.
2. After receiving a measurement report from the UE, the source gNodeB forwards the RSRP of the target cell to the target gNodeB, which then selects an uplink carrier for the UE based on the following rules:
 - If the RSRP of the target cell is greater than or equal to the value of the **NRDUCellCollabServ.RsrpThld** parameter, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NR TDD carrier.
 - If the RSRP of the target cell is less than the value of the **NRDUCellCollabServ.RsrpThld** parameter, the UE is in an area with weak or no NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier.
3. The target gNodeB notifies the source gNodeB of the selected uplink carrier. The source gNodeB sends the uplink carrier information of the target cell to the UE through an RRCReconfiguration message.
4. The UE initiates a random access procedure on the uplink carrier indicated in the RRCReconfiguration message.

4.1.3.2 Simultaneous Use with CA

The gNodeB supports the simultaneous use of super uplink and CA in both the uplink and downlink. When super uplink and CA are both enabled, super uplink is used for UE uplink services while downlink CA is used for UE downlink services.

CA functions that can be used with super uplink include:

- 2CC aggregation

- NR TDD intra-band contiguous CA
- NR TDD+NR FDD inter-band CA
- 3CC aggregation, specifically, the aggregation of two NR TDD carriers and an NR FDD carrier

For details about CA, see *Carrier Aggregation*.

4.1.3.2.1 Supported CA Scenarios and Frequency Bands

Simultaneous Use with 2CC Aggregation

The following 2CC aggregation scenarios are involved:

- NR TDD intra-band contiguous CA
Super uplink, uplink CA, and downlink CA can be enabled at the same time. [Figure 4-8](#) uses a single-period 8:2 slot configuration as an example.
- NR TDD+NR FDD inter-band CA
Super uplink and downlink CA can be enabled at the same time. [Figure 4-9](#) uses a PCC slot configuration of single-period 8:2 as an example.

In scenarios where NR TDD intra-band CA is used, SRS carrier switching can be enabled.

Figure 4-8 Simultaneous use of super uplink, NR TDD intra-band uplink CA, and NR TDD intra-band downlink CA

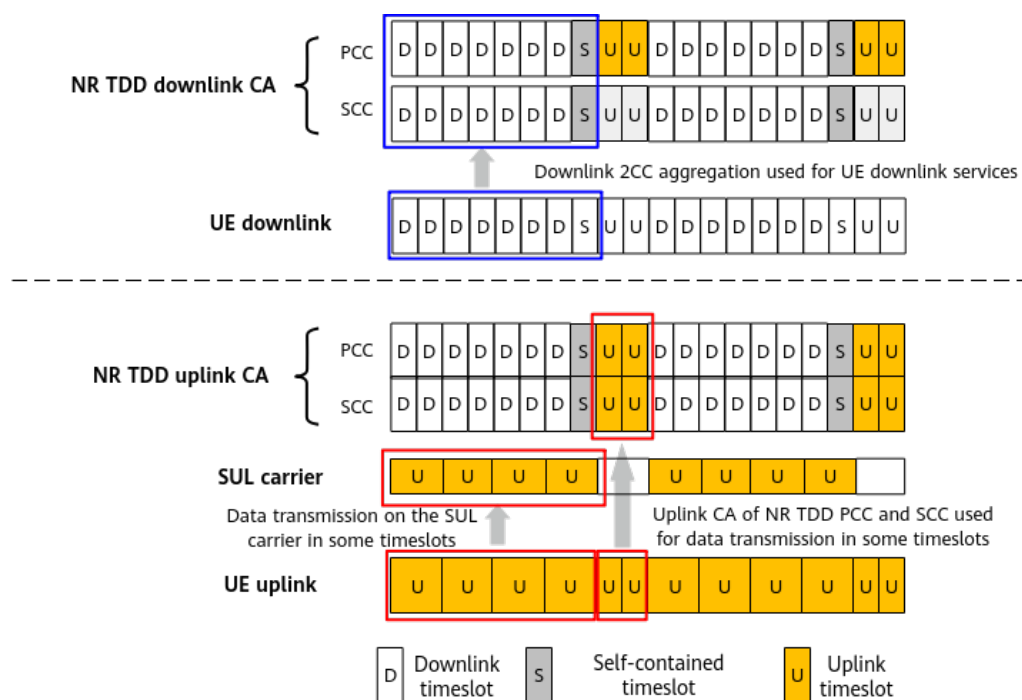
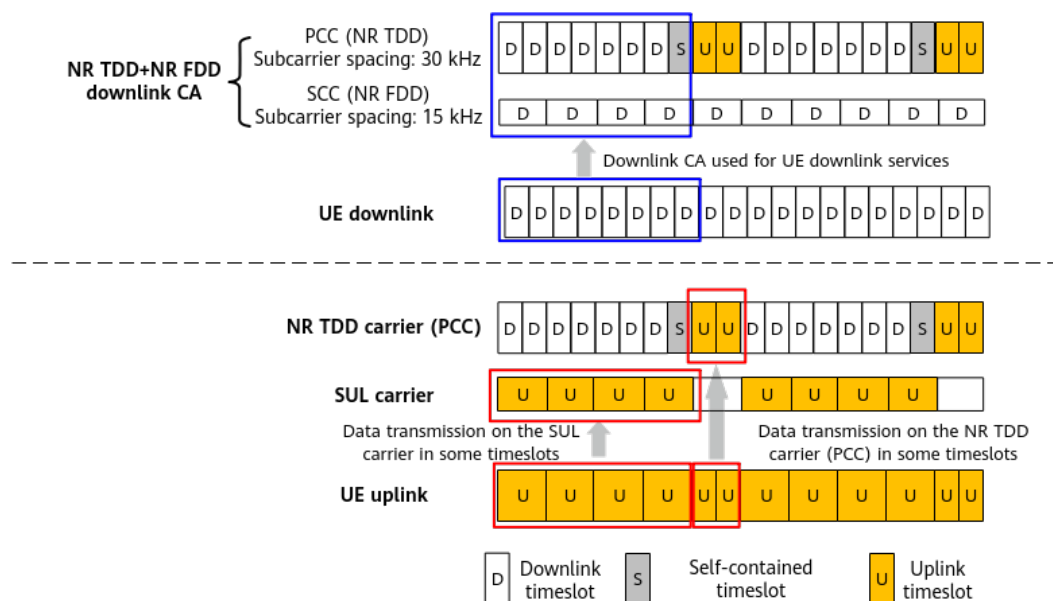


Figure 4-9 Simultaneous use of super uplink and NR TDD+NR FDD inter-band CA



Simultaneous Use with 3CC Aggregation

Super uplink and downlink 3CC aggregation can be enabled at the same time. Using a slot configuration of single-period 8:2, [Figure 4-10](#) shows the simultaneous use of super uplink and the aggregation of two NR TDD carriers and one NR FDD carrier in the downlink.

Figure 4-10 Simultaneous use of super uplink and 3CC aggregation



Supported Frequency Bands and Constraints

The following requirements must be met when super uplink is used with CA:

- gNodeB:
 - NR TDD carrier: The PCC of downlink CA must be the NR TDD carrier with a slot configuration of 4:1 or single- or dual-period 8:2.
 - SUL carrier: The subcarrier spacing must be 15 kHz.
- UE: UEs must support both super uplink and CA. Otherwise, only CA or super uplink takes effect.

Table 4-3 lists frequency band combinations for the simultaneous use of super uplink and downlink CA. The bandwidths supported by the frequency band combinations for the simultaneous use of super uplink and CA are consistent with those defined in 3GPP specifications. For details, see section 5.5C "Configurations for SUL" in 3GPP TS 38.101-1 V18.7.0.

Table 4-3 Frequency band combinations for the simultaneous use of super uplink and downlink CA

Downlink CA Scenario	NR TDD Frequency Band	NR TDD (PCC) Slot Configuration	SCC Slot Configuration	Supported SUL Frequency Band
2CC aggregation: NR TDD intra-band CA	n41	Single- or dual-period 8:2	Single- or dual-period 8:2	n80
2CC aggregation: NR TDD +NR FDD inter-band CA	n41	Single- or dual-period 8:2	N/A	n80
3CC aggregation	NR TDD (n41)+NR TDD (n41)+NR FDD (n28)	Single- or dual-period 8:2	NR TDD: single- or dual-period 8:2 NR FDD: N/A	n80
2CC aggregation: NR TDD intra-band CA	n78	4:1	4:1	n80/n83/n84

Downlink CA Scenario	NR TDD Frequency Band	NR TDD (PCC) Slot Configuration	SCC Slot Configuration	Supported SUL Frequency Band
2CC aggregation: NR TDD +NR FDD inter-band CA	n78	4:1	N/A	n80/n83/n84
2CC aggregation: NR TDD +NR FDD inter-band CA	NR TDD (n78)+NR FDD (n1)	Single- or dual-period 8:2	N/A	n84

Table 4-4 lists frequency band combinations for the simultaneous use of super uplink and uplink CA.

Table 4-4 Frequency band combinations for the simultaneous use of super uplink and uplink CA

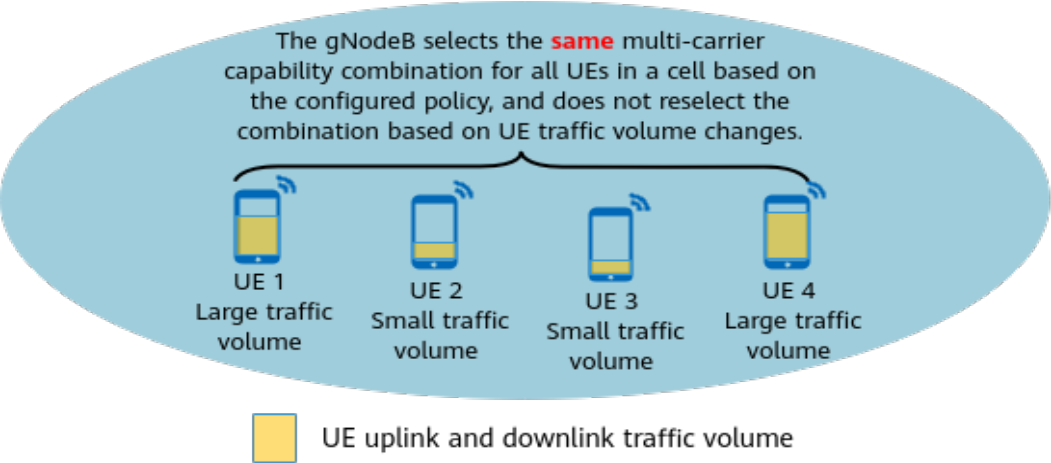
UL CA Scenario	NR TDD Frequency Band	NR TDD (PCC) Slot Configuration	SCC Slot Configuration	Supported SUL Frequency Band
2CC aggregation: NR TDD intra-band CA	n41	Single- or dual-period 8:2	Single- or dual-period 8:2	n80/n81/n83

4.1.3.2.2 SUL and CA Optimal Carrier Selection

When both super uplink and CA are enabled, the gNodeB can perform cell-level or UE-level optimal carrier selection for UEs in a cell.

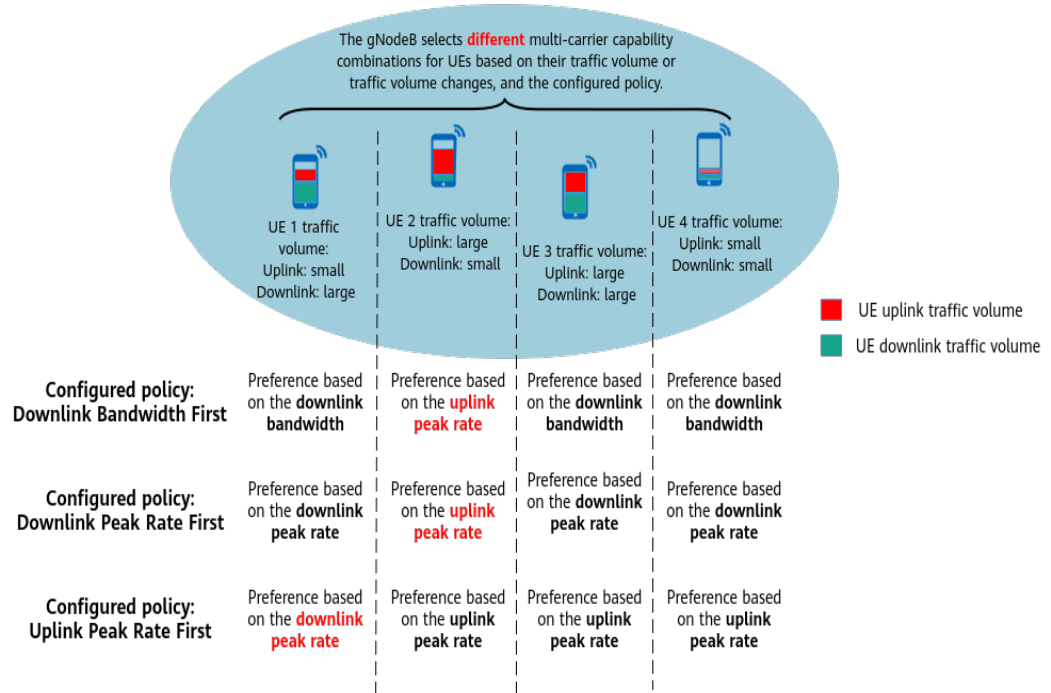
- Cell-level optimal carrier selection: The gNodeB performs carrier selection for all UEs in a cell based on the configured policy, as shown in [Figure 4-11](#).

Figure 4-11 Cell-level optimal carrier selection



- UE-level optimal carrier selection: The gNodeB performs carrier selection for each UE based on its traffic volume and the configured policy. In this way, each UE can obtain the optimal user-perceived rate as required, as shown in Figure 4-12.

Figure 4-12 UE-level optimal carrier selection



Cell-Level Optimal Carrier Selection

When cell-level SUL and CA optimal carrier selection takes effect, uplink intra-band contiguous CA is used near the center of NR TDD cells, maximizing the advantages of high uplink bandwidths in NR TDD. At the edge of NR TDD cells, super uplink is enabled. TDM scheduling in super uplink and the low path loss of the SUL carrier are fully used to optimize the uplink user-perceived rate. Cell-level SUL and CA optimal carrier selection is enabled by selecting the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw**

parameter. Whether a UE is located near the cell center or at the cell edge is determined by the **NRDUCellCollabServ.CarrSelTddSrsRsrpThld** parameter.

- When the gNodeB detects that the SRS RSRP of a UE served by the PCC is greater than the value of the **NRDUCellCollabServ.CarrSelTddSrsRsrpThld** parameter plus 2 dB, it determines that the UE is located near the cell center. In this case, the gNodeB performs carrier selection for the UE based on the value of the **NRCaMgmtConfig.CaBandCombSelectionStrat** parameter:
 - **BANDWIDTH_FIRST**: Indicates that the gNodeB selects the optimal multi-carrier capability combination based on the following multi-carrier factors in sequence: downlink bandwidth, theoretical downlink peak rate, uplink bandwidth, and theoretical uplink peak rate.
 - **PEAK_RATE_FIRST**: Indicates that the gNodeB selects the optimal multi-carrier capability combination based on the following multi-carrier factors in sequence: theoretical downlink peak rate, downlink bandwidth, theoretical uplink peak rate, and uplink bandwidth.
 - **UL_PEAK_RATE_FIRST**: Indicates that the gNodeB selects the optimal multi-carrier capability combination based on the following multi-carrier factors in sequence: theoretical uplink peak rate, uplink bandwidth, theoretical downlink peak rate, and downlink bandwidth.
 - **DEFAULT**: Indicates that the gNodeB selects the optimal multi-carrier capability combination based on the following multi-carrier factors in sequence: number of aggregated carriers in the uplink and theoretical uplink peak rate.

When the **SRS_WT_BASED_CARR_SEL_SW** option of the **NRCaMgmtConfig.CaCompatibilitySwitch** parameter is selected and the **NRCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **BANDWIDTH_FIRST**, **PEAK_RATE_FIRST**, or **UL_PEAK_RATE_FIRST**, the gNodeB also considers the benefits provided by SRS-based weights during multi-carrier capability combination selection. For details, see *Carrier Aggregation*.

- When the gNodeB detects that the SRS RSRP of a UE served by the PCC is less than the value of the **NRDUCellCollabServ.CarrSelTddSrsRsrpThld** parameter minus 2 dB, it determines that the UE is located at the cell edge. In this case, the gNodeB performs carrier selection based on the value of the **NRCaMgmtConfig.CaBandCombSelectionStrat** parameter and the SUL carrier is preferentially selected as the uplink carrier.
- In other cases, the UE maintains the current carrier configuration.

For a UE that supports both the combination of downlink 3CC aggregation and uplink intra-band contiguous 2CC aggregation and that of downlink 2CC aggregation, uplink intra-band contiguous 2CC aggregation, and SUL, uplink optimal carrier selection for SUL and CA is controlled by the **SUL_CA_BASED_UL_CARR_SEL_SW** option of the **NRCaMgmtConfig.CaStrategySwitch** parameter. If this option is selected, both SUL and uplink CA carriers are considered during uplink optimal carrier selection, and uplink intra-band contiguous CA and SUL instead of uplink intra-band contiguous 2CC aggregation is used. If this option is deselected, only uplink CA carriers are considered during uplink optimal carrier selection.

UE-Level Optimal Carrier Selection

The gNodeB performs carrier selection for UEs based on their traffic volumes. With this function, each UE or the same UE with different traffic volume models can use the optimal carrier combination. In this way, each UE can obtain the optimal user-perceived rate as required.

UE-level optimal carrier selection is controlled by the **SERVICE_AWARE_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter.

For a UE that supports both the combination of downlink CA and SUL and that of downlink CA and SRS carrier switching but does not support downlink CA, SUL, and SRS carrier switching at the same time:

- With the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter deselected,
 - Downlink CA and SRS carrier switching are preferentially used when the downlink traffic volume is large and the uplink traffic volume is small. In this case, SRSs can be sent in the PCell and an SCell in turn.
 - Downlink CA and SUL are preferentially used when the downlink traffic volume is small and the uplink traffic volume is large.
 - In other cases, the current carrier configuration is maintained.
- With the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter selected, the gNodeB selects carriers according to the policies described in [Table 4-5](#) and considers SRS carrier switching capability. If a UE supports SUL and no SUL carrier has been added, the gNodeB periodically determines whether to add an SUL carrier based on the uplink or downlink traffic volume of the UE. The period is specified by the **NRDUCellCarrMgmt.SccCfgInterval** parameter.

Table 4-5 Policies on UE-level optimal carrier selection

Uplink Traffic Volume	Downlink Traffic Volume	NRCellCaMgmtConfig.CaBandCombSelectionStrat Set to BANDWIDTH_FIRST	NRCellCaMgmtConfig.CaBandCombSelectionStrat Set to PEAK_RATE_FIRST	NRCellCaMgmtConfig.CaBandCombSelectionStrat Set to UL_PEAK_RATE_FIRST
Small	Large	Preference based on the downlink bandwidth	Preference based on the downlink peak rate	Preference based on the downlink peak rate
Large	Large	Preference based on the downlink bandwidth	Preference based on the downlink peak rate	Preference based on the uplink peak rate
Small	Small	Preference based on the downlink bandwidth	Preference based on the downlink peak rate	Preference based on the uplink peak rate

Uplink Traffic Volume	Downlink Traffic Volume	NRCellCaMgmtConfig.CaBandCombSelectionStrat Set to BANDWIDTH_FIRST	NRCellCaMgmtConfig.CaBandCombSelectionStrat Set to PEAK_RATE_FIRST	NRCellCaMgmtConfig.CaBandCombSelectionStrat Set to UL_PEAK_RATE_FIRST
Large	Small	Preference based on the uplink peak rate	Preference based on the uplink peak rate	Preference based on the uplink peak rate
Note: The NRCellCaMgmtConfig.CaBandCombSelectionStrat parameter specifies the policy on multi-carrier capability combination selection.				

The following is an example. Assume that the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **BANDWIDTH_FIRST**. The uplink peak rate is given priority only when the uplink traffic volume is large and the downlink traffic volume is small. The downlink bandwidth is given priority when other traffic volume models are used.

4.1.3.2.3 SRS Carrier Switching

For a UE that supports downlink CA, SUL, and SRS carrier switching:

- With the **SUL_CA_SRS_SWITCHING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter selected, SRS transmission can use the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R16 or later and downlink 2CC aggregation take effect.
- With the **SUL_CA_2CC_SRS_SWITCHING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter selected, SRS transmission can use the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R16 or later and downlink 3CC aggregation take effect.
- With the **SUL_CA_SRS_SWITCHING_EXT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter selected, SRS transmission can use the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R15 and downlink 2CC or 3CC aggregation take effect.
- With the **SUL_CA_SRS_SWITCHING_SW** and **SUL_CA_SRS_SWITCHING_EXT_SW** options of the **NRDUCellCollabServ.SulAlgoSwitch** parameter deselected, downlink CA and SRS carrier switching are preferentially used when the downlink traffic volume is large and the uplink traffic volume is small, that is, SRSs can be sent in the PCell and an SCell in turn; downlink CA and SUL are preferentially used when the downlink traffic volume is small and the uplink traffic volume is large. In other cases, the current carrier configuration is maintained.

4.1.3.3 Simultaneous Use with MU-MIMO

When super uplink is used together with MU-MIMO, MU-MIMO pairing is performed on the NR TDD carriers for super uplink and non-super uplink UEs.

The scheduling time for super uplink UEs is different from that for non-super uplink UEs on the NR TDD carrier. When simultaneous scheduling of the same air interface resources for these UEs is required and MU-MIMO takes effect, the scheduling time for super uplink UEs and that for non-super uplink UEs must be aligned. Scheduling time alignment for super uplink is supported, and is controlled by the **UL_SCH_ALIGN_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. If this option is selected, PUSCH scheduling for super uplink UEs and that for non-super uplink UEs are aligned in the time domain. If this option is deselected, they are not aligned in the time domain.

4.2 Network Analysis

4.2.1 Benefits

Basic Functions

Super uplink gains are derived from SUL spectrum resources. With super uplink enabled, NR TDD spectrum resources are utilized in TDM mode together with SUL spectrum resources, greatly increasing the uplink throughput of super uplink UEs as well as that of the entire NR cell. This function decreases the proportion of UEs with throughput that falls into a low-throughput range, and increases the proportion of UEs with throughput that falls into a high-throughput range. The preceding benefits can be observed using the methods described in [4.4.4 Network Monitoring](#).

Larger gains result from a lower network interference level (indicated by **N.UL.NI.Avg**), a shorter distance to the main lobe direction of the antenna, and a larger SUL bandwidth. Smaller gains result from a higher network interference level, a longer distance to the main lobe direction of the antenna, and a smaller SUL bandwidth.

If flexible TA adjustment in uplink multi-CC scenarios is enabled, the uplink throughput increases in the uplink multi-CC scenario with the following conditions met: (1) there are negative measured TA values, or the measured TA value and bandwidth differ greatly between CCs; (2) the absolute difference in the measured TA value between CCs does not exceed the configured threshold.

Enhanced Functions

The enhanced functions provide the following gains:

- SUL CoMP: It improves uplink reception performance and throughput for UEs in overlapping areas in an intra-frequency network with continuous coverage. For details, see [4.4.4 Network Monitoring](#).
- Channel decoupling: It enables control channels to be removed from the SUL carrier, thereby increasing the number of RBs that can be scheduled on the PUSCH as well as the uplink UE throughput.
- Symbol-level scheduling: It enables a UE to schedule symbols in the first half timeslot on the SUL carrier. This increases the availability of SUL timeslots for scheduling and improves the UE's uplink throughput.

- Function of updating the uplink transmit power of band combinations: It increases the uplink throughput of CEUs that support R16-compliant power classes.

Simultaneous Use with Other Functions

When super uplink and CA are both enabled, SUL and CA optimal carrier selection facilitates the selection of the optimal carrier combination, increasing the uplink throughput of UEs supporting both super uplink and CA.

When super uplink and uplink CA are both enabled, it is recommended that the **MF_ACTUAL_PUSCH_ILPC_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoExtSwitch** parameter be selected to enable multi-frequency actual-PUSCH-based inner-loop power control to improve the accuracy of data volume distributed to SCells in the uplink. For details about this function, see *Power Control*.

When the **SUL_CA_SRS_SWITCHING_SW**, **SUL_CA_2CC_SRS_SWITCHING_SW**, or **SUL_CA_SRS_SWITCHING_EXT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected, UEs that support the simultaneous use of downlink 2CC aggregation, SUL in compliance with R16 or later, and SRS carrier switching, that of downlink 3CC aggregation, SUL in compliance with R16 or later, and SRS carrier switching, or that of downlink 2CC or 3CC aggregation, SUL in compliance with R15, and SRS carrier switching, can provide the following benefits:

- Compared with UEs for which only downlink CA and SRS carrier switching take effect, the average uplink throughput of UEs that also support super uplink is improved. For details about how to observe the gains, see [4.4.4 Network Monitoring](#).
- Compared with UEs for which only downlink CA and super uplink take effect, the average downlink throughput of UEs that also support SRS carrier switching is improved. The average downlink throughput is indicated by **User Downlink Average Throughput (DU)**.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

The impacts of super uplink on networks are as follows:

- The number of the RBs occupied by SIB1 may slightly increase and the downlink peak throughput slightly decreases because the gNodeB broadcasts the SUL information through SIB1.
- SUL uplink scheduling consumes control channel element (CCE) resources of the NR TDD cell. When CCE resources are insufficient in the NR TDD cell, uplink or downlink CCE resource application is more likely to fail. As a result, the uplink or downlink throughput of the NR TDD cell fluctuates, and the uplink or downlink user-plane delay increases.
- It is recommended that the **UL_DL_CCE_SHARING_SW** or **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter be selected to reduce the probability of CCE resource application failures.

- In an NR TDD cell enabled with uplink MU-MIMO, if scheduling time alignment for super uplink is enabled (that is, the **UL_SCH_ALIGN_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter is selected), the average uplink cell throughput and the **N.CCE.UL.Avail.Equivalent** and **N.CCE.ULDCI.Avail.Equivalent** counter values increase, as does the average delay for non-super-uplink UEs by up to 1%.
- For UEs that switch between carriers in 0 μ s and report the NUL 1T and SUL 1T capabilities over the standard interface, their uplink throughput decreases when UE compatibility handling is enabled with the **NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime** parameter set to a value other than 0 μ s.
- Assume that the **SUL_CA_SRS_SWITCHING_SW**, **SUL_CA_2CC_SRS_SWITCHING_SW**, or **SUL_CA_SRS_SWITCHING_EXT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected. Compared with UEs for which only downlink CA and super uplink take effect, the average uplink throughput of UEs that also support SRS carrier switching decreases by about 7%, as time needs to be reserved for switching SRS transmission between SUL and PCC carriers. The average uplink throughput is indicated by **User Uplink Average Throughput (DU)**.
- For a UE that supports the combination of downlink 3CC aggregation and uplink intra-band 2CC aggregation and that of downlink 2CC aggregation, uplink intra-band contiguous 2CC aggregation, and SUL, if uplink optimal carrier selection for SUL and CA is enabled, the latter combination is preferentially used. This increases the **User Uplink Average Throughput (DU)**. However, the downlink UE throughput decreases because optimal carrier selection introduces gap-assisted inter-frequency measurements and downlink 3CC aggregation cannot take effect.
- As false preamble detection may occur in SUL cells and random access-related counters are measured in NR TDD cells, the measured Msg1-related counter value may be greater than the measured Msg2-related counter value in NR TDD cells. That is, the value of **N.RA.Contention.Att** may be greater than that of **N.RA.Contention.Resp** in contention-based random access scenarios, and the value of **N.RA.Dedicated.Att** may be greater than that of **N.RA.Dedicated.Resp** in non-contention-based random access scenarios, causing a probability of a decrease in the success rate of access to NR TDD cells. If both UL and DL Decoupling (controlled by the **NRDUCellAlgoSwitch.UlDlDecouplingSwitch** parameter) and channel decoupling (controlled by the **SUL_CHANNEL_DECOUPLE_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter) are enabled, there is no such impact.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	NE	Sales Unit
FOFD-030211	Super Uplink	NR0S00SUPC00	Macro gNodeB	per Cell
For the license control item NR0S00SUPC00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter cannot be selected due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.				

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Super uplink is not supported in high-speed cells.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Super uplink is not supported in hyper cells.
Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Super uplink is not supported in combined cells.
Low-frequency NR TDD	Admission control	QOS_FLOW_ADMISSION_SW and QOS_FLOW_PREEMPTION_SW options of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>	Super uplink and admission control cannot be both enabled.
Low-frequency NR TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Super uplink and Fusion Cell cannot be both enabled.
Low-frequency NR TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>MIMO (TDD)</i>	Super uplink and CCE resource optimization cannot be both enabled.
Low-frequency NR TDD	Pattern-free PDCCH rate matching activation rate increase	PDCCH_NO_PAT_RM_RATIO_INC_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	<i>Downlink Scheduling</i>	Super uplink and pattern-free PDCCH rate matching activation rate increase cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Super uplink and CCE partial symbol adaptation cannot be both enabled.
Low-frequency NR TDD	Cell deployment in IAB mode	NRDUCell.DeployPosition set to DEPLOY_IAB	None	Super uplink is not supported in cells deployed in IAB mode.
Low-frequency NR TDD	Uplink assistance	UL_ASSISTANCE_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	None	Super uplink and uplink assistance cannot be both enabled.
Low-frequency NR TDD	Cooperation between joint transmission and pattern-free PDCCH rate matching	JT_PDCCH_NO_PATT_RM_COORD_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	None	Super uplink and cooperation between joint transmission and pattern-free PDCCH rate matching cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Super uplink must be disabled if all the following conditions are met: (1) rate-matching-pattern-configuration-based PDCCH rate matching is enabled, (2) PDCCH uplink-to-downlink CCE ratio adaptation is disabled (by deselecting the UL_DL_CCE_RATIO_ADAPT_SW option of NRDUCellPdcch.PdcchAllocationSwitch), and (3) NRDUCellPdcch.OccupiedRbNum is set to 1 or 2 .
Low-frequency NR TDD	Downlink avoidance enhancement mode 2	DL_AVOID_ENH_MODE2_SW option of the NRDUCellTimeIntrf.TimeIntrfAvoidSw parameter	None	Super uplink and downlink avoidance enhancement mode 2 cannot be both enabled.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequency NR TDD	Power saving BWP	BWP2_SWIT CH option of the NRDUCellU ePwrSaving.BwpPwrSavi ngSw parameter	<i>UE Power Saving</i>	UEs working in BWP2 for power saving do not support SUL CoMP.
Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIA BILITY_BASI C_SW option of the NRDUCellAl goSwitch.HighReliabilit ySwitch parameter	<i>URLLC</i>	The low latency and high reliability function does not take effect for UEs for which super uplink has taken effect.
Low-frequency NR TDD	PUSCH power control	None	<i>Power Control</i>	When super uplink is enabled, the NRDUCellUIPcCon-fig.PhrProhibitTimer parameter configuration does not take effect for UEs for which super uplink has taken effect.
Low-frequency NR TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ _CONNECTE D_MLB_SW option of the NRCelAlgo Switch.Mlb AlgoSwitch parameter selected, and NRCelMlb.MlbTrigger Mode set to SPEC_EFF_B ASED_USER _NUM	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs for which super uplink has taken effect as the UEs to be handed over.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequ ency NR TDD	Device- pipe synergy	SPEC_UE_ID ENTIFY_SW option of the NRDUCellU eCoop.Spec UeCooperat ionSwitch parameter	<i>Device-Pipe Synergy</i>	Device-pipe synergy takes effect only in NR TDD cells, but not in SUL cells.
Low-frequ ency NR TDD	RFSP- based RB resource control	RB_DYNAMI C_CONTROL _SW option of the NRDUCellC ommonSw. RfspAlgoSw itch parameter	<i>Resource Control in NSA Networking</i>	Assume that both RFSP-based RB resource control and super uplink with independent SUL deployment are enabled. If RFSP-based RB resource control is enabled in an SUL cell, it must also be enabled in the associated TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequ ency NR TDD	Uplink multi- frequency coordinati on	MULTI_FRE Q_UL_COOR D_SW option of the NRDUCellFe atureSw.Mu ltiCarrierSw itch parameter	<i>Multi- Frequency Smart Selection</i>	The gNodeB does not select UEs for which super uplink has taken effect.
Low-frequ ency NR TDD	Multi- frequency beam coordinati on	MULTI_FRE Q_BEAM_C OORD_SW option of the NRDUCellFe atureSw.Mu ltiCarrierSw itch parameter	<i>Multi- Frequency Smart Selection</i>	The gNodeB does not select UEs for which super uplink has taken effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequency NR TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavi ngSw parameter	<i>UE Power Saving</i>	When both SUL CoMP and PDCCH skipping are enabled, PDCCH skipping does not take effect for UEs for which super uplink has taken effect.
Low-frequency NR TDD	Massive CA	NRDUCellCa rrMgmt.CaD lMaxCcNum set to DL4CC or DL5CC .	<i>CA Experience Improvemen t</i>	Downlink massive CA does not take effect for UEs for which super uplink has taken effect.
Low-frequency NR TDD	Inter-FR CA	INTER_FR_F R1TOFR2_C A_SW option of the NRDUCellAl goSwitch.Ca AlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-FR CA and super uplink with independent SUL deployment cannot take effect at the same time.
Low-frequency NR TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_S LOT_AGGRE GATION_S W option of the NRDUCellUL Sch.VoiceUL SchSwitch parameter	<i>VoNR</i>	3GPP R15-compliant uplink slot aggregation does not take effect on UEs with super uplink in effect.
Low-frequency NR TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_A GG_SW option of the NRDUCellUL Sch.VoiceUL SchSwitch parameter	<i>VoNR</i>	3GPP R16-compliant uplink slot aggregation does not take effect on UEs with super uplink in effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When QCI-based resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter) is enabled together with super uplink with independent SUL deployment and is enabled in the SUL cell, the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	After both "super uplink" and "flexible experience differentiation" are enabled, the base station estimates the uplink peak data rate of each carrier of a UE. If a UE is configured with an SUL carrier which is not activated, the UE's estimated uplink peak data rate may be inaccurate.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_S option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	When both positioning-dedicated SRS transmission in uplink slots and super uplink are enabled, the SUL cells configured on a specific baseband processing unit must meet the SUL cell specifications supported by the board. Otherwise, positioning-dedicated SRS transmission in uplink slots cannot take effect. For details about the SUL cell specifications supported by different baseband processing units, see the table content where the value of Frequency Band Type of a Cell includes SUL in Product Specifications > BBU Technical Specifications > Technical Specifications of Baseband Processing Units > NR TDD Technical Specifications of Baseband Processing Units in 3900 & 5900 Series Base Station Product Documentation.
Low-frequency NR TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	UEs on the NUL carrier can be switched to BWP2 while UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay of BWP2 UEs on the NUL carrier increases when the SUL carrier is activated.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORKSLICE_SW option of the NRDUCellCommonServiceDiffAIgoExtSwitch parameter	<i>Network Slicing</i>	When user-rate-based dynamic network slice resource management is enabled for an SUL cell where super uplink with independent SUL deployment is in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.
Low-frequency NR TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When user-rate-based dynamic QCI resource management is enabled for an SUL cell where super uplink with independent SUL deployment is in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

For the NR TDD cell involved in super uplink, there are no special requirements for the main control board and baseband processing unit. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

For the SUL cell involved in super uplink:

- Macro base stations support the following main control boards: UMPTe series, UMPTg series, UMPTga series, and UMPTHa series.
- Macro base stations support all baseband processing units that support SUL cell specifications. For details about these boards, see **Product Specifications > BBU Technical Specifications > Technical Specifications of Baseband Processing Units > NR TDD Technical Specifications of Baseband Processing Units** in *3900 & 5900 Series Base Station Product Documentation*. If the value of **Frequency Band Type of a Cell** corresponding to a board includes SUL, the board supports SUL cell specifications. The UBBPfw1 does not support the following functions:
 - Symbol-level scheduling
 - Simultaneous use of super uplink and CA
 - SUL CoMP
- The following constraints exist on the baseband processing units:
 - The SUL cells served by a UBBPfw1 board must be configured with the same bandwidth and number of receive antennas.
 - The associated SUL and NR TDD cells must be deployed on the same baseband processing unit.

RF Modules

Table 4-6 lists the RF modules supported by macro base stations.

Table 4-6 RF modules supported by macro base stations

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n78 (3.5 GHz/3.7 GHz)	Modules supporting 32T32R and 64T64R	32T32R and 64T64R	40 MHz, 50 MHz, 60 MHz, 70 MHz, 80 MHz, 90 MHz, and 100 MHz
n41 (2.6 GHz)	Modules supporting 32T32R and 64T64R	32T32R and 64T64R	60 MHz, 80 MHz, and 100 MHz
n80 (1.8 GHz)	RRU3952m, RRU3953, RRU3953w, RRU3959, RRU3959a, RRU3959w, RRU3962, RRU3971, RRU3971a, and 5000 series RRUs ^a	2R and 4R ^b	15 MHz and 20 MHz
n82 (800 MHz)	RRU3965 and 5000 series RRUs ^a	2R and 4R ^b	10 MHz

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n83 (700 MHz)	RRU3262 and 5000 series RRUs ^a	2R and 4R ^b	10 MHz
n84 (2.1 GHz)	RRU3952, RRU3952m, RRU3953, RRU3958, RRU3959, RRU3959a, RRU3962, RRU3971, and 5000 series RRUs ^a	2R and 4R ^b	15 MHz and 20 MHz
n86 (AWS)	RRU3971 and 5000 series RRUs ^a	2R and 4R ^b	15 MHz and 20 MHz
a: 5000 series RRUs refer to RRUs named RRU5xxx (for example, RRU5909). b: It can be split to serve two cells. Only one cell can be configured as an SUL cell and bound to an NR TDD cell to implement super uplink.			

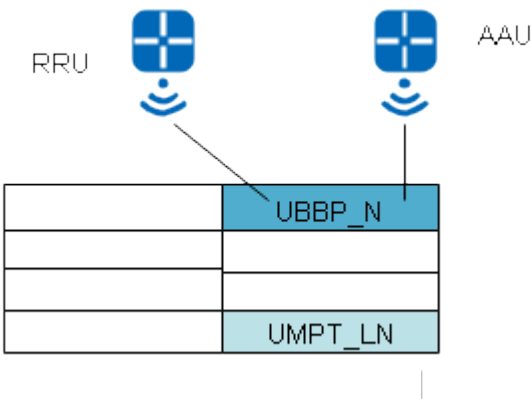
Cells

None

4.3.4 Networking

Networking Requirements of Super Uplink

- The RF modules of the NR TDD and SUL carriers must connect to the same baseband processing unit and provide the same coverage.
 - The following figure shows the networking requirements if the SUL carrier is deployed in an independent frequency band.

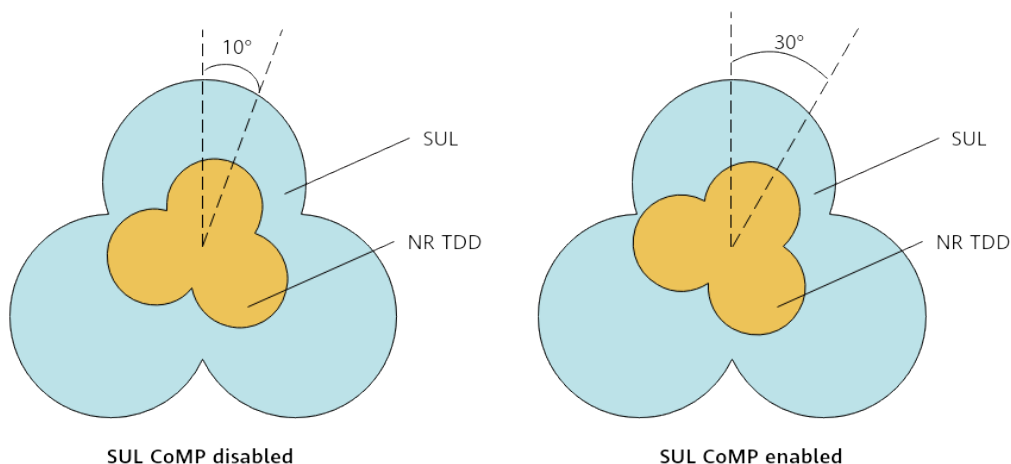


- If the SUL carrier is deployed through LTE FDD and NR Uplink Spectrum Sharing, the networking requirements are the same as those for LTE FDD

and NR Uplink Spectrum Sharing. For details, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

- The SUL carrier is not paired with a downlink carrier. Therefore, the power control and link management of the SUL carrier depend on the downlink measurement of the NR TDD carrier. To maximize the benefits, the SUL carrier coverage must be identical to the corresponding NR TDD carrier. When deploying super uplink, ensure that the antenna azimuth difference between the SUL carrier and corresponding NR TDD carrier is less than 10°. When SUL CoMP is enabled, the SUL carrier and corresponding NR TDD carrier have a maximum antenna azimuth difference of 30°, as shown in [Figure 4-13](#).

Figure 4-13 Azimuth requirements of antennas for the SUL and NR TDD carriers



Networking Requirements when CA Is Enabled

When super uplink is used with CA, an additional UBBP_N is required to carry the SCC in CA.

4.3.5 Others

Requirements on UEs complying with 3GPP R15:

- UEs must support the combinations of NR TDD and SUL bands. If BandCombinationList in the UECapabilityInformation message contains the combinations of NR TDD and SUL bands, UEs support these combinations.
- UEs must support TTI-level switching between NR TDD and SUL bands. If the value of dynamicSwitchSUL in the UECapabilityInformation message is supported, UEs support TTI-level switching between NR TDD and SUL bands. To identify UEs that support TTI-level switching, the **SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter must be selected.

Requirements on UEs complying with 3GPP R16:

- UEs must support the combinations of NR TDD and SUL bands. If the UECapabilityInformation message contains all of the following, super uplink defined in 3GPP R16 is supported.

- BandCombination-UplinkTxSwitch-r16 containing the combinations of NR TDD and SUL bands
- BandCombination-UplinkTxSwitch-r16 containing supportedBandPairListNR-r16
- UEs must support the reporting of the corresponding uplink switching time. If the supportedBandPairListNR-r16 message contains uplinkTxSwitchingPeriod-r16, the corresponding uplink switching time is reported.

Requirements on UEs complying with 3GPP R18 in addition to those on UEs complying with 3GPP R16:

- UEs must support the combinations of NR TDD and SUL bands. If the UECapabilityInformation message contains all of the following, super uplink defined in 3GPP R18 is supported.
 - BandCombination-UplinkTxSwitch-v1800 or BandCombination-UplinkTxSwitch-v1830 containing the combinations of NR TDD and SUL bands
 - BandCombination-UplinkTxSwitch-v1800 containing supportedBandPairListNR-r18
 - switchingPeriodFor2T-r18 indicates that UEs support the combinations of NR 2Tx and SUL 2Tx bands.
 - switchingPeriodFor1T-r18 indicates that UEs support the combinations of NR 2Tx and SUL 1Tx bands, NR 1Tx and SUL 2Tx bands, or NR 1Tx and SUL 1Tx bands.
- UEs must support the reporting of the corresponding uplink switching time. If the supportedBandPairListNR-r18 message contains uplinkTxSwitchingPeriodForBandPair-r18, the corresponding uplink switching time is reported.

4.4 Operation and Maintenance

4.4.1 Precautions

MOs Configured for SUL Cells

As SUL cells only have uplink links, the configuration of these cells involves specific MOs and parameters. In addition to the parameters required for the deployment of super uplink (see [4.4.2 Data Configuration](#)), the following lists the MOs and parameters that take effect for SUL cells.

In the following MOs, all parameters take effect for SUL cells:

- NRDUCellPrach
- NRDUCellQciBearer
- NRDUCellUlpConfig
- NRDUCellUITaConfig
- NRDUCellUeCoop
- NRDUCellSelConfig

- **NRDUCellSpctCloud**

In the MOs described in the tables below, only some parameters take effect for SUL cells.

Table 4-7 Parameters in an **NRDUCellPucch** MO that take effect

Parameter Name	Parameter ID
Format3 CSI-dedicated RB Number	NRDUCellPucch.CsiDedicatedRbNum
Format4 CSI-dedicated RB Number	NRDUCellPucch.Format4CsiDedicatedRbNum
CSI Report Period	NRDUCellPucch.CsiReportPeriod
CSI Resource Algo Switch	NRDUCellPucch.CsiResoureAlgoSwitch
Format1 RB Number	NRDUCellPucch.Format1RbNum
Format3 RB Number	NRDUCellPucch.Format3RbNum
Format4 RB Number	NRDUCellPucch.Format4RbNum
SR Period	NRDUCellPucch.SrPeriod
SR Resource Algo Switch	NRDUCellPucch.SrResoureAlgoSwitch
Note: When channel decoupling is enabled, the parameters in this MO do not take effect for SUL cells.	

Table 4-8 Parameters in an **NRDUCellPusch** MO that take effect

Parameter Name	Parameter ID
Uplink Target IBLER	NRDUCellPusch.UlTargetIbler
SINR Threshold for Waveform Selection	NRDUCellPusch.SinrThldforWaveformSel
Uplink Additional DMRS Position	NRDUCellPusch.UlAdditionalDmr-sPos
Uplink DMRS Type	NRDUCellPusch.UlDmrsType
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch
Initial Uplink SINR Adjustment	NRDUCellPusch.InitUlSinrAdjust
Uplink Preallocation Switch	NRDUCellPusch.UlPreallocationSwitch
Uplink Smart Preallocation Duration	NRDUCellPusch.UlSmartPreallocationDur

Table 4-9 Parameters in an **NRDUCellTrp** MO that take effect

Parameter Name	Parameter ID
CPRI Compression	NRDUCellTrp.CpriCompression
NR DU Cell TRP ID	NRDUCellTrp.NrDuCellTrpId
NR DU Cell ID	NRDUCellTrp.NrDuCellId
Transmit and Receive Mode	NRDUCellTrp.TxRxMode
Baseband Equipment ID	NRDUCellTrp.BasebandEqmId
Baseband Resource Mutual Aid Switch	NRDUCellTrp.BbResMutualAidSw
Frequency Range and Duplex Mode	NRDUCellTrp.FrAndDuplexMode

Table 4-10 Parameters in an **NRDUCellSrs** MO that take effect

Parameter Name	Parameter ID
SRS Period	NRDUCellSrs.SrsPeriod
SRS Algorithm Switch	NRDUCellSrs.SrsAlgoSwitch

Table 4-11 Parameters in an **NRDUCellCoverage** MO that take effect

Parameter Name	Parameter ID
NR DU Cell TRP ID	NRDUCellCoverage.NrDuCellTrpId
NR DU Cell Coverage ID	NRDUCellCoverage.NrDuCellCoverageId
Sector Equipment ID	NRDUCellCoverage.SectorEqmId

Table 4-12 Parameters in an **NRDUCell** MO that take effect

Parameter Name	Parameter ID
Frequency Band	NRDUCell.FrequencyBand
Uplink Bandwidth	NRDUCell.UlBandwidth
Cell Radius	NRDUCell.CellRadius
Subcarrier Spacing	NRDUCell.SubcarrierSpacing
Cyclic Prefix Length	NRDUCell.CyclicPrefixLength
Cell ID	NRDUCell.CellId

Parameter Name	Parameter ID
Physical Cell ID	NRDUCell.PhysicalCellId
Uplink NARFCN	NRDUCell.UlNarfcn
NR DU Cell Networking Mode	NRDUCell.NrDuCellNetworkingMode
Logical Root Sequence Index	NRDUCell.LogicalRootSequenceIndex
PRACH Frequency Start Position	NRDUCell.PrachFreqStartPosition

Table 4-13 Parameter in an **NRDUCellUePwrSaving** MO that takes effect

Parameter Name	Parameter ID	Option
NR DU Cell DRX Algorithm Switch	NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch	BASIC_DRX_SW

Table 4-14 Parameters in an **NRDUCellAlgoSwitch** MO that takes effect

Parameter Name	Parameter ID	Option
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_WAVEFORM_ADAPT_SW
		UL_IBLER_ADAPT_SW
Spectrum Cloudification Switch	NRDUCellAlgoSwitch.SpectrumCloudSwitch	LTE_NR_UL_SPECTRUM_SHARING_SW
Uplink Interference Random Schedule Switch	NRDUCellAlgoSwitch.UlInterfRandomSchSwitch	None
UL and DL Decoupling Switch	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	None
Super Uplink Switch	NRDUCellAlgoSwitch.SuperUplinkSwitch	SUPER_UL_TDM_SCH_SW
UE Link Abnormal Detect Switch	NRDUCellAlgoSwitch.UeLinkAbnormalDetectSwitch	None
CoMP Switch	NRDUCellAlgoSwitch.CompSwitch	INTRA_GNB_UL_COMP_SW
		INTRA_GNB_OVERLAPPING_MEAS_SW
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	TIMING_CARRIER_SHUTDOWN_SW

Parameter Name	Parameter ID	Option
		LNR_SMART_CARRIER_SHUT DOWN_SW
BWP Configuration Policy Switch	NRDUCellAlgoSwitch.BwpC onfigPolicySwitch	INIT_BWP_FULL_BW_SW
Uplink 256QAM Switch	NRDUCellAlgoSwitch.Ul25 6QamSwitch	UL_256QAM_OFF
		UL_256QAM_FIXED
		UL_256QAM_ADAPT

4.4.2 Data Configuration

4.4.2.1 Data Preparation

When enabling this function, ensure that the **NRDUCellUITaConfig.UlTimeAlignmentTimer** parameter for the SUL cell is set to a value consistent with that of the NR TDD cell.

SUL uses the same TA offset (specified by **NRDUCell.TaOffset**) as NR TDD in order to ensure that uplink time alignment is achieved between SUL and NR TDD. If the SUL spectrum is obtained through LTE FDD and NR Uplink Spectrum Sharing, uplink time alignment needs to be achieved between SUL and LTE FDD. In this case, the frame offset must be configured on the LTE side to achieve the alignment of the TA offset. For details about the configuration requirements, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

[Table 4-15](#) and [Table 4-16](#) describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment. For details about these parameters, see *Cell Management*.

Table 4-15 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Super Uplink Switch	NRDUCellAlgoSwitch .SuperUplinkSwitch	Select the SUPER_UL_TDM_SCH_SW option in the NR TDD cell when the feature is enabled.
Policy Switch	NRDUCellOp.PolicyS witch	Select the OP_SUL_SWITCH option of this parameter for the operator that requires super uplink if the gNodeB is shared by multiple operators.
Duplex Mode	NRDUCell.DuplexMo de	Set this parameter to CELL_SUL for the SUL carrier.

Parameter Name	Parameter ID	Setting Notes
Frequency Band	NRDUCell.FrequencyBand	Set this parameter based on the operator's network plan. For details, see section 5.2 "Operating bands" in 3GPP TS 38.104 V15.0.0.
Uplink NARFCN	NRDUCell.UINarfcn	Set this parameter based on the operator's network plan. For details, see section 5.4.2.3 "Channel raster entries for each operating band" in 3GPP TS 38.104 V15.0.0.
Uplink Bandwidth	NRDUCell.UlBandwidth	Set this parameter based on the operator's network plan.
Subcarrier Spacing	NRDUCell.SubcarrierSpacing	Set this parameter to 15KHZ for the SUL carrier.
Logical Root Sequence Index	NRDUCell.LogicalRootSequenceIndex	Set this parameter based on the operator's network plan.
NR DU Cell ID	NRDUCellSulRelation.NrDuCellId	Set this parameter based on the operator's network plan.
SUL Index	NRDUCellSulRelation.SulIndex	Set this parameter based on the operator's network plan.
SUL Type	NRDUCellSulRelation.SulType	Set this parameter to FIX_SUL .
SUL NR DU Cell ID	NRDUCellSulRelation.SulNrDuCellId	This parameter is mandatory when the NRDUCellSulRelation.SulType parameter is set to FIX_SUL . Set this parameter based on the operator's network plan.
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	Select the SPEC_UE_IDENTIFY_SW option.

Parameter Name	Parameter ID	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	Select the UL_SCH_ALIGN_SW option. If this option is selected, PUSCH scheduling for super uplink UEs and that for non-super-uplink UEs are aligned in the time domain. If this option is deselected, they are not aligned in the time domain, and MU-MIMO pairing cannot be performed. When the uplink load of super uplink UEs is high, the uplink throughput of non-super-uplink UEs decreases. In extreme scenarios (for example, the uplink RB usage is close to 100%), non-super-uplink UEs experience service drops.
CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch	Select the UL_MULTI_CC_TA_FLEX_ADJ_SW option.
UL Multi-Carrier Absolute TA Diff Thld	NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld	You are advised to use the recommended value. A smaller value of this parameter indicates a stricter requirement for TA selection based on the CC with a relatively small overall impact. However, if the value is too small, the increase in the uplink cell throughput will be relatively small. A larger value of this parameter indicates a more relaxed requirement for TA selection based on the CC with a relatively small overall impact. However, if the value is too large, there will be a relatively high probability of negative gains in the uplink throughput.
SUL Algorithm Switch	NRDUCellCollabServ.SulAlgorithmSwitch	Select the BAND_COMB_UL_TX_PWR_UDT_SW option.

Parameter Name	Parameter ID	Setting Notes
Max UE Transmit Power	NRDUCellSelConfig. <i>MaxUeTransmitPower</i>	Set this parameter to 29 .
UL Power Control Algorithm Extension Switch	NRDUCellUIPcConfig. <i>ULPwrCtrlAlgoExtSwitch</i>	Select the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume distributed to SCells in the uplink when super uplink and uplink CA are both enabled.

Table 4-16 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
SRS RSRP Threshold	NRDUCellCollabServ. <i>SrsRsrpThld</i>	The default value is recommended.
RSRP Threshold	NRDUCellCollabServ. <i>RsrpThld</i>	The default value is recommended during initial activation. • When the NR uplink load is high, set this parameter to a large value to enable additional UEs to be carried on the NR SUL carrier. • When the NR SUL load is high, set this parameter to a small value to enable additional UEs to be carried on the NR uplink carrier.

Parameter Name	Parameter ID	Setting Notes
SUL Algorithm Switch	NRDUCellCollabServ. <i>SulAlgoSwitch</i>	• Select the SEC_HARMONIC_INTRF_A_VOID_SW option to enable secondary harmonic interference avoidance. • Select the SUL_CHANNEL_DECOUPLE_SW option to enable channel decoupling. • Select the UL_CARR_SEL_FOR_USER_EXP_SW option to enable user-experience-based uplink carrier selection. • Select the INTRA_GNB_SUL_COMP_SW option to enable SUL CoMP. It is recommended that this option be selected when the antenna azimuth difference is greater than 10°. • Select the SCHEDULE_FIRST_HALF_SLOT_SW option to enable symbol-level scheduling for SUL. • Select the SUL_FAULT_TOLERANCE_SW option to enable SUL carrier detection and management. • Select the SUL_UE_POWER_SAVING_SW option to enable SUL UE power saving. • Select the SUL_CA_SRS_SWITCHING_SW option to enable downlink CA, super uplink, and SRS carrier switching for UEs that can support downlink 2CC aggregation, SUL in compliance with R16 or later, and SRS carrier switching at the same time. • Select the SUL_CA_2CC_SRS_SWITCH

Parameter Name	Parameter ID	Setting Notes
		ING_SW option to enable downlink CA, super uplink, and SRS carrier switching for UEs that can support downlink 3CC aggregation, SUL in compliance with R16 or later, and SRS carrier switching at the same time. Select the SUL_CA_SRS_SWITCHING_EXT_SW option to enable downlink CA, super uplink, and SRS carrier switching for UEs that can support downlink 2CC or 3CC aggregation, SUL in compliance with R15, and SRS carrier switching at the same time.
Multi-Carrier Management Switch	NRDUCellCollabServ. <i>MultiCarrMgmtSw</i>	- Select the SMART_CARR_SEL_SW option to enable cell-level SUL and CA optimal carrier selection. - Select the SERVICE_AWARE_CARR_SEL_SW option to enable UE-level SUL and CA optimal carrier selection.
SCC Configuration Interval	NRDUCellCarrMgmt. <i>SccCfgInterval</i>	The default value is recommended.
RSRP Threshold for SSB Selection	NRDUCellPrach.<i>RsrpThldForSsbSelection</i>	For NR TDD cells, the value 29 is recommended when the feature is enabled. The value of this parameter is equal to the sum of NRDUCellSelConfig.MinimumRxLevel and 157.

Parameter Name	Parameter ID	Setting Notes
NUL 1TX to SUL 1TX Switching Time	NRDUCellCollabServ. <i>Nul1TxToSul1TxSwitchTime</i>	It is recommended that this parameter be set based on UE conditions. • A smaller value of this parameter results in more symbols available for the SUL carrier and higher throughput of the SUL carrier. However, super uplink does not take effect for some UEs whose time taken to switch between carriers is larger than the value of this parameter. • A larger value of this parameter results in more UEs for which super uplink takes effect. However, it also results in fewer symbols available for the SUL carrier and lower throughput of the SUL carrier.
SUL SINR Lower Limit	NRDUCellCollabServ. <i>SulSinrLowerLimit</i>	The default value is recommended.
SUL SINR Lower Limit Offset	NRDUCellCollabServ. <i>SulSinrLowerLimitOffset</i>	The default value is recommended.
SUL DTX SINR Lower Limit	NRDUCellCollabServ. <i>SulDtxSinrLowerLimit</i>	The default value is recommended.
Carrier Selection TDD SRS RSRP Threshold	NRDUCellCollabServ. <i>CarrSelTddSrsRsrpThld</i>	The default value is recommended.
Multi-Carrier Capb Comb Select Strat	NRCeCellCaMgmtCon-fig. <i>CaBandCombSelectionStrat</i>	Set this parameter based on the operator's network plan.
Multi-Carrier Compatibility Switch	NRCeCellCaMgmtCon-fig. <i>CaCompatibilitySwitch</i>	It is recommended that the SRS_WT_BASED_CARR_SEL_SW option be selected.

Parameter Name	Parameter ID	Setting Notes
SUL Cov-based HO RSRP Threshold Offset	NRCellMeasConfig.SulCovHoRsrpThldOffset	Set this parameter based on the operator's network plan. • A larger value of this parameter results in a higher probability of triggering event A2 related to coverage-based handover and a lower probability of triggering event A1 related to coverage-based handover. • A smaller value of this parameter results in the opposite effects.
CA Strategy Switch	NRCellCaMgmtConfig.CaStrategySwitch	It is recommended that the SUL_CA_BASED_UL_CARR_SELECT_SW option be selected.
CoMP Configuration Policy	NRDUCellComp.CompConfigPol	Select the UL_2CELL_COMP_BEAR_POLICY_SW option to enable SUL CoMP to take effect for UEs with specific QCLs.
Service Differentiation Algorithm Switch	gNBQciBearer.ServiceDiffAlgoSwitch	Select the COMP_SW option for UEs requiring SUL CoMP for a specific QCL.

4.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

An NR TDD cell must have already been configured before running the following commands. For details, see *Cell Management*. In addition, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

The activation command examples when the SUL carrier is deployed in an independent frequency band are as follows (NR TDD cell ID of 0 and SUL cell ID of 1 used as an example):

```
//Adding RF equipment information for an SUL cell
ADD RRCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=0, HSN=2, HPN=2, CR=AUTO,
```

```
USERDEFRATENEGOSW=OFF;
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=LN, RN="5G06_60", RXNUM=4,
TXNUM=4, MNTMODE=NORMAL, RFDCPWROFFALMDETECTSW=OFF, RFTXSIGNDETECTSW=OFF;
//Adding sector equipment information
ADD SECTOR: SECTORID=1, SECNAME="Sector_0", USERLABEL="for cell", ANTNUM=4, ANT1CN=0,
ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0,
ANT3SRN=60, ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D,
CREATESECTOREQM=FALSE;
ADD SECTOREQM: SECTOREQMID=1, SECTORID=1, ANTCFGMODE=ANTENNAPORT, ANTNUM=4,
ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANTTYPE1=RX_MODE, ANT2CN=0, ANT2SRN=60,
ANT2SN=0, ANT2N=R0B, ANTTYPE2=RX_MODE, ANT3CN=0, ANT3SRN=60, ANT3SN=0, ANT3N=R0C,
ANTTYPE3=RX_MODE, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, ANTTYPE4=RX_MODE;
//Adding an SUL cell
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="SUL Cell", DuplexMode=CELL_SUL, FrequencyBand=N80,
ULNarfcn=344500, ULBandwidth=CELL_BW_15M, SubcarrierSpacing=15KHZ, LogicalRootSequenceIndex=0;
//Adding a TRP for the NR DU cell
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=1, TxRxMode=4R, CpriCompression=NO_COMPRESSION,
FrAndDuplexMode=SUL;
//Adding an NR DU cell coverage item
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=1, SectorEqmId=1;
//Adding an SUL for the NR DU cell by associating the NR TDD cell with the SUL cell
ADD NRDUCELLSULRELATION: NrDuCellId=0, SulIndex=0, SulType=FIX_SUL, SulNrDuCellId=1;
//Turning on the super uplink switch for the NR TDD cell. During the execution of this high-risk command,
the SUL cell may be reestablished. As such, you are advised to only run this command when traffic is light
(for example, in the early morning).
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SuperUplinkSwitch=SUPER_UL_TDM_SCH_SW-1;
//Turning on the specified UE cooperation switch
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1;
//Optional Turning on the super uplink switch for an operator configured for the NR TDD cell in multi-
operator sharing scenarios
MOD NRDUCELLTOP: NrDuCellId=0, OperatorId=0, PolicySwitch=OP_SUL_SWITCH-1;
//Turning on the uplink scheduling time alignment switch. Running the following command will cause the
NR TDD cell to restart.
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-1;
//Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//Setting the absolute TA difference threshold for uplink multi-CC operation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UIMultiCcAbsTaDiffThld=16;
//Enabling the function of updating the uplink transmit power of band combinations
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, SulAlgoSwitch=BAND_COMB_UL_TX_PWR_UDT_SW-1;
//Enabling the high-power UE function on the base station and allowing the maximum transmit power of
UEs to reach 29 dBm
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=29;
//Optional Selecting the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume
distributed to SCells in the uplink when super uplink and uplink CA are both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
```

The activation command examples when the SUL carrier is deployed through LTE FDD and NR Uplink Spectrum Sharing are as follows (NR TDD cell ID of 0 and SUL cell ID of 1 used as an example):

```
//Adding an SUL cell
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="SUL Cell", DuplexMode=CELL_SUL, FrequencyBand=N80,
ULNarfcn=344500, ULBandwidth=CELL_BW_15M, SubcarrierSpacing=15KHZ, LogicalRootSequenceIndex=0;
//Adding a TRP for the NR DU cell
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=1, TxRxMode=4R, CpriCompression=NO_COMPRESSION,
FrAndDuplexMode=SUL;
//Adding an NR DU cell coverage item
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=1, SectorEqmId=1;
//Adding an SUL for the NR DU cell by associating the NR TDD cell with the SUL cell
ADD NRDUCELLSULRELATION: NrDuCellId=0, SulIndex=0, SulType=FIX_SUL, SulNrDuCellId=1;
//Turning on the super uplink switch for the NR TDD cell. During the execution of this high-risk command,
the SUL cell may be reestablished. As such, you are advised to only run this command when traffic is light
(for example, in the early morning).
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SuperUplinkSwitch=SUPER_UL_TDM_SCH_SW-1;
//Turning on the specified UE cooperation switch
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1;
//Optional Turning on the super uplink switch for an operator configured for the NR TDD cell in multi-
operator sharing scenarios
```

```
MOD NRDUCELLPUSCH: NrDuCellId=0, OperatorId=0, PolicySwitch=OP_SUL_SWITCH-1;  
//Turning on the uplink scheduling time alignment switch. Running the following command will cause the  
NR TDD cell to restart.  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-1;  
//Enabling flexible TA adjustment in uplink multi-CC scenarios  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;  
//Setting the absolute TA difference threshold for uplink multi-CC operation  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UIMultiCcAbsTaDiffThld=16;  
//Enabling the function of updating the uplink transmit power of band combinations  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, SulAlgoSwitch=BAND_COMB_UL_TX_PWR_UDT_SW-1;  
//Enabling the high-power UE function on the base station and allowing the maximum transmit power of  
UEs to reach 29 dBm  
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=29;
```

Optimization Command Examples

```
//Configuring the policy on multi-carrier capability combination selection  
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaBandCombSelectionStrat=UL_PEAK_RATE_FIRST,  
CaCompatibilitySwitch=SRS_WT_BASED_CARR_SEL_SW-1;  
//Modifying the SUL for an NR DU cell  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, Nul1TxToSul1TxSwitchTime=MICROSECOND140,  
RsrpThld=-110, SrsRsrpThld=-111,  
SulAlgoSwitch=SEC_HARMONIC_INTRF_AVOID_SW-1&UL_CARR_SEL_FOR_USER_EXP_SW-1&SUL_CHANNEL_  
DECOUPLE_SW-1&INTRA_GNB_SUL_COMP_SW-1&SCHEDULE_FIRST_HALF_SLOT_SW-1&SUL_FAULT_TOLER  
ANCE_SW-1&SUL_UE_POWER_SAVING_SW-1&SUL_CA_SRS_SWITCHING_SW-1&SUL_CA_2CC_SRS_SWITCHIN  
G_SW-1&SUL_CA_SRS_SWITCHING_EXT_SW-1,  
MultiCarrMgmtSw=SERVICE_AWARE_CARR_SEL_SW-1&SMART_CARR_SEL_SW-1, SulDtxSinrLowerLimit=0,  
SulSinrLowerLimit=-80, SulSinrLowerLimitOffset=10, CarrSelTddSrsRsrpThld=-105;  
//Setting the timer specifying the SCC addition interval  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SccCfgInterval=8;  
//Setting the RSRP threshold for SSB selection upon network access  
MOD NRDUCELLPRACH: NrDuCellId=0, RsrpThldForSsbSelection=29;  
//Setting the SUL coverage-based handover RSRP threshold offset  
MOD NRCELLMEASCONFIG: NrCellId=0, SulCovHoRsrpThldOffset=-5;  
//Setting the policy for optimal carrier selection to consider SUL carriers  
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SUL_CA_BASED_UL_CARR_SEL_SW-1;  
//Setting the CoMP configuration policy  
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;  
//Selecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9  
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the uplink scheduling time alignment switch. Running the following command will cause the  
NR TDD cell to restart.  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-0;  
//Turning off the super uplink switch. During the execution of this high-risk command, NR TDD and SUL  
cells are reestablished. As such, you are advised to only run this command when traffic is light (for example,  
in the early morning).  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SuperUplinkSwitch=UL_SCH_ALIGN_SW-0;  
//Disabling flexible TA adjustment in uplink multi-CC scenarios  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;  
//Disabling the policy for optimal carrier selection that considers SUL carriers  
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SUL_CA_BASED_UL_CARR_SEL_SW-0;  
//Deselecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9  
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-0;  
//Restoring the CoMP configuration policy  
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;  
//Optional Deselecting the MF_ACTUAL_PUSCH_ILPC_SW option when super uplink and uplink CA are  
both enabled  
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

Super Uplink

Use either of the following methods to check whether super uplink has taken effect:

- Using MML commands
 - a. Run the **DSP NRDUCELL** command for the SUL cell on the gNodeB. If the value of **NR DU Cell State** is **Normal**, the SUL cell is normal.
 - b. Run the **DSP NRCELLUENUMBER** command on the gNodeB to query the number of UEs enabled with super uplink. If **Cell Super Uplink UE Number** has a non-zero value, super uplink has taken effect.
- Observing counters

When super uplink is enabled and UEs supporting super uplink exist on the network, observe the **N.User.ULEnhTimeFreq.Avg** counter. If it has a non-zero value, super uplink has taken effect. If the counter value is always zero, super uplink has not taken effect.

SUL CoMP

Use either of the following methods to check whether SUL CoMP has taken effect:

- Observing counters

If the value of the **N.IntragNB.UL.CoMP.SchUser.Sum** counter of the SUL cell is not zero, this function has taken effect.
- Tracing signaling

On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance Monitoring > UL CoMP Statistics(Cell) Monitoring**. Check the number of SUL CoMP UEs and the number of RBs they use. If the values are not zero, SUL CoMP has taken effect.

Channel Decoupling

After channel decoupling is enabled, the number of available PUSCH RBs on the SUL carrier increases, which can be observed using the **N.PRB.UL.PUSCH.Avail.Avg** counter.

When the SUL carrier is obtained through LTE and NR uplink spectrum sharing, the **N.PRB.UL.PUSCH.Avail.Avg** counter value includes the number of PRBs allocated to LTE.

Symbol-Level Scheduling

On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > User Common Monitoring** to check whether the number of uplink scheduling times increases. If it does, SUL symbol-level scheduling has taken effect.

SUL UE Power Saving

After the SUL UE power saving function is enabled, check the external CHR event PRIVATE_UL_CARRIER_SWITCH_TRIGGER_REPORT to determine the cause value for uplink carrier change. If the cause value is SUL_DELETE_SUL_UE_POWER_SAVING, the SUL UE power saving function has taken effect.

4.4.4 Network Monitoring

Super Uplink

Super uplink increases the average uplink throughput of SUL UEs in a cell. As a result, the average uplink UE throughput of the entire cell increases, the proportion of UEs with throughput falling into a high-throughput range increases, and the proportion of UEs with throughput falling into a low-throughput range decreases.

- Monitoring SUL UEs in a cell involves the following indicators:
 - The average uplink throughput of SUL UEs in a cell can be obtained using the following formula: $(N.ThpVol.SUL.UL - N.ThpVol.SUL.UL.SmallPkt) / N.ThpTime.SUL.UL.RmvSmallPkt$.
 - The proportion of SUL UEs whose throughput falls into a low-throughput range can be obtained by dividing the number of samples in the low-throughput range by the total number of samples in a cell. A low-throughput range is relative. For example, range 0 is a low-throughput range compared with range 1.
 - The proportion of SUL UEs whose throughput falls into a high-throughput range can be obtained by dividing the number of samples in the high-throughput range by the total number of samples in a cell. A high-throughput range is relative. For example, range 14 is a high-throughput range compared with range 13. [Table 4-17](#) lists the counters used to observe the number of UEs in each rate range.

Table 4-17 Number of samples with the uplink throughput of SUL UEs in each range

Counter ID	Counter Name
1911829662	N.Thp.SUL.UL.Samp.Index0
1911829663	N.Thp.SUL.UL.Samp.Index1
1911829664	N.Thp.SUL.UL.Samp.Index2
1911829665	N.Thp.SUL.UL.Samp.Index3

Counter ID	Counter Name
1911829666	N.Thp.SUL.UL.Samp.Index4
1911829667	N.Thp.SUL.UL.Samp.Index5
1911829668	N.Thp.SUL.UL.Samp.Index6
1911829669	N.Thp.SUL.UL.Samp.Index7
1911829670	N.Thp.SUL.UL.Samp.Index8
1911829671	N.Thp.SUL.UL.Samp.Index9
1911829672	N.Thp.SUL.UL.Samp.Index10
1911829673	N.Thp.SUL.UL.Samp.Index11
1911829674	N.Thp.SUL.UL.Samp.Index12
1911829675	N.Thp.SUL.UL.Samp.Index13
1911829676	N.Thp.SUL.UL.Samp.Index14

- Monitoring the overall performance of a cell involves the following indicators:
 - The average uplink UE throughput in a cell is measured by **User Uplink Average Throughput (DU)**.
 - The proportion of UEs whose throughput falls into a low-throughput range can be obtained by dividing the number of samples in the low-throughput range by the total number of samples in a cell. A low-throughput range is relative (for example, range 0 is a low-throughput range compared with range 1).
 - The proportion of UEs whose throughput falls into a high-throughput range can be obtained by dividing the number of samples in the high-throughput range by the total number of samples in a cell. A high-throughput range is relative. For example, range 14 is a high-throughput range compared with range 13. [Table 4-18](#) lists the counters used to observe the number of UEs in each rate range.

Table 4-18 Number of samples with the uplink throughput in each range in a cell

Counter ID	Counter Name
1911816708	N.Thp.UL.Samp.Index0
1911816709	N.Thp.UL.Samp.Index1
1911816710	N.Thp.UL.Samp.Index2
1911816711	N.Thp.UL.Samp.Index3
1911816712	N.Thp.UL.Samp.Index4
1911816713	N.Thp.UL.Samp.Index5

Counter ID	Counter Name
1911816714	N.Thp.UL.Samp.Index6
1911816715	N.Thp.UL.Samp.Index7
1911816716	N.Thp.UL.Samp.Index8
1911816717	N.Thp.UL.Samp.Index9
1911816937	N.Thp.UL.Samp.Index10
1911816941	N.Thp.UL.Samp.Index11
1911816938	N.Thp.UL.Samp.Index12
1911816939	N.Thp.UL.Samp.Index13
1911816940	N.Thp.UL.Samp.Index14

SUL CoMP

To compare the throughput before and after SUL CoMP is enabled, you are advised to select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter for the NR TDD cell one week before SUL CoMP is enabled so that statistics of throughput-related counters for UEs in overlapping areas can be collected in advance.

SUL CoMP increases the uplink throughput of UEs in overlapping areas that excludes small-packet throughput without decreasing the average uplink UE throughput and the uplink UE throughput that excludes small-packet throughput. You can evaluate the gains of SUL CoMP by comparing the values of the following performance counters before and after this function is enabled. The counter values depend on parameter settings.

To observe the gains of SUL CoMP, you are advised to set the **NRCellComp.OverLappingRsrpOffset** (for the NR FDD cell) and **NRCellComp.UlCompRsrpOffset** (for the NR TDD cell where super uplink UEs are located) parameters to the same value. The reason is as follows: The offset for event A3 in the overlapping area of the NR FDD cells is determined by the **NRCellComp.OverLappingRsrpOffset** parameter. The RSRP difference for event A5 indicating whether SUL CoMP takes effect for super uplink UEs is determined by the **NRCellComp.UlCompRsrpOffset** parameter.

- Uplink throughput of UEs in overlapping areas excluding small-packet throughput = $(\text{N.ThpVol.UL.CellOverlap} - \text{N.ThpVol.UL.SmallPkt.CellOverlap}) / \text{N.ThpTime.UL.RmvSmallPkt.CellOverlap}$
- Uplink UE throughput excluding small-packet throughput = $(\text{N.ThpVol.UL} - \text{N.ThpVol.UE.UL.SmallPkt}) / \text{N.ThpTime.UE.UL.RmvSmallPkt}$
- Average uplink UE throughput = $\text{N.ThpVol.UL} / \text{N.ThpTime.UL}$

Channel Decoupling

Channel decoupling improves the average uplink UE throughput in a cell, increases the proportion of UEs with throughput falling into a high-throughput range, and decreases the proportion of UEs with throughput falling into a low-throughput range. The performance counters are the same as those for super uplink. For details, see [Super Uplink](#).

Symbol-Level Scheduling

See network monitoring for super uplink in [Super Uplink](#).

5 SUL and NR FDD Co-Deployment

This chapter describes the super uplink function with SUL and NR FDD co-deployment. Super uplink described in this chapter refers to super uplink with SUL and NR FDD co-deployment.

5.1 Principles

In SUL and NR FDD co-deployment mode, the SUL function is provided by NR FDD cells. That is, super uplink is implemented using NR TDD cells and NR FDD cells. Super uplink supports flexible mapping between intra-gNodeB and inter-gNodeB NR TDD cells and SUL cells (NR FDD cells). Unless otherwise specified, the descriptions in the following sections apply to both intra-gNodeB and inter-gNodeB deployment.

Table 5-1 lists the combinations of NR TDD, NR FDD, and SUL frequency bands supported in SUL and NR FDD co-deployment mode. For details about the mapping between frequency bands and frequencies, see 3GPP TS 38.104. The bandwidth requirement of each frequency band is described in **RF Modules** in **5.3.3 Hardware**.

Table 5-1 Combinations of NR TDD, NR FDD, and SUL frequency bands supported by super uplink with SUL and NR FDD co-deployment

NR TDD Frequency Band	SUL Frequency Band	NR FDD Frequency Band	Applicable Base Station Model
n78	n80	n3	Macro base station
	n81	n8	Macro base station
	n82	n20	Macro base station
	n83	n28	Macro base station

NR TDD Frequency Band	SUL Frequency Band	NR FDD Frequency Band	Applicable Base Station Model
	n84	n1	Macro and LampSite base stations
n41	n80	n3	LampSite base station
	n81	n8	Macro base station
	n83	n28	Macro base station
	n84	n1	Macro base station
n40	n83	n28	Macro base station
	n82	n20	Macro base station

 NOTE

The frequency band combinations n41+n81 and n40+n83 are only for trial use.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

5.1.1 Basic Functions

Super uplink with SUL and NR FDD co-deployment is enabled by selecting the **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**. *SuperUplinkSwitch* parameter in both NR TDD and NR FDD cells. If the NR TDD and NR FDD cells are served by different gNodeBs, the **INTER_GNB_SUPER_UPLINK_SW** option of the

NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is required to be selected in both NR TDD and NR FDD cells.

If the gNodeB is shared by multiple operators (the **gNBSharingMode.gNBMultiOpSharingMode** parameter is set to a value other than **INDEPENDENT**), operators requiring super uplink must also select the **OP_SUL_SWITCH** option of the **NRDUCellOp.PolicySwitch** parameter for the NR TDD cell.

If two NR TDD cells with different slot configurations need to be paired with an NR FDD cell to implement the super uplink function at the same time, the **SUL_WITH_MULTI_SLOT_CFG_NUL_SW** option of the **NRDUCellCollabCoop.SulAlgoSwitch** parameter must be selected for the NR FDD cell.

The differences between "super uplink with independent SUL deployment" and "super uplink with SUL and NR FDD co-deployment" lie in cell configuration, carrier management, uplink power control, timing advance, and SUL CoMP.

5.1.1.1 Cell Configuration and Carrier Information

Before enabling super uplink with SUL and NR FDD co-deployment, parameters must be configured for NR TDD and NR FDD cells.

NR TDD Cell Configuration

In addition to the basic configurations of the NR TDD cell, parameters (as listed in [Table 5-2](#)) in an **NRDUCellSulRelation** MO must be set to configure the relationship between NR TDD and SUL cells.

Table 5-2 Parameters for configuring SUL relationship specific for the NR TDD cell

Parameter Name	Parameter ID	Description
NR DU Cell ID	NRDUCellSulRelation.NrDuCellId	This parameter indicates the ID of the NR TDD cell.
SUL Index	NRDUCellSulRelation.SulIndex	This parameter identifies a unique SUL instance within the NR TDD cell. Currently, only one SUL carrier is supported.
SUL Type	NRDUCellSulRelation.SulType	This parameter indicates the SUL type when super uplink is enabled. This parameter must be set to DYN_SUL for super uplink with SUL and NR FDD co-deployment.

Parameter Name	Parameter ID	Description
FDD Frequency Band	NRDUCellSulRelation.FddFrequencyBand	This parameter indicates the operating band of the FDD cell on the SUL carrier in super uplink with SUL and NR FDD co-deployment. This parameter must be set to the same value as the NRDUCell.FrequencyBand parameter for the NR FDD cell.
FDD UL NR-ARFCN	NRDUCellSulRelation.FddULNrarfcn	This parameter indicates the uplink NR-ARFCN of the FDD cell on the SUL carrier in super uplink with SUL and NR FDD co-deployment. This parameter is mandatory when the NRDUCellSulRelation.SulType parameter is set to DYN_SUL . This parameter must be set to the same value as the NRDUCell.UlNrarfcn parameter for the NR FDD cell.
FDD UL Bandwidth	NRDUCellSulRelation.FddUlBandwidth	This parameter indicates the uplink bandwidth of the FDD cell on the SUL carrier in super uplink with SUL and NR FDD co-deployment. This parameter must be set to the same value as the NRDUCell.UlBandwidth parameter for the NR FDD cell.

Parameter Name	Parameter ID	Description
FDD UL BWP0 Configuration Mode	NRDUCellSulRelation.FddUlBwp0ConfigMode	This parameter indicates whether FDD uplink BWP0 configuration is automatic or manual for super uplink with SUL and NR FDD co-deployment. - If this parameter is set to AUTO, FDD uplink BWP0 parameters are configured automatically. Automatic configuration applies to NR FDD cells with a standard bandwidth of 20 MHz or lower. It also applies to NR FDD cells with a large bandwidth greater than 20 MHz where spectrum use is exclusive (that is, spectrum is not shared between NR FDD and LTE FDD). - If this parameter is set to MANUAL, FDD uplink BWP0 parameters are configured manually. Manual configuration applies to NR FDD cells with a large bandwidth greater than 20 MHz where spectrum is shared with LTE FDD. It also applies in scenarios where FDD uplink BWP0 must be specially configured.
FDD UL BWP0 Bandwidth	NRDUCellSulRelation.FddUlBwp0Bandwidth	This parameter indicates the number of RBs across the uplink BWP0 bandwidth in super uplink with SUL and NR FDD co-deployment.
FDD UL BWP0 Start PRB	NRDUCellSulRelation.FddUlBwp0StartPrb	This parameter indicates the start frequency-domain position of SUL BWP0 in super uplink with SUL and NR FDD co-deployment. PRB 0 is used as the reference point.
FDD UL Subcarrier Spacing	NRDUCellSulRelation.FddSubcarrierSpacing	This parameter indicates the subcarrier spacing of the FDD cell on the SUL carrier in super uplink with SUL and NR FDD co-deployment. This parameter is mandatory when the NRDUCellSulRelation.SulType parameter is set to DYN_SUL . This parameter must be set to the same value as the NRDUCell.SubcarrierSpacing parameter for the NR FDD cell.

Parameter Name	Parameter ID	Description
FDD Cyclic Prefix Length	NRDUCellSulRelation.FddCyclicPrefixLength	This parameter indicates the cyclic prefix (CP) length of the FDD cell on the SUL carrier in super uplink with SUL and NR FDD co-deployment. Currently, only normal CP is supported. This parameter must be set to the same value as the NRDUCell.CyclicPrefixLength parameter for the NR FDD cell.
FDD LTE NR SPCTSHR Flag	NRDUCellSulRelation.FddLteNrSpctShrFlag	This parameter indicates whether LTE FDD and NR spectrum is shared in the NR FDD cell for super uplink with SUL and NR FDD co-deployment.

NR FDD Cell Configuration

When super uplink with SUL and NR FDD co-deployment is enabled, basic configurations of the NR FDD cell (specified by the **NRDUCell** MO) must be consistent with the parameter settings in the **NRDUCellSulRelation** MO.

Inter-gNodeB Deployment Requirements

If the NR TDD and NR FDD cells are served by different gNodeBs, the **NRCellRelation.InterGnodebSulFlag** parameter must be set to **TRUE** and the delay level must be configured for these cells. The **NRCellRelation.InterGnodebSulFlag** parameter can be automatically or manually configured.

- When the **NRCellAnr.InterGnbCellCfgMode** parameter is set to **AUTO**, the gNodeB can automatically identify inter-gNodeB neighbor relationships and adjust the setting of the **NRCellRelation.InterGnodebSulFlag** parameter, for example, changing its value from **FALSE** to **TRUE** for a neighboring cell.
- When the **NRCellAnr.InterGnbCellCfgMode** parameter is set to **MANUAL**, the inter-gNodeB SUL flag needs to be manually configured using the **NRCellRelation.InterGnodebSulFlag** parameter.

The delay levels of NR TDD and NR FDD cells must be the same. Otherwise, super uplink cannot be enabled. The delay level of an NR TDD cell is specified by the **NRDUCellCollabServ.NulSulMaxInterGnbDelay** parameter, and that of an NR FDD cell is specified by the **NRDUCellCollabCoop.NulSulMaxInterGnbDelay** parameter.

The delay level needs to be determined based on the inter-gNodeB one-way delay. The inter-gNodeB one-way delay (unit: 0.1 microsecond) equals $\max(\text{gNodeB DU eXn User Plane One-Way Delay, gNodeB DU eXn User Plane Two-Way Delay} - \text{gNodeB DU eXn User Plane One-Way Delay})$.

The procedure for querying the values of **gNodeB DU eXn User Plane One-Way Delay** and **gNodeB DU eXn User Plane Two-Way Delay** is as follows:

- Before inter-gNodeB super uplink is enabled, you are advised to manually configure an IP performance monitoring (IP PM) session with the activation direction being uplink to measure the eXn-U transmission link delay. Then, you can run the **DSP IPPMSESSION** command to query the delay.
- After inter-gNodeB super uplink is enabled, the gNodeB automatically creates an IP PM session to measure the eXn-U transmission link delay. In this case, you can run the **DSP GNBUEXNUPINFO** command to query the delay.

After the inter-gNodeB one-way delay is obtained, the delay level needs to be configured based on the following principles:

- If the inter-gNodeB one-way delay is less than or equal to 0.5 ms, **NRDUCellCollabServ.NulSulMaxInterGnbDelay** and **NRDUCellCollabCoop.NulSulMaxInterGnbDelay** need to be set to **DELAY_0DOT5MS**.
- If the inter-gNodeB one-way delay is greater than 0.5 ms and less than or equal to 2 ms, **NRDUCellCollabServ.NulSulMaxInterGnbDelay** and **NRDUCellCollabCoop.NulSulMaxInterGnbDelay** need to be set to **DELAY_2MS**.
- If the inter-gNodeB one-way delay is greater than 2 ms and less than or equal to 4 ms, **NRDUCellCollabServ.NulSulMaxInterGnbDelay** and **NRDUCellCollabCoop.NulSulMaxInterGnbDelay** need to be set to **DELAY_4MS**.

NOTE

Flow control against multi-carrier operations at the gNodeB now applies to the procedure for enabling super uplink. The flow control level is specified by the **gNodeBAlgo.MultiCarrierFlowCtrlLevel** parameter. Multiple flow control levels correspond to one CPU load threshold. When the CPU usage reaches the relevant threshold, super uplink will no longer take effect for newly admitted UEs.

SUL Carrier Information

SUL carrier information for super uplink with SUL and NR FDD co-deployment is the same as that for super uplink with independent SUL deployment. For details, see [SUL Carrier Information](#) in [4.1.1.1 SUL Cell Configuration and Carrier Information](#).

5.1.1.2 Random Access

The random access procedure for SUL and NR FDD co-deployment is the same as that when super uplink with independent SUL deployment is enabled and UL and DL Decoupling is not enabled. For details, see [4.1.1.2 Random Access](#).

5.1.1.3 Carrier Management

When super uplink with SUL and NR FDD co-deployment is enabled, NR TDD and SUL cells are no longer bound through configuration. Instead, they are paired freely based on the NR TDD cell configuration information and measurement events.

SUL Carrier Configuration

The SUL carrier configuration procedure is triggered for one of the following UE types in an NR TDD cell:

- UEs performing a random access procedure
- UEs performing an incoming handover
- UEs performing an incoming RRC connection reestablishment
- UEs performing RRC connection resumption
- Activated UEs that support super uplink but are not configured with super uplink and whose uplink traffic volume meets the triggering condition

The procedure is as follows:

1. The gNodeB delivers measurement configurations related to event A5 for SUL configuration. Event A5 indicates that the signal quality of the serving cell is lower than threshold 1 and the signal quality of a neighboring cell is higher than threshold 2. [Table 5-3](#) describes related parameters.
2. The UE reports the NR FDD cells that meet the threshold for event A5.
3. The **gNodeBAlgo.MultiCarrierMeasWaitTimer** parameter specifies the length of the A5 measurement waiting timer. Assume that the base station has delivered the A5 measurement configurations. If the measurement reports of all frequencies are reported before the timer expires, the base station stops the timer immediately and adds SCells and SUL carriers based on these measurement reports. Otherwise, the base station will not add SCells or SUL carriers until the timer expires, and such addition will be performed only based on the received measurement reports.
4. The gNodeB selects the intra-gNodeB NR FDD cell with the best signal quality as the SUL cell for the NR TDD cell based on the event A5 measurement result.

For the LampSite base station, the SSB SINR of the NR FDD cell carried in the event A5 measurement result also needs to be considered. The NR FDD cell can serve as an SUL cell only when its SSB SINR is greater than the sum of the **NRCellSulFrequency.SulSsbSinrThld** and **NRCellSulFrequency.SulSsbSinrOffset** parameter values.

In scenarios with dense deployment of multiple cells, SUL may not take effect due to a failure to select a suitable NR FDD neighboring cell as an SUL cell. To address this issue, you are advised to enable configuration optimization for SUL UEs (by selecting the **SUL_CONFIG_OPT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter) to expand the range of cells that can be selected as SUL cells, thereby increasing the probability of SUL taking effect.

Inter-gNodeB super uplink and intra-gNodeB super uplink can both take effect. During the SUL carrier configuration procedure, intra-gNodeB SUL cells are preferentially selected.

Table 5-3 Parameters related to event A5 for SUL configuration

Variable	Meaning	Value
Hys	Hysteresis for event A5	1 dB
TimeToTrigger	Time-to-trigger for event A5	320 ms
A5 Thresh1	Threshold 1 for event A5	-30 dBm
A5 Thresh2	Threshold 2 for event A5	Specified by the NRCellSulFrequency.SulInterFreqA5RsrpThld2 parameter
Ofn	Frequency offset for connected mode	0 dB
Ocn	Cell individual offset	Specified by the NRCellRelation.CellIndividualOffset parameter

SUL Carrier Change

When the gNodeB detects that the SRS SINR of the UE on the SUL carrier is less than the value of the **NRDUCellSulRelation.SulCarrChangeSinrThld** parameter, the gNodeB starts SUL carrier reselection. Otherwise, SUL carrier reselection is not triggered.

The procedure is as follows:

1. The gNodeB delivers measurement configurations related to event A5. [Table 5-3](#) describes related parameters.
2. The UE reports the NR FDD cells that meet the threshold for event A5.
3. The gNodeB reselects the intra-gNodeB NR FDD cell with the best downlink signal quality as the SUL cell for the NR TDD cell based on the event A5 measurement result.

For the LampSite base station, the SSB SINR of the NR FDD cell carried in the event A5 measurement result also needs to be considered. The NR FDD cell can serve as an SUL cell only when its SSB SINR is greater than the sum of the **NRCellSulFrequency.SulSsbSinrThld** and **NRCellSulFrequency.SulSsbSinrOffset** parameter values.

SUL Carrier Removal

The gNodeB removes the SUL carrier of a UE when any of the following conditions is met:

- There are consecutive DTXs for the UE on the SUL carrier.
If the gNodeB detects that the SRS SINR of the UE on the SUL carrier is greater than or equal to the value of the

NRDUCellCollabServ.SulDtxSinnrLowerLimit parameter, it determines whether the SUL carrier is abnormal based on the DTX result. If abnormal, the gNodeB removes the SUL carrier. If not, the DTX result is still not conclusive enough to rule out the SUL carrier as abnormal.

- The SRS SINR meets the SUL carrier removal threshold.

If the gNodeB detects that the SRS SINR of the UE on the SUL carrier is less than the value of the **NRDUCellCollabServ.SulSinnrLowerLimit** parameter, it determines that the SUL carrier provides limited coverage or is abnormal and triggers SUL carrier removal.

- The TA difference does not meet the restriction condition.

If the gNodeB finds that the TA difference between SUL and TDD carriers is too large through calculation, it triggers SUL carrier removal.

- (Only for the LampSite base station) The SSB SINR of the NR FDD cell where the SUL carrier is located does not meet the restriction condition.

If the gNodeB detects that the SSB SINR of the NR FDD cell corresponding to the SUL carrier serving the UE is less than the result of the value of the **NRCellSulFrequency.SulSsbSinnrThld** parameter minus that of the **NRCellSulFrequency.SulSsbSinnrOffset** parameter, it determines that the UE is in the overlapping area of the NR FDD cells. In this case, strong neighboring cell interference will exist, triggering SUL carrier removal.

5.1.1.4 Uplink TDM Scheduling

The principles of uplink TDM scheduling for SUL and NR FDD co-deployment are the same as those for independent SUL deployment. For details, see [4.1.1.3 Uplink TDM Scheduling](#).

5.1.1.5 Uplink Power Control

When super uplink with SUL and NR FDD co-deployment is enabled, power control is separately performed on the NR TDD carrier and SUL carrier. The NR FDD cell sends the TPC commands of the SUL carrier to the NR TDD cell and the gNodeB serving the NR TDD cell delivers these power control commands to the UEs.

5.1.1.6 Timing Advance

If super uplink is enabled, NR TDD and SUL carriers share the same TAG.

If inter-gNodeB super uplink is enabled, NR TDD and NR FDD cells are served by different gNodeBs, and the uplink transmission distances from UEs to the two gNodeBs over the air interface are different. In this case, there is a difference in uplink TA between the two inter-gNodeB carriers. If the TA difference between the NR TDD and SUL carriers is too large, uplink performance is affected. Therefore, if the TA difference does not meet the performance requirements, the SUL carrier is removed to ensure that the basic performance of NR TDD is not affected. For details, see [5.1.1.3 Carrier Management](#).

If the number of TAGs supported by a UE is less than that of CCs aggregated for the UE, some of the CCs have to share the TAs of other CCs. Inappropriate TA selection has a noticeable impact on the overall uplink demodulation performance of these CCs. To address this issue, flexible TA adjustment in uplink multi-CC

scenarios is introduced. This function is controlled by the **UL_MULTI_CC_TA_FLEX_ADJ_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

With flexible TA adjustment in uplink multi-CC scenarios enabled, if the number of TAGs supported by a UE is less than that of CCs aggregated for the UE, a TA is selected for all CCs involved in a single TAG according to the following rules, which apply to all TAGs:

- If the absolute value of the TA difference between any two CCs involved in a TAG is less than the value of the **NRDUCellCarrMgmt.UIMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on all the CCs involved in this TAG is selected as the TA for all these CCs.
- If the absolute value of the TA difference between any two CCs involved in a TAG is greater than or equal to the value of the **NRDUCellCarrMgmt.UIMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on the PCC is selected as the TA for all the CCs involved in this TAG.

5.1.1.7 Mobility Management

The principles of mobility management for SUL and NR FDD co-deployment are the same as those for independent SUL deployment. For details, see [4.1.1.6 Mobility Management](#).

5.1.2 Enhanced Functions

5.1.2.1 SUL CoMP

When super uplink with SUL and NR FDD co-deployment is enabled, SUL CoMP is used to increase uplink benefits in the overlapping area of SUL cells. This function only applies to intra-gNodeB super uplink.

SUL CoMP is controlled by the **INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter. In addition, UL CoMP must be enabled in the cells where NR FDD and SUL are co-deployed. UL CoMP is controlled by the **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter.

SUL CoMP is triggered when the SUL carrier is configured or changed. The procedure is as follows:

1. The gNodeB delivers measurement configurations related to event A5. For details about parameters related to event A5, see [5.1.1.3 Carrier Management](#).
2. UEs perform event A5 measurement, and send A5 measurement reports, which include NR FDD neighboring cells.
3. The gNodeB selects candidate UEs for SUL CoMP as follows:
 - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp.CompConfigPol** parameter is deselected, all UEs in the cells can be selected as candidate UEs for SUL CoMP.

- If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp.CompConfigPol** parameter is selected, the gNodeB selects specific UEs as candidate UEs for SUL CoMP. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter is selected.
- 4. The gNodeB selects a serving SUL cell based on **5.1.1.3 Carrier Management**. At the same time, the gNodeB selects candidate SUL CoMP cooperating cells from NR FDD cells that overlap the serving SUL cell. The RSRP difference between each candidate SUL CoMP cooperating cell and the serving SUL cell must be greater than the value of **NRCellComp.UlCompRsrpOffset**.
- 5. The gNodeB measures the uplink SRSs of a UE in each candidate SUL CoMP cooperating cell. Among these cells, the cell in which the RSRP of uplink SRSs sent by the UE is greater than the sum of the RSRP of uplink SRSs in the UE's serving cell and the value of **NRCellComp.UlCompRsrpOffset** is selected as the SUL CoMP cooperating cell. At the same time, the gNodeB determines that SUL CoMP takes effect for the UE.
- 6. For UEs for which SUL CoMP takes effect, the serving SUL cell and the SUL CoMP cooperating cell simultaneously receive the UEs' data on PUSCHs. The received data is combined and demodulated, improving the PUSCH performance of these UEs.

The SUL CoMP cooperating cell preferentially processes the data from the UEs it serves. If this cell does not have sufficient capabilities (for example, the number of PRBs used by this cell plus the number of PRBs required for uplink joint reception exceeds the cell's total number of PRBs), it does not process the UL CoMP UE data.

When the SUL CoMP function in super uplink with SUL and NR FDD co-deployment is enabled with an LTE FDD and NR spectrum sharing function (LTE FDD and NR Flash Dynamic Spectrum Sharing/Hybrid DSS Based on Asymmetric Bandwidth), SUL CoMP can take effect only when both the serving SUL cell and the SUL CoMP cooperating cell are in the LTE FDD and NR spectrum sharing state.

For details about CoMP, see *CoMP*.

5.1.2.2 Symbol-Level Scheduling on the SUL Carrier

Basic principles of symbol-level scheduling on the SUL carrier for SUL and NR FDD co-deployment are the same as those for independent SUL deployment. For details, see **4.1.2.4 Symbol-Level Scheduling on the SUL Carrier**.

5.1.2.3 SUL UE Power Saving

When the uplink traffic volume of super uplink UEs is lower than a specified threshold, the gNodeB will trigger SUL carrier removal to implement SUL UE power saving. When the uplink traffic volume of the UEs is greater than the threshold, the gNodeB will trigger SUL carrier addition for UEs to enter the super uplink state.

SUL UE power saving is controlled by the **SUL_UE_POWER_SAVING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

5.1.2.4 SUL Congestion Control

When super uplink with SUL and NR FDD co-deployment is enabled, the gNodeB automatically determines whether a neighboring SUL cell is congested. When the uplink PRB usage of the neighboring SUL cell is greater than the **NRDUCellSulRelation.MultiFreqULPrbUsageThld** parameter value, the gNodeB determines that the cell is congested and will not select the cell as the SUL cell for UEs in serving cell center areas. The SSB RSRP of these UEs is less than the **NRDUCellCollabServ.RsrpThld** parameter value. SUL congestion control is controlled by the **SUL_CONGEST_CONTROL_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

It is recommended that the SUL congestion control function be used when all of the following conditions are met:

- The uplink PRB usage of the neighboring SUL cell of the serving cell is greater than 50%.
- The bandwidth of the neighboring SUL cell of the serving cell is less than 20 MHz.
- The bandwidth of the serving cell is 100 MHz.

5.1.2.5 SUL CCE Adjustment Policy

SUL scheduling consumes CCE resources of the NR TDD cell. The **NRDUCellPdcch.SulCceAdjPolicy** parameter has been added to specify the SUL CCE adjustment policy for flexible super uplink, specifically, the percentage of SUL CCEs in uplink CCEs.

- If this parameter is set to a value other than **DEFAULT**, the percentage of SUL CCEs in uplink CCEs is fixed at the configured value.
- If this parameter is set to **DEFAULT**, the SUL CCE adjustment policy does not take effect.

5.1.2.6 Function of Updating the Uplink Transmit Power of Band Combinations

To increase the uplink throughput of CEUs that support R16-compliant power classes, the uplink transmit power of band combinations can now be updated. In this case, the maximum uplink transmit power for NR TDD is 29 dBm. This function is controlled by the **BAND_COMB_UL_TX_PWR_UDT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

This function takes effect only when the **NRDUCellSelConfig.MaxUeTransmitPower** parameter is set to a value in the range of 29 to 33.

This function takes effect only when the value of the UE-reported **ue-PowerClass-v1610** or **powerClass-v1610** field is **pc1dot5**. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V17.10.0.

5.1.2.7 Fast SUL

Fast SUL is introduced to improve the uplink throughput in scenarios where inter-base-station super uplink is enabled. This function is controlled by the

FAST_SUL_SW option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter.

It is recommended that fast SUL be enabled in a cell that meets both of the following conditions:

- The cell supports inter-base-station super uplink.
- The number of RRC_CONNECTED UEs in the cell is insufficient to cause a heavy load.

5.1.3 Simultaneous Use with Other Functions

5.1.3.1 Simultaneous Use with CA

Super uplink with SUL and NR FDD co-deployment can be enabled with some uplink and downlink CA functions (including 2CC aggregation and 3CC aggregation) on NR TDD carriers.

For the simultaneous use of super uplink with SUL and NR FDD co-deployment and CA, its basic principles are the same as those for the simultaneous use of super uplink with independent SUL deployment and CA described in [4.1.3.2 Simultaneous Use with CA](#). However, the corresponding frequency band combinations are different, as listed in [Table 5-4](#). The bandwidths supported by the frequency band combinations for the simultaneous use of super uplink and CA are consistent with those defined in 3GPP specifications. For details, see section 5.5C "Configurations for SUL" in 3GPP TS 38.101-1 V18.7.0.

Table 5-4 Frequency band combinations for the simultaneous use with downlink CA

Downlink CA Scenario	NR TDD Frequency Band	NR TDD (PCC) Slot Configuration	SCC Slot Configuration	Supported SUL Frequency Band
2CC aggregation: NR TDD intra-band CA	n41	Single- or dual-period 8:2	Single- or dual-period 8:2	n83
2CC aggregation: NR TDD +NR FDD inter-band CA	n41	Single- or dual-period 8:2	N/A	n83

Downlink CA Scenario	NR TDD Frequency Band	NR TDD (PCC) Slot Configuration	SCC Slot Configuration	Supported SUL Frequency Band
3CC aggregation	NR TDD (n41)+NR TDD (n41)+NR FDD (n28)	Single- or dual-period 8:2	NR TDD: single- or dual-period 8:2 NR FDD: N/A	n83
2CC aggregation: NR TDD intra-band CA	n78	4:1	4:1	n83/n84
2CC aggregation: NR TDD +NR FDD inter-band CA	n78	4:1	N/A	n83/n84
2CC aggregation: NR TDD +NR FDD inter-band CA	NR TDD (n78)+NR FDD (n1)	Single- or dual-period 8:2	N/A	n84

Table 5-5 lists the frequency band combinations for the simultaneous use of super uplink with SUL and NR FDD co-deployment and uplink CA.

Table 5-5 Frequency band combinations for the simultaneous use with uplink CA

Uplink CA Scenario	NR TDD Frequency Band	NR TDD (PCC) Slot Configuration	SCC Slot Configuration	Supported SUL Frequency Band
2CC aggregation: NR TDD intra-band CA	n41	Single- or dual-period 8:2	Single- or dual-period 8:2	n80/n83

SUL and CA optimal carrier selection is supported when super uplink with SUL and NR FDD co-deployment is enabled. The principles are the same as those when independent SUL deployment is used. For details, see [4.1.3.2.2 SUL and CA Optimal Carrier Selection](#).

5.1.3.2 Simultaneous Use with Virtual Grid

In SUL and NR FDD co-deployment mode, super uplink and the virtual grid function can be enabled at the same time. When the **CARR_ADD_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter is selected, the virtual-grid-based RSRP prediction results are used during SUL carrier addition, reducing the loss caused by gap-assisted inter-frequency measurements. For details, see *Virtual Grid-based Multi-Frequency Coordination*.

5.1.3.3 Simultaneous Use with High-speed Railway Superior Experience

When super uplink is enabled in high-speed scenarios, it is recommended that the **NRDUCellDmrs.DmrsTypeAPos** parameter be set to the same value for NR TDD and NR FDD cells. If this parameter is set to different values for NR TDD and NR FDD cells, optimized cooperation between SUL and DMRS must be enabled. This function is controlled by the **RSVDSWPARAM2_BIT15** option of the reserved parameter **NRDUCellRsvdExt01.RsvdSwParam2**.

5.1.3.4 Simultaneous Use with UL and DL Decoupling

When super uplink is used together with UL and DL Decoupling, super uplink takes effect in cell center areas, and UL and DL Decoupling takes effect in cell edge areas. When both of these functions are enabled, random access and mobility management are different compared to when only super uplink is enabled. For details about UL and DL Decoupling, see *UL and DL Decoupling*.

Random Access

During initial RRC connection setup, a UE receives system information from the gNodeB, obtains the *rsrp-ThresholdSSB-SUL* IE in SIB1, measures the downlink RSRP of the NR TDD carrier, and determines the uplink carrier for random access.

The value of *rsrp-ThresholdSSB-SUL* is specified by the **NRDUCellCollabServ.RsrpThld** parameter.

- If the RSRP of an NR cell is greater than or equal to the value of the **NRDUCellCollabServ.RsrpThld** parameter, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NR TDD carrier, and super uplink takes effect.
- If the RSRP of an NR cell is less than the value of the **NRDUCellCollabServ.RsrpThld** parameter, the UE is in an area with weak or no NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier, and UL and DL Decoupling takes effect.

Intra-Cell Mobility Management

When a UE moves in an NR cell, it uses super uplink in cell center areas, and UL and DL Decoupling in cell edge areas. Switching between these functions is performed based on uplink quality.

- If the SRS RSRP of a UE on the NR TDD carrier is greater than the sum of the **NRDUCellCollabServ.SrsRsrpThld** parameter value and hysteresis (2 dB), super uplink takes effect for the UE. In this scenario, the UE transmits uplink data on NR TDD and SUL carriers in TDM mode.
- If the SRS RSRP of a UE on the NR TDD carrier is less than or equal to the result of the **NRDUCellCollabServ.SrsRsrpThld** parameter value minus hysteresis (2 dB), UL and DL Decoupling takes effect for the UE. In this scenario, the UE transmits uplink data on the SUL carrier only.

Inter-Cell Mobility Management

When a UE that supports super uplink moves between NR cells and a handover to the target cell for which super uplink and UL and DL Decoupling are enabled is triggered, the gNodeB instructs the UE to initiate a random access procedure on a TDD carrier through an RRCReconfiguration message. The procedure is as follows:

1. The target gNodeB notifies the source gNodeB of the selected uplink carrier. The source gNodeB sends a UE an RRCReconfiguration message, indicating that the uplink carrier on which random access to the target cell is to be performed is a TDD carrier.
2. The UE initiates a random access procedure on the uplink carrier indicated in the RRCReconfiguration message.

5.2 Network Analysis

5.2.1 Benefits

Basic Functions

When super uplink with SUL and NR FDD co-deployment is enabled, uplink spectrum resources of NR FDD cells are used to achieve better uplink data rates. The following table lists the expected gains for a single super uplink UE in different slot configuration scenarios.

Slot Configuration of the NR TDD Cell	SUL Subcarrier Spacing of the NR FDD Cell	Uplink Data Rate Range
4:1	15 kHz	(A x 100% + B x 44%) to (A x 100% + B x 65%)
Single- or dual-period 8:2	15 kHz	(A x 100% + B x 56%) to (A x 100% + B x 76%)
Note: A and B indicate the uplink data rates of the UE in the NR TDD cell and NR FDD cell, respectively.		

SUL performance gains decrease due to the TA difference between NR TDD and SUL carriers caused by factors such as inter-gNodeB time synchronization deviation and different site locations. As a result, peak data rate gains decrease.

If flexible TA adjustment in uplink multi-CC scenarios is enabled, the uplink throughput increases in the uplink multi-CC scenario with the following conditions met: (1) there are negative measured TA values, or the measured TA value and bandwidth differ greatly between CCs; (2) the absolute difference in the measured TA value between CCs does not exceed the configured threshold.

Enhanced Functions

The enhanced functions provide the following gains:

- SUL CoMP: It improves uplink reception performance and throughput for UEs in overlapping areas in an intra-frequency network with continuous coverage.
- Symbol-level scheduling: It allows scheduling in symbols of the first half timeslot on the SUL carrier. This increases the availability of SUL timeslots for scheduling and improves the uplink UE throughput.
- Function of updating the uplink transmit power of band combinations: It increases the uplink throughput of CEUs that support 3GPP R16-compliant power classes.
- Fast SUL: It enables earlier scheduling for UEs on the SUL carrier, increasing the uplink throughput of inter-gNodeB SUL UEs.

Simultaneous Use with Other Functions

When super uplink and CA are both enabled, SUL and CA optimal carrier selection facilitates the selection of the optimal carrier combination, increasing the uplink throughput of UEs supporting both super uplink and CA.

When super uplink and uplink CA are both enabled, it is recommended that the **MF_ACTUAL_PUSCH_ILPC_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoExtSwitch** parameter be selected to enable multi-frequency actual-PUSCH-based inner-loop power control to improve the accuracy of data volume distributed to SCells in the uplink. For details about this function, see *Power Control*.

When the **SUL_CA_SRS_SWITCHING_SW**, **SUL_CA_2CC_SRS_SWITCHING_SW**, or **SUL_CA_SRS_SWITCHING_EXT_SW** option of the

NRDUCellCollabServ.SulAlgoSwitch parameter is selected, UEs that support the simultaneous use of downlink 2CC aggregation, SUL in compliance with R16 or later, and SRS carrier switching, that of downlink 3CC aggregation, SUL in compliance with R16 or later, and SRS carrier switching, or that of downlink 2CC or 3CC aggregation, SUL in compliance with R15, and SRS carrier switching, can provide the following benefits:

- Compared with UEs for which only downlink CA and SRS carrier switching take effect, the average uplink throughput of UEs that also support super uplink is improved. For details about how to observe the gains, see [4.4.4 Network Monitoring](#).
- Compared with UEs for which only downlink CA and super uplink take effect, the average downlink throughput of UEs that also support SRS carrier switching is improved. The average downlink throughput is indicated by **User Downlink Average Throughput (DU)**.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

Basic functions:

- When super uplink with SUL and NR FDD co-deployment is enabled, only one RRC connection is established between a UE and the network. The "RRC Connected User License" is consumed only in the NR TDD cell. However, each UE occupies a set of hardware resources in both the NR TDD cell and NR FDD cell. Therefore, the actual number of RRC_CONNECTED UEs supported on the network is halved in extreme scenarios (where all UEs on the network support SUL).
- When super uplink with SUL and NR FDD co-deployment is enabled, uplink resources of the NR FDD cell can be fully utilized. Therefore, the uplink PRB usage of the NR FDD cell increases. When the NR FDD cell load is high, the uplink throughput of NR FDD UEs in the cell decreases.
- When super uplink with SUL and NR FDD co-deployment is enabled, uplink resources of the NR FDD cell can be fully utilized. The uplink interference between NR FDD cells increases in the case of a large number of SUL UEs. This decreases the uplink throughput of NR FDD UEs and may increase the service drop rate at the cell edge.
- When super uplink with SUL and NR FDD co-deployment is enabled, event A5 measurement configurations are delivered in the NR FDD cell. Since event A5 measurement configurations require measurement gaps, uplink and downlink throughput of the UE in the NR TDD cell decreases.
- When super uplink with inter-gNodeB SUL and NR FDD co-deployment is enabled and the NR FDD cell load is high, SUL UEs may not benefit from super uplink.
- When super uplink with inter-gNodeB SUL and NR FDD co-deployment is enabled, more cell configuration information is exchanged over the eXn interface between base stations. Therefore, the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset increase.

- The number of the RBs occupied by SIB1 may slightly increase and the downlink peak throughput slightly decreases because the gNodeB broadcasts the SUL information through SIB1.
- SUL uplink scheduling consumes CCE resources of the NR TDD cell. When CCE resources are insufficient in the NR TDD cell, uplink or downlink CCE resource application is more likely to fail. As a result, the uplink or downlink throughput of the NR TDD cell fluctuates, and the uplink or downlink user-plane delay increases.
- It is recommended that the **UL_DL_CCE_SHARING_SW** or **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter be selected to reduce the probability of CCE resource application failures.
- In an NR TDD cell enabled with uplink MU-MIMO, if scheduling time alignment for super uplink is enabled (that is, the **UL_SCH_ALIGN_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter is selected), the average uplink cell throughput and the **N.CCE.UL.Avail.Equivalent** and **N.CCE.ULDCI.Avail.Equivalent** counter values increase, as does the average delay for non-super-uplink UEs by up to 1%.
- The uplink throughput of UEs that switch between carriers in 0 μ s decreases when UE compatibility handling is enabled with the **NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime** parameter set to a value other than 0 μ s.
- When an NR TDD cell with two slot configurations needs to be paired with a medium- or heavy-load NR FDD cell at the same time to implement the super uplink function, the uplink throughput of the NR FDD cell slightly decreases.
- When flow control against multi-carrier operations at the gNodeB is triggered and the CPU usage reaches the load threshold corresponding to the flow control level, super uplink will no longer take effect for newly admitted UEs. As a result, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) decrease. When the **gNodeBAlgo.MultiCarrierFlowCtrlLevel** parameter is set to **HIGH**, the cell service drop rate (**Service Call Drop Rate (CU, Inactive)**) increases as well.
- When configuration optimization for SUL UEs is enabled, the number of SUL UEs (indicated by **N.User.UEnhTimeFreq.Avg**) increases, which may increase the probability of NR TDD CCE resource application failures and decrease the uplink user-perceived rate in NR FDD.

Enhanced functions:

- Enabling the symbol-level scheduling function affects the uplink throughput of the NR FDD cell.
- When the SUL congestion control function is enabled and neighboring SUL cells are congested, SUL carriers may not be added for UEs in the center of the serving cell. In this case, the uplink throughput of UEs at the edge of the serving cell increases.
- When the SUL CCE adjustment policy function is enabled and uplink-to-downlink CCE ratio adaptation (controlled by the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter) is enabled:

- If the proportion of super uplink UEs is low, the uplink user-perceived rates of these UEs increase while those of other UEs in the NR TDD cell are not affected.
- If the proportion of super uplink UEs is high, the probability of uplink CCE resource application failures increases and scheduling opportunities for other UEs in the NR TDD cell decrease.
- When the SUL CCE adjustment policy function is enabled and uplink-to-downlink CCE ratio adaptation is disabled, there is no network impact.

Simultaneous use with the virtual grid function: If the **NRCellSmartMultiCarr.PredAccuForDataThld** or **NRCellSmartMultiCarr.PredAccuForVoiceThld** parameter (specifying the accuracy threshold for using a virtual grid model in CA SCC addition or SUL carrier addition) is set to a value other than 100% and the virtual-grid-based prediction results are used to add SCCs for CA or add SUL carriers, the values of the counters related to carrier addition, change, and removal (such as **N.SUL.SulCarrier.Add.Att**, **N.SUL.SulCarrier.Mod.Att**, and **N.SUL.SulCarrier.Rmv.Att**) fluctuate, affecting uplink user-perceived rates.

Assume that the **SUL_CA_SRS_SWITCHING_SW**, **SUL_CA_2CC_SRS_SWITCHING_SW**, or **SUL_CA_SRS_SWITCHING_EXT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected. Compared with UEs for which only downlink CA and super uplink take effect, the average uplink throughput of UEs that also support SRS carrier switching decreases by about 7%, as time needs to be reserved for switching SRS transmission between SUL and PCC carriers. The average uplink throughput is indicated by **User Uplink Average Throughput (DU)**.

For a UE that supports the combination of downlink 3CC aggregation and uplink intra-band 2CC aggregation and that of downlink 2CC aggregation, uplink intra-band contiguous 2CC aggregation, and SUL, if uplink optimal carrier selection for SUL and CA is enabled, the latter combination is preferentially used. This increases the **User Uplink Average Throughput (DU)**. However, the downlink UE throughput decreases because optimal carrier selection introduces gap-assisted inter-frequency measurements and downlink 3CC aggregation cannot take effect.

Simultaneous use with High-speed Railway Superior Experience: Assume that super uplink is enabled in high-speed scenarios. If optimized cooperation between SUL and DMRS is enabled, the uplink throughput of some SUL UEs is affected. If this function is disabled, the uplink data rates of some SUL UEs are zero.

5.3 Requirements

5.3.1 Licenses

The following feature license is required by super uplink with SUL and NR FDD co-deployment. The number of required license units is calculated based on the NR TDD cells but not NR FDD cells.

Feature ID	Feature Name	Model	NE	Sales Unit
FOFD-030211	Super Uplink	NR0S00SUPC00	Macro gNodeB	per Cell
FOFD-030211	Super Uplink	NR0SSPUPPH01	LampSite gNodeB	per Carrier per pRRU
For the license control item NR0S00SUPC00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter cannot be selected due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter. For the license control item NR0SSPUPPH01: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.				

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	SUL CoMP requires that intra-base-station UL CoMP be enabled in NR FDD cells when super uplink with SUL and NR FDD co-deployment is enabled.
Low-frequency NR TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Distributed massive MIMO must be enabled for NR TDD cells when super uplink is deployed on LampSite base stations.
Low-frequency NR TDD	PDCCH uplink and downlink CCE sharing	UL_DL_CCE_SHARING_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	Only the SUL CCE adjustment policy function requires that the switch for uplink and downlink CCE sharing be turned on.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	The SUL_WITH_MULTISLOT_CFG_NUL_SW option of the NRDUCellCollabCoop.SulAlgoSwitch parameter cannot be selected in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Hyper Cell ^a	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Super uplink is not supported in hyper cells.
Low-frequency NR TDD	Joint Transmission Cell Combination ^b	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTranMode set to JOINT_MODE	<i>Cell Combination</i>	Super uplink is not supported in combined cells in joint transmission mode.
Low-frequency NR TDD	Cell Combination ^c	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTranMode set to BASIC_MODE	<i>Cell Combination</i>	Super uplink is not supported in combined cells.
Low-frequency NR TDD	Cell Combination ^d	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTranMode set to DPS_MODE	<i>Cell Combination</i>	Super uplink is not supported in combined cells.
Low-frequency NR TDD	Secondary Harmonic Avoidance Switch	SEC_HARMONIC_INTRF_AVOID_SW option of the NRDUCellCollabServ.SulAlgoSwitch parameter	None	Super uplink and secondary harmonic interference avoidance cannot be both enabled.
Low-frequency NR TDD	5QI-specific UE-number-based connected mode MLB ^e	INTER_FREQ_CONNECTED_MLB option of the NRCCellAlgoSwitch.MlbAlgoSwitch parameter selected, and NRCCellMlb.MlbTriggerMode set to SPECIFIED_5QI_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	Super uplink and 5QI-specific UE-number-based connected mode MLB cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>MIMO (TDD)</i>	Super uplink and CCE resource optimization cannot be both enabled.
Low-frequency NR TDD	Pattern-free PDCCH rate matching activation rate increase	PDCCH_NO_PAT_RM_RATIO_INC_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	<i>Downlink Scheduling</i>	Super uplink and pattern-free PDCCH rate matching activation rate increase cannot be both enabled.
Low-frequency NR TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Super uplink and CCE partial symbol adaptation cannot be both enabled.
Low-frequency NR TDD	PDCCH multi-symbol resource pool	MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	Super uplink and PDCCH multi-symbol resource pool cannot be both enabled when NRDUCellCoreset.CommonCtrlResRbNum is set to RB96 and NRDUCellSulRelation.SulType is set to FIX_SUL .
Low-frequency NR TDD	Cell deployment in IAB mode	NRDUCell.DeployPosition set to DEPLOY_IAB	None	Super uplink is not supported in cells deployed in IAB mode.
Low-frequency NR TDD	Uplink assistance	UL_ASSISTANCE_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	None	Super uplink and uplink assistance cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Cooperation between joint transmission and pattern-free PDCCH rate matching	JT_PDCCH_NO_PATT_RM_COORD_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	None	Super uplink and cooperation between joint transmission and pattern-free PDCCH rate matching cannot be both enabled.
Low-frequency NR TDD	Downlink avoidance enhancement mode 2	DL_AVOID_ENH_MODE2_SW option of the NRDUCellTimeIntrf.TimeIntrfAvoidSw parameter	None	Super uplink and downlink avoidance enhancement mode 2 cannot be both enabled.
a: Super uplink (with SUL and NR FDD co-deployment) cannot be enabled in hyper cells with NRDUCell.HighSpeedFlag set to LOW_SPEED. b: Joint Transmission Cell Combination cannot be enabled together with super uplink (with SUL and NR FDD co-deployment) in an NR TDD cell. c: Super uplink (with SUL and NR FDD co-deployment) cannot be enabled together with Cell Combination in basic transmission mode in an NR TDD cell with NRDUCell.HighSpeedFlag set to HIGH_SPEED. Super uplink (with SUL and NR FDD co-deployment) cannot be enabled in combined cells working in basic mode with the NRDUCell.HighSpeedFlag parameter set to HIGH_SPEED or LOW_SPEED. d: Super uplink (with SUL and NR FDD co-deployment) cannot be enabled together with Cell Combination in DPS transmission mode in an NR FDD cell with NRDUCell.HighSpeedFlag set to LOW_SPEED. e: The mutually exclusive relationship exists only when super uplink is deployed on LampSite base stations.				

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSwitch parameter	<i>UE Power Saving</i>	UEs working in BWP2 for power saving do not support SUL CoMP.
Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SWITCH option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	The low latency and high reliability function does not take effect for UEs for which super uplink has taken effect.
Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for super uplink UEs.
Low-frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARRIER_SHUTDOWN_SWITCH or INTER_GNB_MULTI_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for UEs for which super uplink has taken effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: INTRA_GNB_MULTI_CARRIER_SD_SWITCH option of the NRDUCellAlgoSwitch.h.PowerSavingSwitch parameter selected (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARRIER_SD_SWITCH option of the NRDUCellAlgoSwitch.h.PowerSavingSwitch parameter selected (inter-base-station multi-carrier scenarios) FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.h.PowerSavingExtSwi	<i>Energy Conservation and Emission Reduction</i>	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for super uplink UEs.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
		<i>tch</i> parameter For basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected		
Low-frequency NR TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCCellAlgoSwitch.MlbAlgoSwitch parameter selected, and NRCCellMlb.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs for which super uplink has taken effect as the UEs to be handed over.
Low-frequency NR TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCCellAlgoSwitch.MlbAlgoSwitch parameter selected, and NRCCellMlb.MlbTriggerMode set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs for which super uplink has taken effect as the UEs to be handed over.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Super uplink cannot be enabled when rate-matching-pattern-configuration-based PDCCH rate matching is enabled and the NRDUCellPdcch.OccupiedRbNum parameter is set to 1 or 2 .
Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	When Cell Combination is enabled in an NR TDD cell, the average uplink throughput (indicated by Cell Uplink Average Throughput (DU)) of the super uplink cell slightly decreases during the primary TRP change.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with super uplink, the amount of uplink spectrum resources that can be allocated to implement super uplink decreases if the LTE FDD or NR traffic volume increases. As a result, the uplink throughput of the SUL carrier decreases. Super uplink cannot take effect in an LTE cell when LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with one of the following functions: uplink semi-persistent scheduling, eMTC, or compact bandwidth. This is because these functions periodically or continuously occupy spectrum resources. When the NR TDD and NR FDD cells enabled with super uplink are configured on the same baseband processing unit, super uplink (with SUL and NR FDD co-deployment) cannot be enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter and LTE_NR_FDD_SPCT_SHR_ASYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	When Hybrid DSS Based on Asymmetric Bandwidth is enabled together with super uplink, the amount of uplink spectrum resources that can be allocated to implement super uplink decreases if the LTE FDD or NR traffic volume increases. As a result, the uplink throughput of the SUL carrier decreases. Super uplink cannot take effect in an LTE cell when Hybrid DSS Based on Asymmetric Bandwidth is enabled together with one of the following functions: uplink semi-persistent scheduling, eMTC, or compact bandwidth. This is because these functions periodically or continuously occupy spectrum resources. When the NR TDD and NR FDD cells enabled with super uplink are configured on the same baseband processing unit, super uplink (with SUL and NR FDD co-deployment) cannot be enabled together with Hybrid DSS Based on Asymmetric Bandwidth.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Uplink multi-frequency coordination	MULTI_FREQ_UL_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select UEs for which super uplink has taken effect.
Low-frequency NR TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select UEs for which super uplink has taken effect.
Low-frequency NR TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	When both SUL CoMP and PDCCH skipping are enabled, PDCCH skipping does not take effect for UEs for which super uplink has taken effect.
Low-frequency NR TDD	Massive CA	NRDUCellCarMgmt.CaDLMaxCcNum set to DL4CC or DL5CC .	<i>CA Experience Improvement</i>	Downlink massive CA does not take effect for the UEs for which super uplink takes effect.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Super uplink (with SUL and NR FDD co-deployment) is not supported in hyper cells with the NRDUCell.HighSpeedFlag parameter set to HIGH_SPEED .

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Super uplink (with SUL and NR FDD co-deployment) is not supported in combined cells working in DPS mode with the NRDUCell.HighSpeedFlag parameter set to HIGH_SPEED .
Low-frequency NR TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R15-compliant uplink slot aggregation does not take effect on UEs with super uplink in effect.
Low-frequency NR TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R16-compliant uplink slot aggregation does not take effect on UEs with super uplink in effect.
Low-frequency NR TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePowerSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	UEs with super uplink in effect do not switch to a sparse search space set group (SSSG).
Low-frequency NR TDD	Dynamic uplink slot aggregation	REDCAP_UL_COVERAGE_ENH_SW option of the NRDUCellFeatureSw.UeCapBasedFunction parameter	<i>RedCap</i>	For an NR TDD cell, super uplink does not take effect for RedCap UEs with dynamic uplink slot aggregation in effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RedCap SSSG switching	REDCAP_SSSG_SWITCHING_SW and REDCAP_SSSG_SWITCH_EXT_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>RedCap</i>	UEs with super uplink in effect do not switch to a sparse SSSG.
Low-frequency NR TDD	RedCap DRX adaptation	REDCAP_DRX_ADAPTATION_SW and REDCAP_DRX_ADAPTATION_EXT_SW options of the NRDUCellUePwrSaving.NrDUCellDrxAlgorithmSwitch parameter	<i>RedCap</i>	Super uplink does not take effect for UEs with inter-base-station super uplink in effect. UEs with super uplink in effect exit the RedCap DRX adaptation function before inter-base-station super uplink takes effect for those UEs.
Low-frequency NR TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	After both "super uplink" and "flexible experience differentiation" are enabled, the base station estimates the uplink peak data rate of each carrier of a UE. If a UE is configured with an SUL carrier which is not activated, the UE's estimated uplink peak data rate may be inaccurate.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_US LOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	When both positioning-dedicated SRS transmission in uplink slots and super uplink are enabled, the SUL cells configured on a specific baseband processing unit must meet the SUL cell specifications supported by the board. Otherwise, positioning-dedicated SRS transmission in uplink slots cannot take effect. For details about the SUL cell specifications supported by different baseband processing units, see the table content where the value of Frequency Band Type of a Cell includes SUL in Product Specifications > BBU Technical Specifications > Technical Specifications of Baseband Processing Units > NR TDD Technical Specifications of Baseband Processing Units in <i>3900 & 5900 Series Base Station Product Documentation</i>.
Low-frequency NR TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	UEs on the NUL carrier can be switched to BWP2 while UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay of BWP2 UEs on the NUL carrier increases when the SUL carrier is activated.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	After RF module fast deep dormancy takes effect, SUL scheduling cannot be performed for super uplink UEs.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Requirements on baseband processing units:

- Macro base stations:
 - For NR TDD cells, all NR TDD-capable baseband processing units can be used.
 - For NR FDD cells, all NR FDD-capable baseband processing units can be used.
- LampSite base stations:
 - For NR TDD cells, baseband processing units must support distributed massive MIMO. For details, see *Distributed Antenna Solutions (Low-Frequency TDD)*.
 - For NR FDD cells, all NR FDD-capable baseband processing units can be used.

RF Modules

Table 5-6 lists the RF modules supported by macro base stations.

Table 5-6 RF modules supported by macro base stations

Frequency Band	Supported Module	Bandwidth Requirement
n78 (3.5/3.7 GHz)	Modules supporting 32T32R and 64T64R	Same as the bandwidths supported by the RF modules. For details, see related technical specifications in <i>3900 & 5900 Series Base Station Product Documentation</i> .
n40 (2.3 GHz)	Modules supporting 32T32R and 64T64R	
n41 (2.6 GHz)	Modules supporting 32T32R and 64T64R	
n3/n80 (1.8 GHz)	Modules supporting NR FDD	10 MHz, 15 MHz, and 20 MHz
n20/n82 (800 MHz)	Modules supporting NR FDD	5 MHz, 10 MHz, 15 MHz, and 20 MHz
n28/n83 (700 MHz)	Modules supporting NR FDD	10 MHz, 15 MHz, 20 MHz, and 30 MHz
n1/n84 (2.1 GHz)	Modules supporting NR FDD	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz
n8/n81 (900 MHz)	Modules supporting NR FDD	10 MHz, 15 MHz, and 20 MHz

Table 5-7 lists the RF modules supported by LampSite base stations.

Table 5-7 RF modules supported by LampSite base stations

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n78 (3.5/3.7 GHz)	pRRU56xx series, pRRU596x series ^a , pRRU5973, and pRRU5973H	4T4R	20 MHz, 40 MHz, 60 MHz, 80 MHz, and 100 MHz
n41 (2.6 GHz)	pRRU596x series	4T4R	60 MHz, 80 MHz, and 100 MHz
n3/n80 (1.8 GHz)	pRRU56xx series and pRRU596x series	2R	10 MHz, 15 MHz, and 20 MHz

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n1/n84 (2.1 GHz)	pRRU56xx series and pRRU596x series	2R	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz
a: x in pRRU56xx series and pRRU596x series can be a digit or a combination of a digit and a letter (for example, pRRU5961).			

5.3.4 Networking

NR FDD and NR TDD cells can be deployed in the same BBU or different BBUs. In the case that these cells are deployed in the same BBU, they can be configured on the same baseband processing unit or different baseband processing units.

The networking requirements of inter-BBU super uplink are as follows:

- Paired NR TDD and NR FDD sites must be routed to each other over an inter-BBU eXn link. For details about eXn interface configurations, see *eXn Self-Management*.
- The actual maximum one-way transmission delay between sites cannot exceed the transmission delay configured in the gNodeB. For details, see [5.1.1 Basic Functions](#). Otherwise, the inter-gNodeB SUL relationship cannot be established.

When the transmission between sites is interrupted or the transmission delay exceeds the configured delay due to network congestion, the gNodeB imposes a 10-minute penalty on the peer gNodeB. Within 10 minutes, no inter-gNodeB SUL relationship can be established for any cell served by the gNodeB under penalty. For cells for which the inter-gNodeB SUL relationship has been established, the SUL carriers configured for the involved UEs are automatically deleted. After 10 minutes, if the inter-gNodeB transmission recovers and the transmission delay is less than the configured delay, the inter-gNodeB SUL relationship can be reestablished for all cells served by the gNodeB under penalty.

- The distance between NR TDD and NR FDD sites cannot exceed 1000 m. If the distance is beyond this range, super uplink performance cannot be ensured. The inter-site distance refers to the distance between the antennas of the NR TDD cell where UEs camp and the NR FDD cell corresponding to the SUL carrier.
- The time synchronization precision of NR TDD and NR FDD sites must be within $\pm 1.5 \mu\text{s}$. If the time synchronization precision does not meet the requirement, super uplink does not take effect.
- The following base stations are compatible with this function:
 - Both NR FDD and NR TDD cells are served by macro base stations.
 - Both NR FDD and NR TDD cells are served by LampSite base stations.
 - The NR TDD cell is served by a macro base station, and the NR FDD cell is served by a LampSite base station.

- The NR TDD cell is served by a LampSite base station, and the NR FDD cell is served by a macro base station.

5.3.5 Cells

If inter-gNodeB super uplink is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the **NRDUCellCollabCoop.NulSulMaxInterGnbDelay** parameter cannot be set to **DELAY_0DOT5MS**.

If **NRDUCellSulRelation.SulType** is set to **DYN_SUL**, **NRDUCellGigaBand.LnrULRbEstCorrectCoeff** must be set to **0**.

5.3.6 Others

In SUL and NR FDD co-deployment mode, UEs support NR FDD and A5 inter-frequency measurements.

Requirements on UEs complying with 3GPP R15:

- UEs must support the combinations of NR TDD and SUL bands. If BandCombinationList in the UECapabilityInformation message contains the combinations of NR TDD and SUL bands, UEs support these combinations.
- UEs must support TTI-level switching between NR TDD and SUL bands. If the value of dynamicSwitchSUL in the UECapabilityInformation message is supported, UEs support TTI-level switching between NR TDD and SUL bands. To identify UEs that support TTI-level switching, the **SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter must be selected.

Requirements on UEs complying with 3GPP R16:

- UEs must support the combinations of NR TDD and SUL bands. If the UECapabilityInformation message contains all of the following, super uplink defined in 3GPP R16 is supported.
 - BandCombination-UplinkTxSwitch-r16 containing the combinations of NR TDD and SUL bands
 - BandCombination-UplinkTxSwitch-r16 containing supportedBandPairListNR-r16
- UEs must support the reporting of the corresponding uplink switching time. If the supportedBandPairListNR-r16 message contains uplinkTxSwitchingPeriod-r16, the corresponding uplink switching time is reported.

Requirements on UEs complying with 3GPP R18 in addition to those on UEs complying with 3GPP R16:

- UEs must support the combinations of NR TDD and SUL bands. If the UECapabilityInformation message contains all of the following, super uplink defined in 3GPP R18 is supported.
 - BandCombination-UplinkTxSwitch-v1800 or BandCombination-UplinkTxSwitch-v1830 containing the combinations of NR TDD and SUL bands
 - BandCombination-UplinkTxSwitch-v1800 containing supportedBandPairListNR-r18

- switchingPeriodFor2T-r18 indicates that UEs support the combinations of NR 2Tx and SUL 2Tx bands.
- switchingPeriodFor1T-r18 indicates that UEs support the combinations of NR 2Tx and SUL 1Tx bands, NR 1Tx and SUL 2Tx bands, or NR 1Tx and SUL 1Tx bands.
- UEs must support the reporting of the corresponding uplink switching time. If the supportedBandPairListNR-r18 message contains uplinkTxSwitchingPeriodForBandPair-r18, the corresponding uplink switching time is reported.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

For super uplink, the NR TDD carrier and SUL carrier use TDM scheduling and share the same TAG. Therefore, it is required that the subframe and uplink time of the NR TDD cell be aligned with those of the NR FDD cell. Before activating the NR TDD and NR FDD cells, ensure that the frame offset and TA offset of the NR TDD cell are the same as those of the NR FDD cell.

- The frame offset is specified by the **gNodeBParam.FrameOffset** or **gNBFreqBandConfig.FrameOffset** parameter. If both parameters are configured, the value of the **gNBFreqBandConfig.FrameOffset** parameter is used.
- The TA offset is specified by the **NRDUCell.TaOffset** parameter.

If the configurations of the NR TDD cell and NR FDD cell need to be modified, you are advised to retain the configurations of the NR TDD cell, modify the TA offset of the NR FDD cell, and then modify the frame offset of the NR FDD cell. If the frame offset and TA offset of the NR TDD cell cannot be the same as those of the NR FDD cell, contact Huawei engineers to confirm the values.

Table 5-8 and **Table 5-9** describe the parameters used for function activation in NR TDD and NR FDD cells, respectively. This section does not describe parameters related to cell establishment. For details about these parameters, see *Cell Management*.

Table 5-8 Parameters used for function activation in the NR TDD cell

Parameter Name	Parameter ID	Setting Notes
Super Uplink Switch	NRDUCellAlgoSwitch <i>.SuperUplinkSwitch</i>	Select the FLEX_SUPER_UPLINK_SW option. Select the FLEX_SUPER_UPLINK_SW option to enable intra-gNodeB super uplink. Select the FLEX_SUPER_UPLINK_SW and INTER_GNB_SUPER_UPLINK_SW options to enable inter-gNodeB super uplink.
NR DU Cell ID	NRDUCellSulRelation . <i>NrDuCellId</i>	Set this parameter based on the network plan.
SUL Type	NRDUCellSulRelation . <i>SulType</i>	Set this parameter to DYN_SUL for SUL and NR FDD co-deployment.
FDD Frequency Band	NRDUCellSulRelation . <i>FddFrequencyBand</i>	Set this parameter to the same value as the NRDUCell.FrequencyBand parameter for the NR FDD cell.
FDD UL NR-ARFCN	NRDUCellSulRelation . <i>FddULNrarfcn</i>	Set this parameter to the same value as the NRDUCell.UINrarfcn parameter for the NR FDD cell.
FDD UL Bandwidth	NRDUCellSulRelation . <i>FddULBandwidth</i>	Set this parameter to the same value as the NRDUCell.ULBandwidth parameter for the NR FDD cell.
FDD UL BWP0 Configuration Mode	NRDUCellSulRelation . <i>FddULBwp0ConfigMode</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
FDD UL BWP0 Bandwidth	NRDUCellSulRelation.FddUlBwp0Bandwidth	This parameter needs to be specified only when the NRDUCellSulRelation.FddUlBwp0ConfigMode parameter is set to MANUAL . Set this parameter based on the BWP0 bandwidth of the NR FDD cell queried by using the DSP NRDUCELLBWP command.
FDD UL BWP0 Start PRB	NRDUCellSulRelation.FddUlBwp0StartPrb	This parameter needs to be specified only when the NRDUCellSulRelation.FddUlBwp0ConfigMode parameter is set to MANUAL . Set this parameter based on the start PRB of the BWP0 bandwidth of the NR FDD cell queried by using the DSP NRDUCELLBWP command.
FDD UL Subcarrier Spacing	NRDUCellSulRelation.FddSubcarrierSpacing	Set this parameter to the same value as the NRDUCell.SubcarrierSpacing parameter for the NR FDD cell.
FDD Cyclic Prefix Length	NRDUCellSulRelation.FddCyclicPrefixLength	Set this parameter to the same value as the NRDUCell.CyclicPrefixLength parameter for the NR FDD cell.
FDD LTE NR SPCTSHR Flag	NRDUCellSulRelation.FddLteNrSptShrFlag	Set this parameter to YES when spectrum sharing is performed between NR FDD and LTE FDD cells. Set this parameter to NO when spectrum sharing is not performed between NR FDD and LTE FDD cells.
SUL Carrier Change SINR Threshold	NRDUCellSulRelation.SulCarrChangeSnrThld	The default value is recommended.
SUL Inter-freq A5 RSRP Threshold 2	NRCellSulFrequency.SulInterFreqA5RsrpThld2	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
NUL-SUL Maximum Inter-gNodeB Delay	NRDUCellCollabServ.NulSulMaxInterGnbDelay	The value of the NRDUCellCollabServ.NulSulMaxInterGnbDelay parameter of the NR TDD cell must be the same as that of the NRDUCellCollabCoop.NulSulMaxInterGnbDelay parameter of the NR FDD cell. Otherwise, super uplink cannot take effect for the corresponding NR TDD and NR FDD cells.
Inter-gNodeB SUL Flag	NRCellRelation.InterGnodebSulFlag	In inter-gNodeB scenarios, set this parameter to TRUE .
Inter-gNodeB Cell Config Mode	NRCellAnr.InterGnbCellCfgMode	The default value is recommended. If this parameter is set to AUTO , this parameter must be set to AUTO for the NR FDD cell.
SUL Algorithm Switch	NRDUCellCollabServ.SulAlgoSwitch	- The SCHEDULE_FIRST_HALF_SLOT_SW option must be selected to enable symbol-level scheduling on the SUL carrier. The SCHEDULE_FIRST_HALF_SLOT_SW option of the NRDUCellCollabCoop.SulAlgoSwitch parameter must also be selected for the corresponding NR FDD cell. Otherwise, super uplink cannot take effect for the corresponding NR TDD and NR FDD cells. - The SUL_UE_POWER_SAVING_SW option must be selected to enable SUL UE power saving. - The SUL_CONGEST_CONTROL_SW option must be selected to enable SUL congestion control.

Parameter Name	Parameter ID	Setting Notes
Multiple Frequency Uplink PRB Usage Threshold	NRDUCellSulRelation.MultiFreqULPrbUsageThld	The default value is recommended.
RSRP Threshold	NRDUCellCollabServ.RsrpThld	When SUL congestion control is enabled, the value -105 is recommended.
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	Select the SPEC_UE_IDENTIFY_SW option.
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	Select the UL_SCH_ALIGN_SW option.
Multi-Carrier Measurement Wait Timer	gNodeBAlgo.MultiCarrierMeasWaitTimer	You are advised to use the recommended value. The smaller the value of this parameter, the shorter the measurement waiting time. In scenarios with a large number of candidate frequencies, however, if the value is too small, SCells may not be configured because no measurement reports are received, or better SCells may not be configured because no measurement reports on better candidates are received. As a result, the user-perceived rates of CA UEs decrease. A larger value of this parameter indicates a longer measurement waiting time and results in a higher probability of configuring better SCells but a longer time taken for SCell configuration.

Parameter Name	Parameter ID	Setting Notes
SUL SSB SINR Threshold	NRCellSulFrequency.SulSsbSinrThld	Set this parameter based on the network plan. The smaller the value of this parameter, the higher the probability that SUL carriers are added and the higher the proportion of times SUL takes effect. However, interference between SUL carriers increases and the throughput of SUL UEs decreases. The larger the value of this parameter, the lower the probability that SUL carriers are added and the lower the proportion of times SUL takes effect. However, interference between SUL carriers decreases and the throughput of SUL UEs increases.
SUL SSB SINR Offset	NRCellSulFrequency.SulSsbSinrOffset	Set this parameter based on the network plan. A smaller value of this parameter results in a shorter interval between SUL carrier addition and removal and a higher probability of ping-pong SUL carrier addition and removal. A larger value of this parameter results in a lower probability of ping-pong SUL carrier addition and removal. However, SUL carriers are not promptly removed, interference between SUL carriers increases, and the throughput of SUL UEs decreases.

Parameter Name	Parameter ID	Setting Notes
Multi-Carrier Management Switch	NRDUCellCollabServ. <i>MultiCarrMgmtSw</i>	- Select the SMART_CARR_SEL_SW option to enable cell-level SUL and CA optimal carrier selection. - Select the SERVICE_AWARE_CARR_SEL_SW option to enable UE-level SUL and CA optimal carrier selection.
SCC Configuration Interval	NRDUCellCarrMgmt. <i>SccCfgInterval</i>	The default value is recommended.
Carrier Selection TDD SRS RSRP Threshold	NRDUCellCollabServ. <i>CarrSelTddSrsRsrpThld</i>	The default value is recommended.
Multi-Carrier Capb Comb Select Strat	NRCeIlCaMgmtCon- <i>fig.CaBandCombSele</i> <i>ctionStrat</i>	Set this parameter based on the network plan.
SUL Cov-based HO RSRP Threshold Offset	NRCeIlMeasConfig. <i>SulCovHoRsrpThldOff-</i> <i>set</i>	Set this parameter based on the network plan. - A larger value of this parameter results in a higher probability of triggering event A2 related to coverage-based handover and a lower probability of triggering event A1 related to coverage-based handover. - A smaller value of this parameter results in the opposite effects.
Multi-Frequency Algorithm Switch	NRCeIlAlgoSwitch. <i>MultiFreqAlgoSwitch</i>	Select the CARR_ADD_MEAS_QTY_PRED_SW option if virtual grids need to be used in intra-NR SUL carrier addition.

Parameter Name	Parameter ID	Setting Notes
SUL CCE Adjustment Policy	NRDUCellPdcch.SulCceAdjPolicy	Set this parameter based on the network plan. • If this parameter is set to DEFAULT, network performance is not affected. • A smaller value of this parameter results in a smaller number of CCE resources available to the SUL carrier, a smaller probability of super uplink UEs being scheduled on the SUL carrier, and a larger probability of other UEs in the NR TDD cell being scheduled. • A larger value of this parameter results in increases in the probability of super uplink UEs being scheduled on the SUL carrier and the user-perceived rate of these UEs. When there are a large number of super uplink UEs, the probability of other UEs in the NR TDD cell being scheduled decreases.
PDCCH Algorithm Switch	NRDUCellPdcch.PdccbAlgoSwitch	Select the UL_DL_CCE_SHARING_SW option.
Multi-carrier Flow Control Level	gNodeBAlgo.MultiCarrierFlowCtrlLevel	Use the recommended value.
CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	Select the UL_MULTI_CC_TA_FLEX_ADJ_SW option.

Parameter Name	Parameter ID	Setting Notes
UL Multi-Carrier Absolute TA Diff Thld	NRDUCellCarrMgmt. <i>ULMultiCcAbsTaDiffThld</i>	You are advised to use the recommended value. A smaller value of this parameter indicates a stricter requirement for TA selection based on the CC with a relatively small overall impact. However, if the value is too small, the increase in the uplink cell throughput will be relatively small. A larger value of this parameter indicates a more relaxed requirement for TA selection based on the CC with a relatively small overall impact. However, if the value is too large, there will be a relatively high probability of negative gains in the uplink throughput.
SUL Algorithm Switch	NRDUCellCollabServ. <i>SulAlgoSwitch</i>	Select the BAND_COMB_UL_TX_PWR_UDT_SW option.
Max UE Transmit Power	NRDUCellSelConfig. <i>MaxUeTransmitPower</i>	Set this parameter to 29 .
Service Differentiation Algorithm Switch	NRDUCellQciBearer. <i>ServiceDiffAlgoSwitch</i>	It is recommended that the FAST_SUL_SW option be selected for a cell that meets the following conditions: • The cell supports inter-base-station super uplink. • The number of RRC_CONNECTED UEs in the cell is insufficient to cause a heavy load.
CA Strategy Switch	NRCeCellCaMgmtConfig. <i>CaStrategySwitch</i>	It is recommended that the SUL_CA_BASED_UL_CARR_SELECT_SW option be selected.
UL Power Control Algorithm Extension Switch	NRDUCellULPcConfig. <i>ULPwrCtrlAlgoExtSwitch</i>	Select the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume distributed to SCells in the uplink when super uplink and uplink CA are both enabled.

Parameter Name	Parameter ID	Setting Notes
SUL Algorithm Switch	NRDUCellCollabServ. <i>SulAlgoSwitch</i>	Select the SUL_CONFIG_OPT_SW option to expand the range of cells that can be added as neighboring cells for SUL UEs in a cell and increase the probability of SUL taking effect for the UEs in scenarios with a short inter-site distance, for example, in densely populated urban areas.

Table 5-9 Parameters used for function activation in the NR FDD cell

Parameter Name	Parameter ID	Setting Notes
Super Uplink Switch	NRDUCellAlgoSwitch <i>.SuperUplinkSwitch</i>	Select the FLEX_SUPER_UPLINK_SW option. Select the FLEX_SUPER_UPLINK_SW option to enable intra-gNodeB super uplink. Select the FLEX_SUPER_UPLINK_SW and INTER_GNB_SUPER_UPLINK_SW options to enable inter-gNodeB super uplink.
NUL-SUL Maximum Inter-gNodeB Delay	NRDUCellCollabCoop <i>.NulSulMaxInterGnbDelay</i>	The value of the NRDUCellCollabCoop.NulSulMaxInterGnbDelay parameter of the NR FDD cell must be the same as that of the NRDUCellCollabServ.NulSulMaxInterGnbDelay parameter of the NR TDD cell. Otherwise, super uplink cannot take effect for the corresponding NR FDD and NR TDD cells.
Inter-gNodeB SUL Flag	NRCellRelation.<i>InterGnodebSulFlag</i>	In inter-gNodeB scenarios, set this parameter to TRUE .

Parameter Name	Parameter ID	Setting Notes
Inter-gNodeB Cell Config Mode	NRCellAnr.<i>InterGnbCellCfgMode</i>	The default value is recommended. If this parameter is set to AUTO , this parameter must be set to AUTO for the NR TDD cell.
SUL Algorithm Switch	NRDUCellCollabCoop.<i>SulAlgoSwitch</i>	The SCHEDULE_FIRST_HALF_SLOT_SW option must be selected to enable symbol-level scheduling on the SUL carrier. The SCHEDULE_FIRST_HALF_SLOT_SW option of the NRDUCellCollabServ.<i>SulAlgoSwitch</i> parameter must also be selected for the corresponding NR TDD cell. Otherwise, super uplink cannot take effect for the corresponding NR TDD and NR FDD cells. Select the SUL_WITH_MULTI_SLOT_CFG_NUL_SW option when an SUL carrier needs to be bound to NUL carriers with different slot configurations.
CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.<i>CaEnhancedAlgoSwitch</i>	Select the UL_MULTI_CC_TA_FLEX_ADJ_SW option.

Parameter Name	Parameter ID	Setting Notes
UL Multi-Carrier Absolute TA Diff Thld	NRDUCellCarrMgmt. <i>ULMultiCcAbsTaDiffThld</i>	You are advised to use the recommended value. A smaller value of this parameter indicates a stricter requirement for TA selection based on the CC with a relatively small overall impact. However, if the value is too small, the increase in the uplink cell throughput will be relatively small. A larger value of this parameter indicates a more relaxed requirement for TA selection based on the CC with a relatively small overall impact. However, if the value is too large, there will be a relatively high probability of negative gains in the uplink throughput.
CoMP Configuration Policy	NRDUCellComp. <i>CompConfigPol</i>	Select the UL_2CELL_COMP_BEAR_POL_SW option to enable SUL CoMP to take effect for UEs with specific QCLs.
Service Differentiation Algorithm Switch	gNBQciBearer. <i>ServiceDiffAlgoSwitch</i>	Select the COMP_SW option for UEs requiring SUL CoMP for a specific QCL.
UL Power Control Algorithm Extension Switch	NRDUCellUIPcConfig. <i>ULPwrCtrlAlgoExtSwitch</i>	Select the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume distributed to SCells in the uplink when super uplink and uplink CA are both enabled.
Reserved Switch Parameter 2	NRDUCellRsvdExt01. <i>RsvdSwParam2</i>	Select the RSVDSWPARAM2_BIT15 option of this parameter when the position of the start symbol for the front-loaded DMRS is set to different values for NR TDD and NR FDD cells in high-speed scenarios.

5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

An NR TDD cell must have already been configured before running the following commands. For details, see *Cell Management*. In addition, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

The following is an example of activation commands for intra-gNodeB super uplink. Assume that:

- The NR TDD cell ID is 100 and the gNodeB ID is 750074.
- The NR FDD cell ID is 120 and the gNodeB ID is 750074.
- The NR TDD frequency band is n78, the SUL frequency band is n84, and the NR FDD frequency band is n1.

```
//NR TDD cell configuration:
//Adding a neighbor relationship between NR TDD and NR FDD cells
ADD NRCELLRELATION: NrCellId=100, Mcc="460", Mnc="08", gNBId=750074, CellId=120;
//Adding the frequency of the NR FDD cell that serves as the neighboring cell of the NR TDD cell
ADD NRCELLFREQLATION: NrCellId=100, SsbFreqPos=5309, FrequencyBand=N1,
SubcarrierSpacing=15KHZ;
//Adding an SUL frequency for the NR TDD cell
ADD NRCELLSULFREQUENCY: NrCellId=100, SulFreqIndex=0, SsbFreqPos=5309,
SulInterFreqA5RsrpThld2=-110;
//Setting the A5 measurement waiting timer
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Adding the association between the NR TDD cell and the SUL frequency band
ADD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, FddFrequencyBand=N1,
FddULNrarfcn=386000, FddULBandwidth=FDD_UL_BW_20M, FddULBwp0ConfigMode=AUTO,
FddSubcarrierSpacing=15KHZ, FddCyclicPrefixLength=NCP;
//((Optional) Configuring the TA offset for the NR TDD cell if the TA offset configuration does not meet the
requirements
MOD NRDUCELL: NrDuCellId=100, FrequencyBand=N78, TaOffset=25600TC;
//((Optional) Configuring the frame offset for the NR TDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N78, FrameOffset=0;
//Turning on the super uplink switch for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=100, SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-1;
//Turning on the UE cooperation configuration switch for the NR TDD cell
MOD NRDUCELLUECOOP: NrDuCellId=100, SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1;
//Turning on the uplink scheduling time alignment switch for the NR TDD cell. Running the following
command will cause the NR TDD cell to restart.
MOD NRDUCELLPUSCH: NrDuCellId=100, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-1;
//((Optional) Setting the FddLteNrSpctShrFlag parameter for the NR TDD cell when spectrum sharing is
performed between NR FDD and LTE FDD cells
MOD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, FddLteNrSpctShrFlag=YES;
//((Optional) Modifying the SUL SSB SINR threshold and offset for the NR TDD cell on LampSite base
stations
MOD NRCELLSULFREQUENCY: NrCellId=100, SulFreqIndex=0, SulSsbSinrThld=DB10, SulSsbSinrOffset=20;
//((Optional) Setting the SUL CCE adjustment policy
MOD NRDUCELLPDCCH: NrDuCellId=100, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=100, SulCceAdjPolicy=FIX50;
//((Optional) Setting the level of flow control against multi-carrier operations at the gNodeB
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=100, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//Setting the absolute TA difference threshold for uplink multi-CC operation
MOD NRDUCELLCARRMGMT: NrDuCellId=100, UlMultiCcAbsTaDiffThld=16;
//Enabling the function of updating the uplink transmit power of band combinations
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=BAND_COMB_UL_TX_PWR_UDT_SW-1;
```

```
//Enabling the high-power UE function on the base station and allowing the maximum transmit power of
UEs to reach 29 dBm
MOD NRDUCELLSELCONFIG: NrDuCellId=100, MaxUeTransmitPower=29;
//Setting the policy for optimal carrier selection to consider SUL carriers
MOD NRCELLCAMGMTCONFIG: NrCellId=100, CaStrategySwitch=SUL_CA_BASED_UL_CARR_SEL_SW-1;
//((Optional) Selecting the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume
distributed to SCells in the uplink when super uplink and uplink CA are both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=100, ULPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//((Optional) Enabling configuration optimization for SUL UEs to expand the range of cells that can be
added as neighboring cells for SUL UEs in a cell and increase the probability of SUL taking effect for the
UEs in scenarios with a short inter-site distance, for example, in densely populated urban areas
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=SUL_CONFIG_OPT_SW-1;
//NR FDD cell configuration:
//Adding a neighbor relationship between NR FDD and NR TDD cells
ADD NRCELLRELATION: NrCellId=120, Mcc="460", Mnc="08", gNBId=750074, CellId=100;
//((Optional) Configuring the TA offset for the NR FDD cell if the TA offset configuration does not meet the
requirements
MOD NRDUCELL: NrDuCellId=120, FrequencyBand=N1, TaOffset=25600TC;
//((Optional) Configuring the frame offset for the NR FDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N1, FrameOffset=0;
//Turning on the super uplink switch for the NR FDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-1;
//Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=120, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//Setting the absolute TA difference threshold for uplink multi-CC operation
MOD NRDUCELLCARRMGMT: NrDuCellId=120, UIMultiCcAbsTaDiffThld=16;
//((Optional) Selecting the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume
distributed to SCells in the uplink when super uplink and uplink CA are both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=120, ULPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//((Optional) Enabling optimized cooperation between SUL and DMRS when NRDUCellDmrs.DmrsTypeApos
is set to different values for NR TDD and NR FDD cells in high-speed scenarios
MOD NRDUCELLRSVDEXT01: NrDuCellId=120, RsvdSwParam2=RSVDSWPARAM2_BIT15-1;
```

The following is an example of activation commands for inter-gNodeB super uplink. Assume that:

- The NR TDD cell ID is 100 and the gNodeB ID is 750074.
- The IDs of NR FDD cells are 220 and 221, and the gNodeB ID is 750382.
- The NR TDD frequency band is n41, the SUL frequency band is n83, and the NR FDD frequency band is n28.

An NR TDD cell must have already been configured before running the following commands. For details, see *Cell Management*.

```
//NR TDD cell configuration:
//Adding an external neighboring NR FDD cell
ADD NREXTERNALNCELL: Mcc="460", Mnc="08", gNBId=750382, CellId=220, PhysicalCellId=220, Tac=7,
SsbFreqPos=1955, NrNetworkingOption=SA;
ADD NREXTERNALNCELL: Mcc="460", Mnc="08", gNBId=750382, CellId=221, PhysicalCellId=221, Tac=7,
SsbFreqPos=1955, NrNetworkingOption=SA;
//Adding a neighbor relationship between NR TDD and NR FDD cells and setting the Inter-gNodeB SUL
Flag to TRUE. For this command, the Inter-gNodeB SUL Flag is manually configured.
MOD NRCELLRELATION: NrCellId=100, Mcc="460", Mnc="08", gNBId=750382, CellId=220,
InterGnodebSulFlag=TRUE;
MOD NRCELLRELATION: NrCellId=100, Mcc="460", Mnc="08", gNBId=750382, CellId=221,
InterGnodebSulFlag=TRUE;
//((Optional) Running the following command when the Inter-gNodeB SUL Flag is automatically configured:
MOD NRCELLANR: NrCellId=100, InterGnbCellCfgMode=AUTO;
//Adding an NR TDD cell frequency relationship
ADD NRCELLFREQRELATION: NrCellId=100, SsbFreqPos=1955, FrequencyBand=N28,
SubcarrierSpacing=15KHZ;
//Add an SUL frequency for the NR TDD cell
ADD NRCELLSULFREQUENCY: NrCellId=100, SulFreqIndex=0, SsbFreqPos=1955;
//Setting the A5 measurement waiting timer
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Adding the association between the NR TDD cell and the SUL frequency band
```

```

ADD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, FddFrequencyBand=N28,
FddULNrarfcn=143000, FddULBandwidth=FDD_UL_BW_20M, FddULBwp0ConfigMode=AUTO,
FddSubcarrierSpacing=15KHZ, FddCyclicPrefixLength=NCP;
//Modifying the inter-gNodeB delay of the NR TDD and NR FDD cells
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, NulSulMaxInterGnbDelay= DELAY_2MS;
//(Optional) Configuring the TA offset for the NR TDD cell if the TA offset configuration does not meet the
requirements
MOD NRDUCELL: NrDuCellId=100, FrequencyBand=N41, TaOffset=25600TC;
//(Optional) Configuring the frame offset for the NR TDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N41, FrameOffset=0;
//Turning on the inter-gNodeB super uplink switch for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=100,
SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-1&INTER_GNB_SUPER_UPLINK_SW-1;
//Turning on the uplink scheduling time alignment switch for the NR TDD cell. Running the following
command will cause the NR TDD cell to restart.
MOD NRDUCELLPUSCH: NrDuCellId=100, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-1;
//(Optional) Setting the SUL CCE adjustment policy
MOD NRDUCELLPDCCH: NrDuCellId=100, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=100, SulCceAdjPolicy=FIX50;
//(Optional) Setting the level of flow control against multi-carrier operations at the gNodeB
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=100, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//Setting the absolute TA difference threshold for uplink multi-CC operation
MOD NRDUCELLCARRMGMT: NrDuCellId=100, UlMultiCcAbsTaDiffThld=16;
//Enabling the function of updating the uplink transmit power of band combinations
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=BAND_COMB_UL_TX_PWR_UDT_SW-1;
//Enabling the high-power UE function on the base station and allowing the maximum transmit power of
UEs to reach 29 dBm
MOD NRDUCELLSELCONFIG: NrDuCellId=100, MaxUeTransmitPower=29;
//(Optional) Setting the FddLteNrSptShrFlag parameter for the NR TDD cell when spectrum sharing is
performed between NR FDD and LTE FDD cells
MOD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, FddLteNrSptShrFlag=YES;
//(Optional) Enabling fast SUL
MOD NRDUCELLQCIBEARER: NrDuCellId=100, Qci=6, ServiceDiffAlgoSwitch=FAST_SUL_SW-1;
//Setting the policy for optimal carrier selection to consider SUL carriers
MOD NRCELLCAMGMTCONFIG: NrCellId=100, CaStrategySwitch=SUL_CA_BASED_UL_CARR_SEL_SW-1;
//(Optional) Selecting the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume
distributed to SCells in the uplink when super uplink and uplink CA are both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=100, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//(Optional) Enabling configuration optimization for SUL UEs to expand the range of cells that can be
added as neighboring cells for SUL UEs in a cell and increase the probability of SUL taking effect for the
UEs in scenarios with a short inter-site distance, for example, in densely populated urban areas
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=SUL_CONFIG_OPT_SW-1;

//NR FDD cell configuration:
//Adding an external neighboring NR TDD cell
ADD NREXTERNALNCELL: Mcc="460", Mnc="08", gNBId=750074, CellId=100, PhysicalCellId=100, Tac=7,
SsbFreqPos=6312, NrNetworkingOption=SA;
//Adding a neighbor relationship between NR FDD and NR TDD cells and setting the Inter-gNodeB SUL
Flag to TRUE. For this command, the Inter-gNodeB SUL Flag is manually configured.
ADD NRCELLRELATION: NrCellId=220, Mcc="460", Mnc="08", gNBId=750074, CellId=100,
InterGnodebSulFlag=TRUE;
ADD NRCELLRELATION: NrCellId=221, Mcc="460", Mnc="08", gNBId=750074, CellId=100,
InterGnodebSulFlag=TRUE;
//(Optional) Running the following command when the Inter-gNodeB SUL Flag is automatically configured:
MOD NRCELLANR: NrCellId=220, InterGnbCellCfgMode=AUTO;
MOD NRCELLANR: NrCellId=221, InterGnbCellCfgMode=AUTO;
//Turning on the inter-gNodeB super uplink switch for the NR FDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=220,
SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-1&INTER_GNB_SUPER_UPLINK_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=221,
SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-1&INTER_GNB_SUPER_UPLINK_SW-1;
//Modifying the inter-gNodeB delay of the NR TDD and SUL cells
MOD NRDUCELLCOLLABCOOP: NrDuCellId=220, NulSulMaxInterGnbDelay= DELAY_2MS;
MOD NRDUCELLCOLLABCOOP: NrDuCellId=221, NulSulMaxInterGnbDelay= DELAY_2MS;
//(Optional) Configuring the TA offset for the NR FDD cell if the TA offset configuration does not meet the
requirements

```



```
MOD NRDUCELL: NrDuCellId=220, FrequencyBand=N28, TaOffset=25600TC;
MOD NRDUCELL: NrDuCellId=221, FrequencyBand=N28, TaOffset=25600TC;
//(Optional) Configuring the frame offset for the NR FDD cell if the frame offset configuration does not
meet the requirements
MOD GNBREQBANDCONFIG: FrequencyBand=N28, FrameOffset=0;
//Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=220, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=221, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//Setting the absolute TA difference threshold for uplink multi-CC operation
MOD NRDUCELLCARRMGMT: NrDuCellId=220, UIMultiCcAbsTaDiffThld=16;
MOD NRDUCELLCARRMGMT: NrDuCellId=221, UIMultiCcAbsTaDiffThld=16;
//(Optional) Selecting the MF_ACTUAL_PUSCH_ILPC_SW option to improve the accuracy of data volume
distributed to SCells in the uplink when super uplink and uplink CA are both enabled
MOD NRDUCELLULPCONFIG: NrDuCellId=220, ULPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
MOD NRDUCELLULPCONFIG: NrDuCellId=221, ULPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//(Optional) Enabling optimized cooperation between SUL and DMRS when NRDUCellDmrs.DmrsTypeAPos
is set to different values for NR TDD and NR FDD cells in high-speed scenarios
MOD NRDUCELLRSVDEXT01: NrDuCellId=220, RsvdSwParam2=RSVDSWPARAM2_BIT15-1;
MOD NRDUCELLRSVDEXT01: NrDuCellId=221, RsvdSwParam2=RSVDSWPARAM2_BIT15-1;
```

Optimization Command Examples

```
//(Optional) Running the following commands when both super uplink and CA are enabled and SUL and
CA optimal carrier selection is required
//Configuring the policy on multi-carrier capability combination selection
MOD NRCELLCAMGMTCONFIG: NrCellId=100, CaBandCombSelectionStrat=UL_PEAK_RATE_FIRST;
//Configuring multi-carrier management switches
MOD NRDUCELLCOLLABSERV: NrDuCellId=100,
MultiCarrMgmtSw=SERVICE_AWARE_CARR_SEL_SW-1&SMART_CARR_SEL_SW-1,
CarrSelTddSrsRsrpThld=-105;
//Setting the timer specifying the SCC addition interval
MOD NRDUCELLCARRMGMT: NrDuCellId=100, SccCfgInterval=8;
//(Optional) Enabling symbol-level scheduling, SUL UE power saving, and SUL congestion control
MOD NRDUCELLCOLLABSERV: NrDuCellId=100,
SulAlgoSwitch=SUL_UE_POWER_SAVING_SW-1&SCHEDULE_FIRST_HALF_SLOT_SW-1&SUL_CONGEST_CONT
ROL_SW-1;
//(Optional) Setting the multiple frequency uplink PRB usage threshold
MOD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL,
MultiFreqULPrbUsageThld=50;
//Modifying the RSRP threshold
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, RsrpThld=-105;
//Setting the SUL coverage-based handover RSRP threshold offset
MOD NRCELLMEASCONFIG: NrCellId=100, SulCovHoRsrpThldOffset=-5;
//Setting the threshold for triggering an SUL carrier change
MOD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, SulCarrChangeSinrThld=-2;
//(Optional) Enabling measurement quantity prediction for carrier addition
MOD NRCELLALGOSWITCH: NrCellId=100, MultiFreqAlgoSwitch=CARR_ADD_MEAS_QTY_PRED_SW-1;
//(Optional) Enabling an SUL carrier to be bound to NUL carriers with different slot configurations
MOD NRDUCELLCOLLABCOOP: NrDuCellId=220, SulAlgoSwitch=SUL_WITH_MULTI_SLOT_CFG_NUL_SW-1;
MOD NRDUCELLCOLLABCOOP: NrDuCellId=221, SulAlgoSwitch=SUL_WITH_MULTI_SLOT_CFG_NUL_SW-1;
//Setting the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=100, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;
//Selecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

The following is an example of deactivation commands for intra-gNodeB super uplink.

```
//Turning off the uplink scheduling time alignment switch for the NR TDD cell. Running the following
command will cause the NR TDD cell to restart.
MOD NRDUCELLPUSCH: NrDuCellId=100, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-0;
```

```
//Turning off the super uplink switch for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=100, SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-0;
//Disabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=100, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;
//Disabling the policy for optimal carrier selection that considers SUL carriers
MOD NRCELLCAMGMTCONFIG: NrCellId=100, CaStrategySwitch=SUL_CA_BASED_UL_CARR_SEL_SW-0;
//Optional Deselecting the MF_ACTUAL_PUSCH_ILPC_SW option when super uplink and uplink CA are
both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=100, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Optional Disabling configuration optimization for SUL UEs
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=SUL_CONFIG_OPT_SW-0;
//Turning off the super uplink switch for the NR FDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-0;
//Disabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=120, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;
//Deselecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-0;
//Restoring the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=120, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;
//Optional Deselecting the MF_ACTUAL_PUSCH_ILPC_SW option when super uplink and uplink CA are
both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=120, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Optional Disabling optimized cooperation between SUL and DMRS in high-speed scenarios
MOD NRDUCELLRSVDEXT01: NrDuCellId=120, RsvdSwParam2=RSVDSWPARAM2_BIT15-0;
```

The following is an example of deactivation commands for inter-gNodeB super uplink.

```
//Turning off the uplink scheduling time alignment switch for the NR TDD cell. Running the following
command will cause the NR TDD cell to restart.
MOD NRDUCELLPUSCH: NrDuCellId=100, PuschPerformanceSwitch=UL_SCH_ALIGN_SW-0;
//Turning off the super uplink switch for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=100,
SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-0&INTER_GNB_SUPER_UPLINK_SW-0;
//Disabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=100, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;
//Optional Disabling fast SUL
MOD NRDUCELLQCIBEARER: NrDuCellId=100, Qci=6, ServiceDiffAlgoSwitch=FAST_SUL_SW-0;
//Disabling the policy for optimal carrier selection that considers SUL carriers
MOD NRCELLCAMGMTCONFIG: NrCellId=100, CaStrategySwitch=SUL_CA_BASED_UL_CARR_SEL_SW-0;
//Optional Deselecting the MF_ACTUAL_PUSCH_ILPC_SW option when super uplink and uplink CA are
both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=100, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Optional Disabling configuration optimization for SUL UEs
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=SUL_CONFIG_OPT_SW-0;
//Turning off the super uplink switch for the NR FDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=220,
SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-0&INTER_GNB_SUPER_UPLINK_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=221,
SuperUplinkSwitch=FLEX_SUPER_UPLINK_SW-0&INTER_GNB_SUPER_UPLINK_SW-0;
//Disabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=220, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;
MOD NRDUCELLCARRMGMT: NrDuCellId=221, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;
//Deselecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-0;
//Restoring the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=220, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=221, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;
//Optional Deselecting the MF_ACTUAL_PUSCH_ILPC_SW option when super uplink and uplink CA are
both enabled
MOD NRDUCELLULPCCONFIG: NrDuCellId=220, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
MOD NRDUCELLULPCCONFIG: NrDuCellId=221, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Optional Disabling optimized cooperation between SUL and DMRS in high-speed scenarios
MOD NRDUCELLRSVDEXT01: NrDuCellId=220, RsvdSwParam2=RSVDSWPARAM2_BIT15-0;
MOD NRDUCELLRSVDEXT01: NrDuCellId=221, RsvdSwParam2=RSVDSWPARAM2_BIT15-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observing Counters

When super uplink with SUL and NR FDD co-deployment is enabled and UEs supporting super uplink exist on the network, observe counters in [Table 5-10](#). If they both have a non-zero value, super uplink has taken effect. If the counter value is always zero, super uplink has not taken effect.

Table 5-10 Counters used to verify activation of super uplink

Counter ID	Counter Name	Description
1911820757	N.User.UEnhTimeFreq.Avg	Observe the counter of the NR TDD cell.
1911829385	N.User.SUL.UL.Avg	Observe the counter of the NR FDD cell.

To check whether inter-gNodeB super uplink has taken effect, observe counters in [Table 5-11](#). If they both have a non-zero value, inter-gNodeB super uplink has taken effect. If the counter value is always zero, inter-gNodeB super uplink has not taken effect.

Table 5-11 Counters used to verify activation of inter-gNodeB super uplink

Counter ID	Counter Name	Description
1911829388	N.User.IntergNB.UEnhTimeFreq.Avg	Observe the counter of the NR TDD cell.
1911829384	N.User.IntergNB.SUL.UL.Avg	Observe the counter of the NR FDD cell.

When super uplink with SUL and NR FDD co-deployment is enabled, if either of the counters listed in [Table 5-12](#) produces a non-zero value, virtual-grid-based RSRP prediction results are used during SUL carrier addition. If the values of the counters are both zero, virtual-grid-based RSRP prediction results are not used during SUL carrier addition.

Table 5-12 Counters used to verify the simultaneous use of super uplink and virtual grid

Counter ID	Counter Name	Description
1911840080	N.SUL.SulCarrier.VirtualGrid.Add.Att	Observe the counter of the NR TDD cell.
1911840081	N.SUL.SulCarrier.VirtualGrid.Add.Succ	Observe the counter of the NR FDD cell.

Tracing Signaling

Start a Uu interface tracing task on the gNodeB. If the spCellConfigDedicated field in the RRCReconfiguration message contains the *supplementaryUplink* IE, super uplink has taken effect.

5.4.3 Network Monitoring

Super Uplink

Super uplink increases the average uplink throughput of SUL UEs in a cell. As a result, the average uplink UE throughput of the entire cell increases, the proportion of UEs with throughput falling into a high-throughput range increases, and the proportion of UEs with throughput falling into a low-throughput range decreases.

- Monitoring SUL UEs in a cell involves the following indicators:
 - The average uplink throughput of SUL UEs in a cell can be obtained using the following formula: $(N.ThpVol.SUL.UL - N.ThpVol.SUL.UL.SmallPkt) / N.ThpTime.SUL.UL.RmvSmallPkt$.
 - The proportion of SUL UEs whose throughput falls into a low-throughput range can be obtained by dividing the number of samples in the low-throughput range by the total number of samples in a cell. A low-throughput range is relative (for example, range 0 is a low-throughput range compared with range 1).
 - The proportion of SUL UEs whose throughput falls into a high-throughput range can be obtained by dividing the number of samples in the high-throughput range by the total number of samples in a cell. A high-throughput range is relative (for example, range 14 is a high-throughput range compared with range 13). [Table 5-13](#) lists the counters used to observe the number of UEs in each throughput range.

Table 5-13 Number of samples with the uplink throughput of SUL UEs in each range

Counter ID	Counter Name
1911829662	N.Thp.SUL.UL.Samp.Index0

Counter ID	Counter Name
1911829663	N.Thp.SUL.UL.Samp.Index1
1911829664	N.Thp.SUL.UL.Samp.Index2
1911829665	N.Thp.SUL.UL.Samp.Index3
1911829666	N.Thp.SUL.UL.Samp.Index4
1911829667	N.Thp.SUL.UL.Samp.Index5
1911829668	N.Thp.SUL.UL.Samp.Index6
1911829669	N.Thp.SUL.UL.Samp.Index7
1911829670	N.Thp.SUL.UL.Samp.Index8
1911829671	N.Thp.SUL.UL.Samp.Index9
1911829672	N.Thp.SUL.UL.Samp.Index10
1911829673	N.Thp.SUL.UL.Samp.Index11
1911829674	N.Thp.SUL.UL.Samp.Index12
1911829675	N.Thp.SUL.UL.Samp.Index13
1911829676	N.Thp.SUL.UL.Samp.Index14

- Monitoring the overall performance of a cell involves the following indicators:
 - The average uplink UE throughput in a cell is measured by **User Uplink Average Throughput (DU)**.
 - The proportion of UEs whose throughput falls into a low-throughput range can be obtained by dividing the number of samples in the low-throughput range by the total number of samples in a cell. A low-throughput range is relative (for example, range 0 is a low-throughput range compared with range 1).
 - The proportion of UEs whose throughput falls into a high-throughput range can be obtained by dividing the number of samples in the high-throughput range by the total number of samples in a cell. A high-throughput range is relative (for example, range 14 is a high-throughput range compared with range 13). [Table 5-14](#) lists the counters used to observe the number of UEs in each throughput range.

Table 5-14 Number of samples with the uplink throughput in each range in a cell

Counter ID	Counter Name
1911816708	N.Thp.UL.Samp.Index0
1911816709	N.Thp.UL.Samp.Index1
1911816710	N.Thp.UL.Samp.Index2

Counter ID	Counter Name
1911816711	N.Thp.UL.Samp.Index3
1911816712	N.Thp.UL.Samp.Index4
1911816713	N.Thp.UL.Samp.Index5
1911816714	N.Thp.UL.Samp.Index6
1911816715	N.Thp.UL.Samp.Index7
1911816716	N.Thp.UL.Samp.Index8
1911816717	N.Thp.UL.Samp.Index9
1911816937	N.Thp.UL.Samp.Index10
1911816941	N.Thp.UL.Samp.Index11
1911816938	N.Thp.UL.Samp.Index12
1911816939	N.Thp.UL.Samp.Index13
1911816940	N.Thp.UL.Samp.Index14

SUL CoMP

To compare the throughput before and after SUL CoMP is enabled, you are advised to select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter for the NR TDD cell one week before SUL CoMP is enabled so that statistics of throughput-related counters for UEs in overlapping areas can be collected in advance.

SUL CoMP increases the uplink throughput of UEs in overlapping areas that excludes small-packet throughput without decreasing the average uplink UE throughput and the uplink UE throughput that excludes small-packet throughput. You can evaluate the gains of SUL CoMP by comparing the values of the following performance counters before and after this function is enabled. The counter values depend on parameter settings.

To observe the gains of SUL CoMP, you are advised to set the **NRCellComp.OverLappingRsrpOffset** (for the NR FDD cell) and **NRCellComp.UlCompRsrpOffset** (for the NR TDD cell where super uplink UEs are located) parameters to the same value. The reason is as follows: The offset for event A3 in the overlapping area of the NR FDD cells is determined by the **NRCellComp.OverLappingRsrpOffset** parameter. The RSRP difference for event A5 indicating whether SUL CoMP takes effect for super uplink UEs is determined by the **NRCellComp.UlCompRsrpOffset** parameter.

- Uplink throughput of UEs in overlapping areas excluding small-packet throughput = $(\text{N.ThpVol.UL.CellOverlap} - \text{N.ThpVol.UL.SmallPkt.CellOverlap}) / \text{N.ThpTime.UL.RmvSmallPkt.CellOverlap}$
- Uplink UE throughput excluding small-packet throughput = $(\text{N.ThpVol.UL} - \text{N.ThpVol.UE.UL.SmallPkt}) / \text{N.ThpTime.UE.UL.RmvSmallPkt}$

- Average uplink UE throughput = $\text{N.ThpVol.UL} / \text{N.ThpTime.UL}$

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *LTE FDD and NR SUL Dynamic Spectrum Sharing*
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
- *Cell Management*
- *5G Networking and Signaling*
- *Power Control*
- *CoMP*
- *UE Power Saving*
- *URLLC*
- *Carrier Aggregation*
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *UL and DL Decoupling*
- *Mobility Load Balancing in Connected Mode*
- *SA Mobility Management in Connected Mode*
- *Energy Conservation and Emission Reduction*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Device-Pipe Synergy*
- *Admission Control*
- *Downlink Scheduling*
- *eXn Self-Management*
- *Resource Control in NSA Networking*
- *Multi-Frequency Smart Selection*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Fusion Cell (Low-Frequency TDD)*
- *MIMO (TDD)*
- *VoNR*

- *CA Experience Improvement*
- *Channel Management*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Positioning*
- *RedCap*
- *QCI-based Resource Control*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Service-differentiated Experience Guarantee*
- *Multi-RAT Coordinated Energy Saving*
- *Network Slicing*
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception"